Force Control and Human Interaction with the FR3 manipulator

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Dedication

Special thanks to my family for supporting me throughout the years.

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I would like to thank the University of Ioannina and, in particular, my supervisor, Assistant Professor Konstantinos Vlachos, for his guidance and support during the preparation of this thesis.

Abstract

This thesis investigates advanced force-aware control strategies for the Franka Research 3 (FR3), a 7-degree-of-freedom collaborative robotic manipulator, with a primary focus on safe physical human–robot interaction and robot-assisted physiotherapy. Using the robot's integrated joint-torque sensors and the Franka Control Interface (FCI), a torque-based impedance control framework is developed to provide compliant and responsive interaction with the user.

The work also exploits the kinematic redundancy of the FR3 through differential kinematics and null-space projection, allowing secondary objectives such as posture optimization and obstacle avoidance without interfering with the main end-effector task.

A major motivation of the thesis is the use of the robot as an assistive system for rehabilitation exercises. Based on physiotherapy-inspired upper-limb movements, shoulder and elbow exercises were implemented to examine whether the robot can safely guide motion, detect user resistance, adapt its behavior, and provide assistance when required. The robot is intended to support, rather than replace, the physiotherapist. The rehabilitation-oriented movements were selected with the support of informal discussions and practical demonstrations provided by individuals familiar with physiotherapy exercises. Their input was used to guide the choice and adaptation of representative shoulder and elbow movements for implementation with the FR3.

The proposed methods are evaluated through real-world experiments involving contact tasks, external disturbances, human-guided motion, and rehabilitation-oriented exercises. The results demonstrate the potential of force-aware compliant control to improve the safety, adaptability, and effectiveness of physical human–robot interaction.

Artificial intelligence tools were used as supporting tools for software development, debugging, and the preparation and linguistic refinement of the thesis text, while all technical decisions, experimental procedures, and final results were reviewed and validated by the author.

Extended Abstract

This diploma thesis focuses on the development and implementation of advanced interaction-control strategies for the Franka Research 3 (FR3) robotic manipulator, with particular emphasis on safe physical human-robot interaction and robot-assisted physiotherapy. In contrast to traditional industrial robots, which are usually based on rigid position control, rehabilitation-oriented human-robot collaboration requires robotic systems that are physically aware, compliant, and capable of adapting their behavior according to the forces applied by the user. The FR3 was selected as the reference platform because it integrates torque sensors in all seven joints, offering high dynamic transparency and making it particularly suitable for force-sensitive interaction.

The primary motivation of this work is to investigate whether the FR3 can operate as an assistive robotic system during the execution of physiotherapy exercises. The robot is not intended to replace the physiotherapist, but rather to support the user during repetitive rehabilitation movements by guiding the desired motion, detecting physical resistance, and adapting the level of assistance accordingly. The design of the rehabilitation-oriented experiments was inspired by practical demonstrations and discussions concerning upper-limb exercises commonly used for individuals recovering from neurological conditions, such as stroke. Based on this motivation, representative shoulder and elbow movements were selected and adapted for implementation with the robotic system.

The theoretical background of the thesis is based on differential kinematics and on the Elementary Transform Sequence (ETS) method for describing the kinematic chain. The Jacobian matrix, denoted by J, plays a central role because it maps task-space velocities to joint velocities and, through force-velocity duality, maps end-effector forces to joint torques:

$$\tau = J^T F. \tag{1}$$

Special emphasis is placed on the redundancy of the 7-DOF manipulator. Through null-space projection,

$$N = I - J^\dagger J, \tag{2}$$

secondary objectives, such as obstacle avoidance and posture optimization, can be executed without significantly disturbing the primary end-effector interaction task. This

capability is particularly important in human-centered applications, where the robot must maintain the required end-effector behavior while simultaneously preserving safe and suitable joint configurations.

The thesis implements a torque-based impedance control framework. The manipulator is modeled as a virtual mass-spring-damper system, allowing its apparent mechanical behavior to be adjusted and enabling compliant interaction during physical contact. The Franka Control Interface (FCI) is used to execute high-frequency control commands at 1 kHz, providing the responsiveness required for smooth interaction with the human user.

The proposed framework is evaluated experimentally on the real FR3 platform through a series of tests involving free-space motion, contact interaction, disturbance response, hand-guiding, and rehabilitation-oriented movements. Particular attention is given to two physiotherapy-inspired upper-limb exercises: a vertical shoulder movement and an elbow flexion-extension movement. These experiments investigate whether the robot can follow a predefined trajectory while simultaneously detecting user-applied resistance, reducing or pausing its motion when necessary, and providing assistance when the user requires support.

The experimental results demonstrate that the combination of force-aware impedance control, torque sensing, and redundancy management enables the FR3 to interact safely and adaptively with a human participant. The robot is able to respond to external disturbances, maintain controlled physical interaction, and modify its behavior according to the measured resistance during the rehabilitation-oriented exercises. In addition, the use of the FR3's integrated joint-torque sensors reduces the dependence on external wrist force/torque sensors, thereby decreasing the hardware complexity of the system.

Overall, this work demonstrates the potential of the FR3 as an assistive platform for physical human-robot interaction and rehabilitation-oriented applications. The integration of compliant torque control, force-aware adaptation, and kinematic redundancy provides a flexible framework for the safe and repeatable execution of physiotherapy exercises, while also establishing a basis for future developments in adaptive and personalized robot-assisted rehabilitation.

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Chapter 1

Introduction

1.1 Motivation

The field of industrial robotics is undergoing a significant paradigm shift. For many decades, robotic manipulators were designed as rigid, high-precision machines operating in structured environments and physically separated from humans by safety barriers. However, the emergence of Industry 4.0 and human-robot collaboration has created the need for robotic systems that can safely share their workspace with people and interact with them through physical contact.

The primary motivation of this thesis is to investigate how a robotic manipulator can operate safely in situations where human contact is not considered an error, but rather an essential part of the task. This is particularly relevant in applications such as robot-assisted physiotherapy, where the robot must physically interact with the user, guide rehabilitation movements, detect resistance, and adapt its behavior according to the forces applied by the human.

Traditional position-controlled robots generally lack the dynamic transparency and compliance required for this type of interaction. For this reason, this work focuses on force-aware and compliant control strategies that allow the robot to respond smoothly and safely to external forces while still maintaining the desired task performance. The overall motivation is therefore to move from rigid automation toward adaptive and physically interactive robotic systems that can support humans safely and effectively.

1.2 State of the Art of Control Algorithms

The state of the art in robotics is defined by the transition from rigid automation to physically aware autonomy. Stable and sensitive physical interaction with the environment is one of the defining characteristics of advanced manipulation. A robot that can regulate its applied force and adapt its apparent compliance can operate more

safely and effectively in collaborative and uncertain environments.

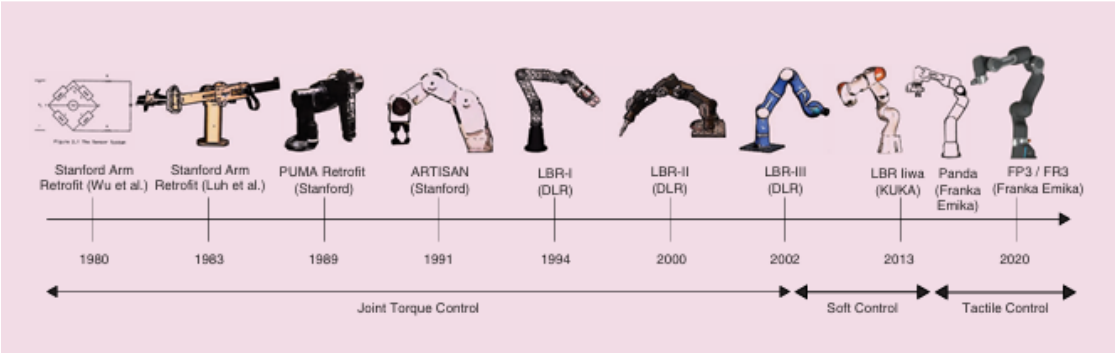

Figure 1.1: Important development steps in joint-torque control, from early theoretical foundations to modern product implementations.

Figure 1.1 depicts important development steps in joint-torque control, beginning with early theoretical work and reaching modern product implementations. These developments laid the foundations for soft and tactile control in advanced industrial robots. The mathematical framework for controlling the interaction between the FR3 and its environment can be categorized according to how the controller handles the relationship between motion and force.

1.3 Goals

The main goal of this thesis is to study, implement, and experimentally evaluate force-control methods for the Franka Research 3 manipulator, with emphasis on safe physical interaction between the robot, the environment, and a human operator. The work focuses not only on the theoretical design of force-control algorithms, but also on the practical implementation of a complete experimental workflow using the Franka Control Interface, `libfranka` and real robot experiments.

The specific objectives of this thesis are:

1. **Study of the FR3 platform and its force-control capabilities:** to understand the kinematic, dynamic, sensing, and control characteristics of the Franka Research 3, especially its seven-degree-of-freedom structure, integrated joint torque sensing, external wrench estimation, and torque-level control through the Franka Control Interface.

2. **Implementation of direct real-time control using `libfranka`:** to establish direct communication with the real FR3 without ROS, read the robot state, access force and torque estimates, send control commands, and execute safe real-time experiments through the Franka Control Interface.

3. **Development and testing of force-control strategies:** to implement and compare the main interaction-control approaches used in this thesis, namely Cartesian impedance control, Cartesian admittance control, and Cartesian hybrid force/position control.

4. **Evaluation of compliant human–robot interaction:** to investigate how the FR3 behaves during physical interaction, including soft contact, external human pushes, hand-guiding behavior, and variable stiffness response.

5. **Execution of real robot experiments:** to perform a set of practical experiments, including free-space joint motion, contact approach with soft objects, constant force pressing, human disturbance response, hand-guiding, pick-and-place motion, variable stiffness impedance, physiotherapy-inspired interaction, and contact with different materials.

Overall, the thesis aims to demonstrate that the FR3 can be used as a force-aware collaborative robot platform for safe contact tasks and human–robot interaction. The final objective is not only to design controllers theoretically, but also to validate them through a complete practical pipeline on real hardware execution.

Chapter 2

Mathematical Model

2.1 Robot Configuration

To address the challenges of force-sensitive interaction, this research uses the Franka Research 3, a state-of-the-art 7-DOF tactile robot platform. Unlike conventional collaborative robots that estimate forces only through motor currents, the FR3 is equipped with integrated strain-gauge torque sensors in all seven joints.

This hardware architecture provides the following advantages:

- **Redundancy:** the 7-DOF anthropomorphic configuration provides an internal degree of freedom, or null space, which is essential for obstacle avoidance and posture optimization.
- **High-bandwidth control:** through the Franka Control Interface, the robot supports a 1 kHz control loop, which is a prerequisite for stable and smooth impedance control.
- **Force sensitivity:** the ability to sense and command torques directly allows the FR3 to perform delicate tasks that are difficult for purely position-controlled manipulators.

Dynamic transparency describes the ability of the FR3 to compensate for its own weight, friction, and dynamic effects so that interaction with a human user feels as natural as possible. In this thesis, the term unstructured environment refers to a workspace that can contain humans, objects, obstacles, and uncertain contact conditions.

A precise kinematic model is required in order to control the FR3. The Denavit-Hartenberg convention has traditionally been used for modeling serial robotic chains. However, this thesis adopts the Elementary Transform Sequence (ETS) representation. Unlike Denavit-Hartenberg parameters, which can be counter-intuitive and constrained by specific frame alignments, ETS describes the robot as a sequence of elementary

translations and rotations,

$$T_x,\ T_y,\ T_z,\ R_x,\ R_y,\ R_z. \tag{2.1}$$

This representation gives a transparent description of the FR3's 7-DOF anthropomorphic structure and facilitates the integration of link-side torque sensing into the kinematic chain.

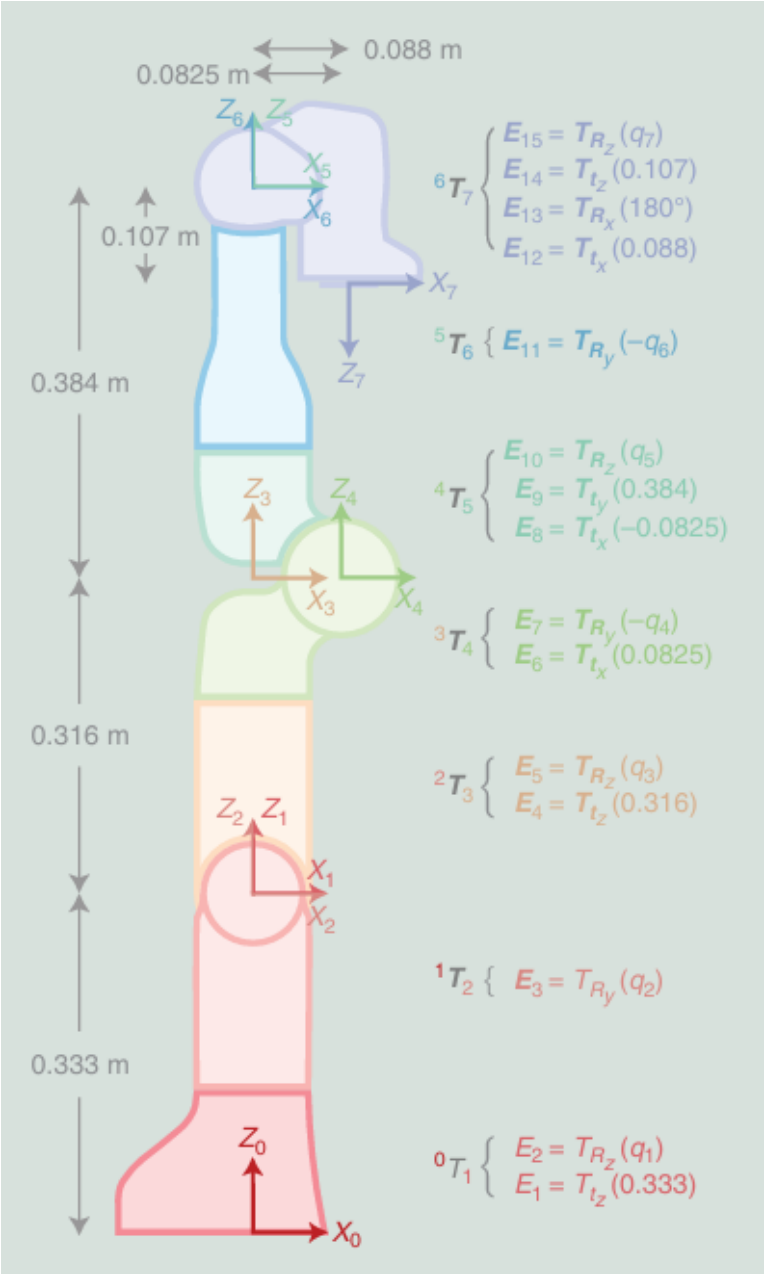

Figure 2.1: Elementary Transform Sequence of a 7-DOF Franka Emika Panda/FR3-type serial-link manipulator in its zero-angle configuration.

2.2 Forward and Inverse Kinematics

Kinematics describes the motion of a robotic manipulator without directly considering the forces and torques that generate this motion. For the Franka Research 3, which is a serial robotic manipulator with seven revolute joints, kinematics is essential because it provides the mathematical relationship between the joint variables and the pose of the end-effector. This relationship is used in trajectory generation, motion planning, Cartesian control, impedance control, admittance control, and hybrid force/position control.

The joint configuration of the FR3 is represented by the vector:

$$q = \begin{bmatrix} q_1 & q_2 & q_3 & q_4 & q_5 & q_6 & q_7 \end{bmatrix}^T \in \mathbb{R}^7$$

where q_i is the position of joint i. Since all seven joints of the FR3 are revolute, each joint variable represents an angular displacement.

Forward kinematics maps the joint-space coordinates $q \in \mathbb{R}^7$ to the pose of the end-effector in operational space. The end-effector pose is usually represented as an element of the Special Euclidean group $SE(3)$, which contains both position and orientation:

$${}^0T_e(q) = \begin{bmatrix} {}^0R_e(q) & {}^0p_e(q) \\ 0 & 1 \end{bmatrix} \in SE(3)$$

where ${}^0R_e(q)$ is the rotation matrix of the end-effector with respect to the base frame, and ${}^0p_e(q)$ is the Cartesian position of the end-effector.

Using the Elementary Transform Sequence (ETS) framework, the pose of the end-effector is computed as the product of elementary homogeneous transformation matrices from the base frame to the flange frame:

$${}^0T_e(q) = T_1(q_1)T_2(q_2)\cdots T_7(q_7)T_e$$

Each transformation $T_i(q_i)$ describes the rotation or translation associated with a link or joint of the robot. The final transformation T_e represents the fixed transformation from the last robot link to the end-effector or tool frame. This formulation is especially useful because it gives a clear and systematic description of the serial kinematic chain of the FR3.

Forward kinematics is used when the joint angles are known and the end-effector pose must be computed. For example, during an experiment, the robot continuously measures its joint positions through its encoders. By applying the forward kinematics model, the controller can compute the current Cartesian pose of the end-effector. This is required in Cartesian motion control, contact experiments, impedance control, and

force/position hybrid control.

Inverse kinematics solves the opposite problem. Given a desired end-effector pose, inverse kinematics determines the joint configuration that allows the robot to reach that pose. For a desired Cartesian pose $T_d \in SE(3)$, the inverse kinematics problem can be written as:

$$^0T_e(q) = T_d$$

or equivalently:

$$f(q) = x_d$$

where $f(q)$ represents the forward kinematics function and x_d is the desired task-space pose.

For the FR3, inverse kinematics is more complex than for a non-redundant 6-DOF manipulator because the FR3 has seven degrees of freedom. A general pose in 3D space requires six variables: three for position and three for orientation. Since the FR3 has one additional degree of freedom, the inverse-kinematics problem generally has infinitely many solutions. This redundancy allows the robot to reach the same end-effector pose with different elbow configurations.

In practice, inverse kinematics is often solved numerically. A common approach is to linearize the relationship between joint motion and Cartesian motion using the Jacobian matrix:

$$\Delta x \approx J(q)\Delta q$$

where Δx is the Cartesian pose error and Δq is the required joint correction. The joint update can then be computed iteratively until the Cartesian error becomes sufficiently small.

For redundant robots, a common solution is based on the pseudoinverse of the Jacobian:

$$\Delta q = J^\dagger(q)\Delta x$$

where $J^\dagger(q)$ is the Moore–Penrose pseudoinverse. However, near singular configurations, the pseudoinverse may produce very large joint updates. For this reason, this thesis considers numerical methods such as damped least squares (DLS), which improves stability near singularities.

The DLS joint velocity solution is:

$$\dot{q} = J^T \left(JJ^T + \lambda^2 I \right)^{-1} v$$

where λ is the damping coefficient and v is the desired Cartesian velocity or pose correction. The damping term $\lambda^2 I$ prevents numerical instability when the Jacobian becomes close to singular.

The damping coefficient controls the trade-off between accuracy and stability. A small value of λ makes the solution behave similarly to the standard pseudoinverse, giving accurate tracking when the robot is far from singularities. A larger value of λ reduces excessive joint velocities near singularities, but it may also reduce Cartesian tracking accuracy.

In an iterative inverse-kinematics algorithm, the joint configuration is updated as:

$$q_{k+1} = q_k + \Delta q_k$$

where:

$$\Delta q_k = J^T(q_k) \left(J(q_k) J^T(q_k) + \lambda^2 I \right)^{-1} \Delta x_k$$

and Δx_k is the pose error at iteration k. The iteration stops when the pose error is below a predefined threshold or when the maximum number of iterations has been reached.

For the FR3, inverse kinematics is not only used to reach a desired position. It is also useful for choosing a safe and efficient posture. Because the robot is redundant, the controller can select among multiple possible joint configurations by considering additional criteria, such as avoiding joint limits, avoiding obstacles, improving manipulability, reducing joint motion, or keeping the elbow in a preferred region of the workspace.

Therefore, forward and inverse kinematics provide the foundation for all higher-level control strategies used in this thesis. Forward kinematics allows the robot to estimate where its end-effector is, while inverse kinematics allows the controller to compute how the joints should move in order to achieve a desired Cartesian task.

2.3 The Jacobian Matrix

The Jacobian matrix is the fundamental operator of differential kinematics. While forward kinematics relates joint positions to end-effector pose, the Jacobian relates joint velocities to end-effector velocity. For the FR3, the Jacobian is especially important because it is used not only for motion control, but also for force control and torque mapping.

The differential kinematic relation is:

$$v = J(q)\dot{q}$$

where v is the Cartesian velocity, $J(q)$ is the geometric Jacobian, and $\dot{q}$ is the vector of joint velocities. For a seven-degree-of-freedom manipulator such as the FR3, the joint velocity vector is:

$$\dot{q} = \begin{bmatrix} \dot{q}_1 & \dot{q}_2 & \dot{q}_3 & \dot{q}_4 & \dot{q}_5 & \dot{q}_6 & \dot{q}_7 \end{bmatrix}^T \in \mathbb{R}^7$$

The Cartesian velocity of the end-effector is usually expressed as a six-dimensional twist:

$$v = \begin{bmatrix} v_x \\ v_y \\ v_z \\ \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} \in \mathbb{R}^6$$

where v_x, v_y, v_z are the linear velocity components and $\omega_x, \omega_y, \omega_z$ are the angular velocity components. Therefore, for the FR3, the Jacobian has dimension:

$$J(q) \in \mathbb{R}^{6\times 7}$$

The Jacobian can be separated into two parts:

$$J(q) = \begin{bmatrix} J_v(q) \\ J_\omega(q) \end{bmatrix}$$

where $J_v(q)$ maps joint velocities to linear end-effector velocity, and $J_\omega(q)$ maps joint velocities to angular end-effector velocity.

For a revolute joint i, the contribution of the joint to the Jacobian can be written as:

$$J_{v_i} = z_{i-1} \times (p_e - p_{i-1})$$

$$J_{\omega_i} = z_{i-1}$$

where z_{i-1} is the axis of rotation of joint i, p_{i-1} is the position of joint i, and p_e is the position of the end-effector. This shows that each joint contributes to the end-effector velocity depending on its axis and its position relative to the end-effector.

The Jacobian is used in several important ways. First, it is used for velocity control. If a desired Cartesian velocity v_d is given, the corresponding joint velocity can be computed using a pseudoinverse:

$$\dot{q} = J^{\dagger}(q) v_d$$

Second, the Jacobian is used in inverse kinematics, where small Cartesian pose corrections are converted into joint corrections. Third, it is used in force control through the principle of force–velocity duality.

In force control, an external wrench applied at the end-effector can be mapped to joint torques using the transpose of the Jacobian:

$$\tau = J^T(q) F$$

where:

$$F = \begin{bmatrix} F_x & F_y & F_z & M_x & M_y & M_z \end{bmatrix}^T$$

is the Cartesian wrench acting on the end-effector. The vector F contains three force components and three moment components. The resulting joint torque vector is:

$$\tau = \begin{bmatrix} \tau_1 & \tau_2 & \tau_3 & \tau_4 & \tau_5 & \tau_6 & \tau_7 \end{bmatrix}^T$$

This relationship is very important for the FR3 because the robot supports torque-level control. If a desired Cartesian interaction force is specified, the controller can compute the required joint torques using the Jacobian transpose. This is the basic principle used in Cartesian impedance control and hybrid force/position control.

The Jacobian is also used to estimate the effect of external forces. If an external force acts at the end-effector, it creates torques at the joints. By comparing measured joint torques with model-predicted torques, the FR3 can estimate external interaction forces and torques. This makes the Jacobian an essential link between the robot's internal torque sensors and the Cartesian force-control task.

Another important use of the Jacobian is the analysis of manipulability. The manipulability of a robot describes how easily the end-effector can move in different Cartesian directions. A common manipulability measure is:

$$w(q) = \sqrt{\det\left(J(q) J^T(q)\right)}$$

When $w(q)$ is large, the robot can move well in many Cartesian directions. When $w(q)$ approaches zero, the robot is close to a singularity. For the FR3, manipulability is useful for selecting safe postures and avoiding configurations where motion or force control becomes difficult.

Therefore, the Jacobian matrix is central to the control framework of this thesis. It connects joint-space motion to Cartesian motion, Cartesian forces to joint torques, and redundancy resolution to task performance. Without the Jacobian, it would not

be possible to formulate Cartesian impedance control, admittance control, hybrid force/position control, or null-space posture optimization.

2.4 Redundancy and Null-Space Projection

A robotic manipulator is kinematically redundant when it has more degrees of freedom than the number of variables required to describe a given task. In terms of the spaces defined above, a manipulator is intrinsically redundant when the dimension of the task space is smaller than the dimension of the joint space, i.e., $m < n$. Redundancy is task-dependent: the same manipulator may be redundant for one task and non-redundant for another.

The Franka Research 3 is a redundant manipulator because it has seven degrees of freedom, while a general end-effector pose in three-dimensional space requires six variables: three for position and three for orientation. Therefore, for a full Cartesian pose task:

$$n = 7, \qquad m = 6$$

and:

$$n > m$$

This means that the FR3 has one extra degree of freedom. This extra degree of freedom does not change the end-effector pose directly, but it allows the robot to modify its internal posture while keeping the end-effector at the same position and orientation.

Redundancy makes a robotic manipulator more dexterous and adaptable. The human arm is a familiar example, with approximately seven main degrees of freedom: three at the shoulder, one at the elbow, and three at the wrist. If the position and orientation of the hand are fixed, only six degrees of freedom are required, so the elbow can still move because of the additional degree of freedom. This makes obstacle avoidance and comfortable posture selection possible.

Similarly, the FR3 can use its redundant degree of freedom to move the elbow or change its posture while keeping the end-effector fixed. This is useful in several situations:

- avoiding obstacles in the workspace,
- avoiding joint limits,
- avoiding self-collisions,
- improving manipulability,

- keeping a comfortable elbow posture,
- reducing unnecessary joint motion,
- increasing safety during human–robot interaction.

When a manipulator is redundant, the Jacobian matrix has more columns than rows. For the FR3:

$$J(q) \in \mathbb{R}^{6\times 7}$$

As a result, the differential kinematic equation:

$$v = J(q)\dot{q}$$

has infinitely many solutions for $\dot{q}$, provided that the desired Cartesian velocity v is feasible. The Moore–Penrose pseudoinverse gives the minimum-norm solution:

$$\dot{q} = J^{\dagger}v$$

However, because the robot is redundant, additional joint motion can be added without changing the end-effector velocity. The general solution is:

$$\dot{q} = J^{\dagger}v + \left(I - J^{\dagger}J\right)\dot{q}_0$$

The first term:

$$J^{\dagger}v$$

performs the primary Cartesian task. The second term:

$$\left(I - J^{\dagger}J\right)\dot{q}_0$$

lies in the null space of the Jacobian and therefore does not affect the end-effector velocity.

The matrix:

$$N = I - J^{\dagger}J$$

is called the null-space projection matrix. It projects any arbitrary joint velocity vector $\dot{q}_0$ into the null space of the Jacobian. This means that:

$$J(q)N = 0$$

and therefore:

$$J(q)N\dot{q}_0 = 0$$

This proves that the null-space motion does not contribute to the end-effector velocity. It only changes the internal configuration of the robot.

The vector $\dot{q}_0$ can be chosen according to a secondary objective. For example, to avoid joint limits, a potential function can be defined that becomes large when a joint approaches its limit. The robot can then move in the direction that reduces this potential while still performing the main Cartesian task.

A common form is:

$$\dot{q}_0 = -k_0 \nabla H(q)$$

where $H(q)$ is a secondary cost function and k_0 is a positive gain. The complete redundancy-resolution law becomes:

$$\dot{q} = J^{\dagger} v - \left(I - J^{\dagger} J\right) k_0 \nabla H(q)$$

If $H(q)$ represents joint-limit avoidance, the null-space term moves the joints away from their limits. If $H(q)$ represents obstacle avoidance, the null-space term moves the robot body away from obstacles. If $H(q)$ represents posture optimization, the null-space term drives the robot toward a preferred configuration.

For the purposes of this thesis, redundancy is especially important because force-control tasks usually define a primary task at the end-effector. For example, during surface contact, the robot may need to maintain a desired force at the tool while moving along a surface. At the same time, the robot can use its null space to keep a safer posture, avoid joint limits, or improve manipulability. This is one of the main advantages of using a 7-DOF manipulator such as the FR3 instead of a non-redundant 6-DOF industrial robot.

In physical human–robot interaction, redundancy also improves safety. If a human approaches the robot, the FR3 can modify its elbow configuration without disturbing the end-effector task. This allows the robot to react more naturally in shared workspaces. Therefore, null-space projection is not only a mathematical tool, but also a practical mechanism for safer and more adaptive robotic behavior.

2.5 Singularities

Singularities are configurations of a robotic manipulator where the robot loses the ability to move or apply force in one or more Cartesian directions. Mathematically, a singularity occurs when the Jacobian matrix loses rank. For the FR3, the Jacobian has

dimension:

$$J(q) \in \mathbb{R}^{6\times 7}$$

In a regular configuration, the Jacobian has full row rank:

$$\text{rank}(J) = 6$$

This means that the robot can generate arbitrary instantaneous Cartesian motion of the end-effector in all six task-space directions. A singularity occurs when:

$$\text{rank}(J) < 6$$

In that case, the robot loses at least one independent Cartesian motion direction.

Singularities are important because they affect both motion control and force control. In motion control, a desired Cartesian velocity may require extremely large joint velocities near a singularity. This can be seen from the pseudoinverse solution:

$$\dot{q} = J^{\dagger} v$$

When the Jacobian is close to singular, the pseudoinverse may contain very large values. As a result, even a small desired Cartesian velocity v can produce large joint velocities. This is unsafe and may violate joint velocity limits.

In force control, singularities also cause problems. The relation between Cartesian wrench and joint torque is:

$$\tau = J^{T}(q) F$$

Near a singularity, the robot may lose the ability to generate or regulate forces in certain Cartesian directions. This can reduce the quality of impedance control, admittance control, and hybrid force/position control. Therefore, singularities must be considered not only in motion planning, but also in interaction control.

A useful way to analyze singularities is through the singular value decomposition of the Jacobian:

$$J = U \Sigma V^{T}$$

where Σ contains the singular values:

$$\sigma_1, \sigma_2, \ldots, \sigma_6$$

The robot is close to a singularity when the smallest singular value approaches zero:

$$\sigma_{\min} \to 0$$

A common measure of manipulability is:

$$w(q) = \sqrt{\det\left(JJ^T\right)}$$

This can also be expressed as the product of the singular values:

$$w(q) = \sigma_1 \sigma_2 \cdots \sigma_6$$

When $w(q)$ is close to zero, the robot is close to a singular configuration. Another useful measure is the condition number:

$$\kappa(J) = \frac{\sigma_{\max}}{\sigma_{\min}}$$

A large condition number means that the Jacobian is ill-conditioned and the robot is close to a singularity.

For a 7-DOF manipulator such as the FR3, singularities can occur in several situations. One common case is when the arm is close to being fully stretched. In such a configuration, the robot may lose the ability to move the end-effector in certain directions without producing very large joint velocities. Another possible case is wrist alignment, where several joint axes become aligned and the robot loses an independent orientation direction. Singular behavior can also appear near joint limits, workspace boundaries, or configurations where the elbow and wrist geometry reduce the available motion directions.

Although the FR3 is redundant, redundancy does not eliminate singularities completely. The additional degree of freedom helps the robot avoid singular configurations by changing its internal posture, but the robot can still become singular if it reaches an unfavorable configuration. Therefore, redundancy should be used actively through null-space control or motion planning to stay away from singular regions.

Several methods can be used to handle singularities in practice. The first method is to use damped least squares instead of the pure pseudoinverse:

$$\dot{q} = J^T \left(JJ^T + \lambda^2 I\right)^{-1} v$$

The damping term $\lambda^2 I$ prevents the solution from becoming excessively large near singularities. This improves numerical stability and helps keep joint velocities within safe limits.

The second method is manipulability maximization. A secondary objective can be added in the null space in order to move the robot away from singular configurations:

$$\dot{q} = J^{\dagger} v + \left(I - J^{\dagger} J \right) \dot{q}_0$$

where:

$$\dot{q}_0 = k_m \nabla w(q)$$

and $w(q)$ is the manipulability measure. This causes the robot to move in a direction that increases manipulability, without disturbing the primary end-effector task.

The third method is to limit Cartesian velocities and joint velocities. Even if the controller computes a feasible command, the command should be saturated to avoid unsafe motion. This is particularly important in real-time experiments with the FR3, because large velocity or torque commands may trigger safety stops.

The fourth method is to plan trajectories that avoid problematic postures. In MoveIt or other motion-planning frameworks, the robot can be guided through configurations that avoid joint limits, self-collisions, and singular regions. In direct `libfranka` control, the same idea can be implemented by checking manipulability or singular values during motion.

In the context of force control, singularity avoidance is especially important. During impedance control, the robot must map Cartesian force commands to joint torques. During admittance control, the robot must convert measured forces into motion commands. During hybrid force/position control, the robot must regulate force in some directions and position in others. If the robot is near a singularity, these mappings become less reliable and may lead to unstable or inaccurate behavior.

For this reason, the experiments in this thesis should be performed in configurations that keep the FR3 away from singularities. The arm should not be fully stretched, the wrist should not be placed in an extreme alignment, and the joints should remain away from their mechanical limits. The redundancy of the FR3 can then be used to keep the robot in a safe and well-conditioned posture while the end-effector performs the primary task.

In summary, singularities represent configurations where the robot loses motion or force capability in one or more Cartesian directions. For the FR3, singularity analysis is essential because the robot is used for force-control and human-interaction tasks. By monitoring the Jacobian, using damped least squares, maximizing manipulability, applying null-space posture control, and enforcing safe velocity and torque limits, the controller can reduce the risk of singular behavior and improve the stability of real robot experiments.

2.6 Joint Space and Operational Space

Joint space describes the state of the manipulator through its joint variables,

$$q \in \mathbb{R}^n, \tag{2.2}$$

subject to joint limits and dynamic constraints such as velocity, acceleration, and torque limits. Operational space, also called task space, describes the desired position and orientation of the end-effector in $SE(3)$. The two spaces are connected by forward kinematics and differential kinematics:

$$x = f(q), \qquad \dot{x} = J(q)\dot{q}. \tag{2.3}$$

Joint-space control is natural for low-level controllers because motors and sensors are located at the joints. For the FR3, a 7-DOF serial manipulator, $q \in \mathbb{R}^7$. Joint-space descriptions are especially suitable for torque control, trajectory tracking, and dynamic algorithms such as the Recursive Newton-Euler Algorithm (RNEA) and computed-torque control. Joint limits, velocities, and accelerations are critical because they determine not only where the robot can move but also how it can move safely, without excessive jerk or actuator overload.

The workspace is the set of end-effector poses that can be reached while respecting the robot's kinematic structure, joint limits, and collision constraints. In a 7-DOF manipulator such as the FR3, redundancy enables multiple inverse-kinematics solutions for the same end-effector pose. This improves dexterity and supports criteria such as avoiding joint limits, avoiding obstacles, avoiding singularities, avoiding self-collision, and maintaining a comfortable elbow posture.

For the FR3, inverse-kinematics solutions are usually selected through optimization criteria such as:

- minimizing joint displacement,
- maximizing manipulability, for example through the Yoshikawa index,
- avoiding joint limits,
- avoiding obstacles and self-collisions.

2.7 Dynamic Analysis

The dynamic model of a robotic manipulator is important for motion simulation, mechanical analysis, and the design of control algorithms. The computation of forces and torques required for typical motions provides useful information for the design of

joints, transmission systems, and actuators. In this thesis, the equations of motion of the FR3 are presented using the Newton-Euler approach, which is computationally efficient and expresses the model recursively along the open kinematic chain of the robot.

Deriving a complete dynamic model by hand for a 7-DOF manipulator such as the FR3 would be time-consuming and prone to errors. Therefore, in practical implementations, symbolic software and robot-dynamics libraries are commonly used. Dynamic analysis describes the mathematical relationship between motion variables $(q, \dot{q}, \ddot{q})$ and the joint torques τ required to execute the motion, taking into account masses, inertia tensors, gravity, friction, and external contacts.

The main symbols used in this section are:

- $q \in \mathbb{R}^n$: vector of joint variables; for the FR3, $n = 7$.
- $\dot{q}$ and $\ddot{q}$: joint velocities and joint accelerations.
- $\tau \in \mathbb{R}^n$: joint torques or generalized actuation forces.
- g: gravitational acceleration.
- T: kinetic energy; U: potential energy.
- $M(q)$: inertia matrix.
- $C(q, \dot{q})\dot{q}$: centrifugal and Coriolis terms.
- $G(q)$: gravity vector.
- $\tau_f(\dot{q})$: friction terms.
- $J(q)$: Jacobian matrix, also useful for mapping external forces.

2.7.1 Equations of Motion

The equations of motion of a robotic manipulator in joint space can be obtained either through the Lagrange method or through the Recursive Newton-Euler Algorithm. The Lagrange method is systematic and independent of the chosen coordinate frame, while RNEA produces the model in a recursive form that is computationally efficient.

RNEA is an efficient method for computing inverse dynamics in serial robotic manipulators. Instead of deriving large nonlinear expressions, it computes the required joint torques directly from $(q, \dot{q}, \ddot{q})$ using two recursive passes: a forward pass for the angular and linear velocities and accelerations of the links, and a backward pass for transferring forces and moments toward the base. Its complexity is linear in the number of degrees of freedom, $O(n)$, which makes it suitable for real-time applications such as

feedforward torque control, gravity compensation, and computed-torque control on the FR3.

In general form, the dynamics can be written as

$$\tau = M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) + \tau_f + J^T F_{\text{ext}}. \tag{2.4}$$

RNEA computes τ recursively along the robot chain. It is based on the Newton-Euler laws:

$$F = ma, \tag{2.5}$$
$$N = I\alpha + \omega \times (I\omega), \tag{2.6}$$

where F is force, N is moment, m is mass, a is linear acceleration, I is the inertia tensor, α is angular acceleration, and ω is angular velocity.

For each link, the algorithm computes angular velocities and accelerations, the linear acceleration of the center of mass, the force F_i, and the moment N_i required by the link. The calculations are performed in two stages:

1. **Forward recursion:** from the base to the end-effector, computing kinematic quantities such as ω, α, and a.

2. **Backward recursion:** from the end-effector back to the base, computing link forces, link moments, and finally the joint torques τ.

For a revolute joint i with axis z_i, the transformations from link $i-1$ to link i can be described by a rotation matrix R_i and a displacement vector p_i. The base conditions are usually set as

$$\omega_0 = 0, \qquad \alpha_0 = 0, \qquad a_0 = -g. \tag{2.7}$$

For every link i, a typical forward recursion can be expressed as

$$\omega_i = R_i^T \omega_{i-1} + z_i \dot{q}_i, \tag{2.8}$$
$$\alpha_i = R_i^T \alpha_{i-1} + z_i \ddot{q}_i + \omega_i \times (z_i \dot{q}_i), \tag{2.9}$$
$$a_i = R_i^T \left(a_{i-1} + \alpha_{i-1} \times p_i + \omega_{i-1} \times (\omega_{i-1} \times p_i)\right), \tag{2.10}$$
$$a_{c,i} = a_i + \alpha_i \times r_{c,i} + \omega_i \times (\omega_i \times r_{c,i}), \tag{2.11}$$

where $r_{c,i}$ is the vector from the origin of frame i to the center of mass of link i.

The internal force and moment of link i are

$$F_i = m_i a_{c,i}, \tag{2.12}$$
$$N_i = I_{c,i}\alpha_i + \omega_i \times (I_{c,i}\omega_i). \tag{2.13}$$

The backward recursion combines the forces and moments from the subsequent links:

$$f_i = R_{i+1} f_{i+1} + F_i, \tag{2.14}$$

$$n_i = N_i + R_{i+1} n_{i+1} + r_{c,i} \times F_i + p_{i+1} \times (R_{i+1} f_{i+1}). \tag{2.15}$$

For a revolute joint, the joint torque is obtained by projecting the moment onto the joint axis:

$$\tau_i = n_i^T z_i. \tag{2.16}$$

For a prismatic joint, the generalized force would be

$$\tau_i = f_i^T z_i. \tag{2.17}$$

2.7.2 Forward Dynamics and Inverse Dynamics

The Newton-Euler method provides the relationship between joint torques, possible end-effector forces, and the motion of the robot. The forward-dynamics problem consists of determining the joint accelerations $\ddot{q}(t)$ for $t > t_0$ caused by given joint torques $\tau(t)$ and possible end-effector forces $h_e(t)$, when the initial positions $q(t_0)$ and velocities $\dot{q}(t_0)$ are known. The joint velocities and positions are obtained by integrating the nonlinear differential equations of motion.

The inverse-dynamics problem consists of determining the joint torques $\tau(t)$ required to produce a desired motion, defined by q, $\dot{q}$, and $\ddot{q}$, while also considering any external forces acting on the end-effector. Inverse dynamics is useful for trajectory planning and control because, once the desired joint trajectory is known, it provides the torques that must be applied to the joints in order to generate the desired motion.

For a robot with n joints, forward dynamics typically requires $O(n^2)$ operations, while inverse dynamics using RNEA requires $O(n)$ operations. This computational efficiency is one of the reasons RNEA is commonly used in real-time robotic control.

To achieve smooth interaction, the controller must account for the robot dynamics. A commonly used form of the Euler-Lagrange model is

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau_{\text{cmd}} - \tau_{\text{ext}}, \tag{2.18}$$

where $M(q)$ is the inertia matrix, $C(q, \dot{q})\dot{q}$ contains Coriolis and centrifugal effects, and $g(q)$ is the gravity vector. The high-quality dynamic model of the FR3 enables gravity compensation, allowing the arm to float and providing the dynamic transparency required for sensitive human-robot interaction.

2.8 Force Sensors

The measurement of a force or torque is commonly reduced to the measurement of the deformation caused by that force or torque on an elastic element with known properties. Therefore, an indirect force measurement can be achieved by measuring very small displacements or strains. The basic component of many force sensors is the strain gauge, which uses the change in electrical resistance of a conductor when it is deformed.

Force sensing is a fundamental requirement in robotic systems that interact physically with their environment. In classical industrial manipulators, the robot often operates mainly through position control, and contact with the environment is either avoided or treated as an error. In collaborative and contact-rich robotics, however, contact forces become part of the task itself. For example, when a robot presses on a surface, grasps an object, slides an object on a table, or reacts to a human push, the controller must be able to estimate and regulate the physical interaction between the robot and the environment.

In general, a force sensor converts mechanical deformation into an electrical signal. A common sensing principle is based on strain gauges, where a small deformation of an elastic element changes its electrical resistance. From this deformation, the applied force or torque can be estimated. Traditional robotic force sensing is often implemented using a six-axis force/torque sensor mounted between the robot flange and the end-effector. Such a sensor measures three force components and three torque components at the wrist of the robot.

In the Franka Research 3, force sensing is implemented in a more integrated way. The FR3 is a seven-degree-of-freedom robotic manipulator equipped with link-side torque sensors in all seven joints. This means that each joint can directly measure the torque transmitted through the mechanical structure of the arm. As a result, the robot can detect external interactions not only at the end-effector, but also along the body of the robot. This is important for human-robot interaction, because a human may touch or push the robot at any link, not only at the tool.

The use of joint torque sensors gives the FR3 a form of proprioceptive force sensing. Proprioception means that the robot senses interaction through its own internal joints, rather than only through an external force sensor. The measured joint torque vector can be written as:

$$\tau_J = \begin{bmatrix} \tau_1 & \tau_2 & \tau_3 & \tau_4 & \tau_5 & \tau_6 & \tau_7 \end{bmatrix}^T$$

where τ_i is the measured torque at joint i. In the Franka Control Interface and libfranka, this quantity is available through the robot state as:

`state.tau_J`

which represents the measured link-side joint torque sensor signals in Nm. The external torque estimate is also available as:

`state.tau_ext_hat_filtered`

This value represents a filtered estimate of the external torques acting on the robot joints. Conceptually, the external torque estimate is obtained by comparing the measured joint torques with the torques expected from the robot dynamic model. If the robot model predicts the torque needed for gravity, inertia, Coriolis effects, and commanded motion, then any significant difference between the expected torque and the measured torque can indicate an external interaction. This can be expressed as:

$$\hat{\tau}_{ext} = \tau_J - \tau_{model}$$

where:

$$\tau_{model} = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q)$$

Here, $M(q)$ is the inertia matrix, $C(q, \dot{q})\dot{q}$ represents Coriolis and centrifugal effects, and $g(q)$ represents gravity. Therefore, the external torque estimate is not a completely independent measurement like a standalone sensor. It is a model-based estimate that depends on the accuracy of the robot model, payload configuration, friction compensation, and sensor calibration.

For the purposes of this thesis, this sensing architecture is highly useful because it allows force-control experiments without adding a separate wrist force/torque sensor. The FR3 can estimate external forces and torques through its internal sensors and dynamic model, which reduces hardware complexity and keeps the end-effector lighter. This is especially useful in tasks such as hand-guiding, compliant motion, surface contact, obstacle detection, and human-robot interaction.

However, the internal torque-sensing approach also has limitations. The robot can detect that an external interaction exists, but the exact point of contact on the robot body is not always directly known. In addition, the force estimate may be affected by modeling errors, unmodeled friction, cable forces, payload errors, or small offsets in the estimated bias. For this reason, practical force-control experiments usually require bias compensation, filtering, conservative gains, and safety limits.

In this thesis, the FR3 torque and force estimates were used for reading external interaction forces, detecting contact with the environment, implementing compliant behavior, applying normal force during surface manipulation, monitoring external joint torques, and evaluating the safety of human-robot interaction. Therefore, the FR3's integrated torque sensing is not only a measurement feature, but also a central part of the control architecture used in this work.

2.9 Wrench Force Sensors

When the end-effector of a robot is in contact with the environment, a wrench force sensor can measure the three components of force and the three components of torque with respect to a coordinate frame attached to the sensor or end-effector. This information is useful for contact detection, force regulation, assembly tasks, polishing, grinding, and other operations that require controlled interaction with the environment.

A wrench is a six-dimensional vector that combines the three components of force and the three components of torque acting on a rigid body. In robotics, the wrench applied at the end-effector is usually written as:

$$W = \begin{bmatrix} F_x \\ F_y \\ F_z \\ M_x \\ M_y \\ M_z \end{bmatrix}$$

where F_x, F_y, and F_z are the Cartesian force components, measured in Newtons, and M_x, M_y, and M_z are the moment components, measured in Newton-meters. A classical six-axis force/torque sensor mounted at the wrist of a robot directly measures this wrench at the interface between the robot and the end-effector.

In the FR3, the external wrench can be estimated without necessarily mounting an additional wrist force/torque sensor. Through libfranka, the robot state provides the estimated external wrench acting on the stiffness frame K. Two important variables are:

`state.O_F_ext_hat_K`

and:

`state.K_F_ext_hat_K`

The first one, `state.O_F_ext_hat_K`, is the estimated external wrench acting on the stiffness frame and expressed relative to the robot base frame O. The second one, `state.K_F_ext_hat_K`, is the same estimated external wrench expressed relative to the stiffness frame K. In both cases, the vector contains six values with units:

$$[N, N, N, Nm, Nm, Nm]$$

This distinction between frames is important. When the wrench is expressed in the base frame O, the force directions are interpreted according to the fixed robot base

coordinate system. When the wrench is expressed in the stiffness frame K, the force directions are interpreted according to the local frame used for Cartesian impedance behavior. For example, if the robot performs a surface-contact task, the vertical force component may be easier to interpret in the base frame if the table normal is aligned with the base z-axis.

The relation between a Cartesian wrench and joint torques is given by the Jacobian transpose:

$$\tau_{ext} = J(q)^T W$$

where $J(q)$ is the geometric Jacobian of the robot. This equation shows how a force or torque applied at the end-effector produces torques at the joints. Conversely, if external joint torques are estimated, the robot can use its kinematic and dynamic model to estimate the corresponding Cartesian wrench.

For force-control applications, the wrench estimate is extremely useful because it allows the controller to regulate contact forces in task space. For example, in a surface manipulation task, the controller may regulate only the vertical force component F_z, while still controlling the horizontal position of the end-effector. This leads to a hybrid force/position control structure:

$$W = \begin{bmatrix} F_x^{pos} \\ F_y^{pos} \\ F_z^{force} \\ M_x^{ori} \\ M_y^{ori} \\ M_z^{ori} \end{bmatrix}$$

In this structure, F_x and F_y can be generated by Cartesian position control, F_z can be generated by force feedback, and the moment components can be used to maintain the desired end-effector orientation.

For the FR3 experiments in this thesis, the most important wrench component was the vertical force F_z, because the robot had to maintain contact with a table while sliding an object. The measured raw force was read from:

$$F_{z,raw} = \texttt{state.O_F_ext_hat_K[2]}$$

However, raw force readings may contain offsets due to gravity compensation, payload configuration, sensor drift, or the initial contact condition. For this reason, bias compensation was used before force control:

$$F_{z,bias} = \frac{1}{N} \sum_{i=1}^{N} F_{z,raw,i}$$

Then the compensated force was computed as:

$$F_{z,\Delta} = F_{z,raw} - F_{z,bias}$$

Depending on the sign convention and the direction of contact, the measured contact force was written as:

$$F_{z,contact} = s_c F_{z,\Delta}$$

where s_c is a sign coefficient selected experimentally. In the implemented experiments, downward contact with the table produced a negative change in the raw F_z signal, so the sign coefficient was chosen as:

$$s_c = -1$$

This allowed the controller to interpret a downward pressing contact as a positive normal contact force.

The desired normal force was then compared with the measured contact force:

$$e_F = F_{z,target} - F_{z,contact}$$

and the commanded vertical force was computed using a feedback term:

$$F_{z,cmd} = K_f e_F - D_f \dot{z}$$

where K_f is the force-control gain and D_f is the damping term used to reduce oscillations during contact. Finally, the Cartesian wrench command was converted into joint torques using:

$$\tau_{cmd} = J(q)^T W_{cmd} + C(q, \dot{q})$$

where $C(q, \dot{q})$ represents Coriolis compensation.

In conclusion, although the FR3 does not require an external six-axis wrist force/torque sensor for the experiments of this thesis, it can still provide wrench information through its internal joint torque sensors and model-based estimation. This makes the robot especially suitable for force control, impedance control, hand-guiding, collision detection, and safe physical human-robot interaction. The wrench estimate is therefore a key measurement for transforming the FR3 from a purely motion-controlled manipulator into a force-aware collaborative robotic system.

Chapter 3

Force Control and Sensing

A basic requirement for the success of a robotic task is the ability to handle the interaction between the manipulator and its environment. The most effective quantity for describing the state of this interaction is the contact force at the end-effector. Large contact forces are generally undesirable because they can overstress the robot, damage the object, or create unsafe conditions for nearby humans.

3.1 Manipulator Interaction with the Environment

The control of interaction between a robotic manipulator and its environment is critical for many practical tasks in which the end-effector must manipulate an object or operate on a surface, such as polishing, cutting, machining, or assembly. During interaction, the environment imposes constraints on the geometric paths that the end-effector can follow. This situation is called constrained motion, and a pure position-control strategy may fail because small position errors can create large contact forces.

Interaction-control strategies are commonly classified into two categories: indirect force control and direct force control. In indirect force control, the force is regulated through motion control without an explicit force-feedback loop. In direct force control, the controller includes force feedback and can regulate the contact force toward a desired value. Compliance control and impedance control belong to the first category and are studied in detail in this thesis.

The contact force is the quantity that best describes the interaction state. The availability of force measurements improves the performance of interaction control. Planning errors can increase the contact force and cause the end-effector to deviate from the desired trajectory. The controller may then react by attempting to reduce the position error, which can further increase the contact force until actuator saturation or damage occurs. This problem becomes more likely when the environment is stiff and the position controller is very accurate. The drawback can be reduced by ensuring compliant behavior during contact.

3.2 Force Control

This thesis focuses mainly on methods that implement indirect force control, because the interaction force is controlled through the desired end-effector motion and impedance behavior. The interaction between the robot and the environment is affected directly by the compliance of the environment and by the compliance or impedance of the manipulator. A force-control system usually includes a control law based not only on force measurements but also on velocity and position measurements.

The effectiveness of such methods depends on the geometry of the contact and on the specific interaction case. A force-control strategy is meaningful only in those directions of the operational space in which interaction forces can actually appear between the manipulator and the environment.

3.3 Compliance Control

Compliance control is the part of robot control that prevents the manipulator from behaving as a perfectly rigid mechanism during contact. Instead of resisting all external forces, the robot yields, absorbs energy, and interacts safely and stably with humans, objects, and surfaces. This capability is one of the key strengths of the FR3 because the robot is designed for collaborative operation and torque control.

A rigid manipulator attempts to hold its commanded position almost independently of external forces. During contact, this behavior can become dangerous or lead to large loads and instability. A compliant manipulator, on the other hand, allows its motion or position to depend on the external force, as if a virtual spring and damper existed between the tool and the target. The goal is safe contact, stable behavior, soft interaction, and improved adaptation to uncertainty.

The three main categories of compliance control are impedance control, admittance control, and hybrid force/position control. In admittance control, the controller converts an external force into a desired motion according to a virtual dynamic model:

$$M_a\ddot{x}_c + D_a\dot{x}_c + K_a x_c = F_{\text{ext}}, \tag{3.1}$$

and then defines the desired position as

$$x_d = x_{\text{ref}} + x_c. \tag{3.2}$$

Admittance control is commonly preferred when the robot is mainly position-controlled and compliance must be generated externally. In torque-controlled robots such as the FR3, impedance control is often more natural, although admittance control can also be implemented.

3.4 Impedance Control

The FR3 is fundamentally a torque-controlled manipulator, which distinguishes it from traditional position-controlled industrial robots. Through the Franka Control Interface, the robot allows direct torque commands at each joint at a frequency of 1 kHz.

In this research, an impedance-control strategy is implemented. The goal is to regulate the relationship between the end-effector pose error and the interaction wrench. The controller is modeled as a virtual mass-spring-damper system. A typical Cartesian impedance behavior is expressed as

$$M_d(\ddot{x} - \ddot{x}_d) + D_d(\dot{x} - \dot{x}_d) + K_d(x - x_d) = F_{\text{ext}}, \tag{3.3}$$

where x is the actual Cartesian pose of the tool, x_d is the desired pose, F_{ext} is the external wrench, M_d is the virtual inertia, D_d is the virtual damping, and K_d is the virtual stiffness.

A common torque command for Cartesian impedance control can be written as

$$\tau_{\text{cmd}} = J^T \left(K(x_d - x) + D(\dot{x}_d - \dot{x})\right) + g(q) + C(q, \dot{q})\dot{q}. \tag{3.4}$$

This approach ensures that the robot behaves compliantly during human contact and effectively transforms the manipulator into a programmable soft mechanism that can yield to external disturbances.

The interpretation of the controller parameters is intuitive. A larger stiffness matrix K_d makes the robot more rigid and less compliant, while a smaller K_d makes it softer and more compliant. The damping matrix D_d reduces oscillations and helps stabilize contact. This makes impedance control particularly suitable for surface-contact tasks such as polishing, grinding, and assembly.

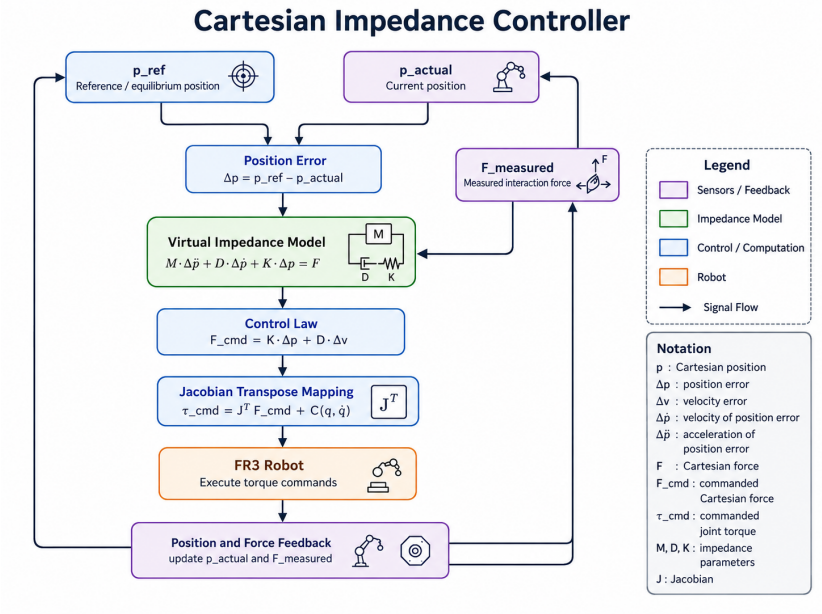

Figure 3.1: Cartesian impedance controller block diagram.

The above block diagram presents a Cartesian impedance controller. This controller does not command the robot to behave as a perfectly rigid position-controlled system. Instead, it defines a desired dynamic relationship between the Cartesian position error and the interaction force.

The main idea is to make the end-effector behave like a virtual mechanical system composed of a mass, a spring, and a damper. The general impedance model is:

$$M\Delta\ddot{p} + D\Delta\dot{p} + K\Delta p = F$$

where M is the virtual inertia matrix, D is the virtual damping matrix, K is the virtual stiffness matrix, and F is the interaction force. The position deformation is defined as:

$$\Delta p = p_{ref} - p_{actual}$$

where p_{ref} is the desired reference position and p_{actual} is the measured current position of the end-effector.

In the block diagram, the controller starts from a desired Cartesian reference position:

$$p_{ref}$$

This reference is not necessarily treated as a strict position command. Instead, it is interpreted as an equilibrium position. If no external force is applied, the robot tends to remain close to this equilibrium. If an external force is applied, the robot reacts according to the selected stiffness and damping values.

A simplified Cartesian impedance control law can be written as:

$$F_{cmd} = K(p_{ref} - p_{actual}) + D(\dot{p}_{ref} - \dot{p}_{actual})$$

If the desired velocity is zero, this becomes:

$$F_{cmd} = K(p_{ref} - p_{actual}) - D\dot{p}_{actual}$$

The generated Cartesian force command is then mapped into joint torques using the Jacobian transpose:

$$\tau_{cmd} = J^T(q)F_{cmd} + C(q, \dot{q})$$

where $C(q, \dot{q})$ represents Coriolis compensation. In practical implementations, gravity compensation and torque-rate saturation may also be included for safety and stability.

The impedance controller is very important for the FR3 because the robot supports torque-level control and has integrated torque sensors in its joints. This allows the robot to behave compliantly during contact. If the stiffness K is high, the robot behaves more rigidly and strongly resists displacement. If the stiffness K is low, the robot behaves more softly and allows larger displacement under the same external force. The damping D is used to reduce oscillations and improve contact stability.

The impedance controller is suitable for tasks where the robot must maintain stable contact with the environment while still reacting safely to external disturbances. Examples include surface following, physical human-robot interaction, polishing, pressing, and compliant manipulation.

3.5 Admittance Control

Admittance control is complementary to impedance control. In admittance control, the measured external force is used as an input to a virtual dynamic system that generates a desired motion. The robot then follows that motion through its position or velocity controller. This approach is useful for stiff, high-precision manipulators that do not accept direct torque commands. Since the FR3 supports torque control, impedance control is the main focus of this thesis, but admittance control remains an important alternative for systems with different low-level control architectures.

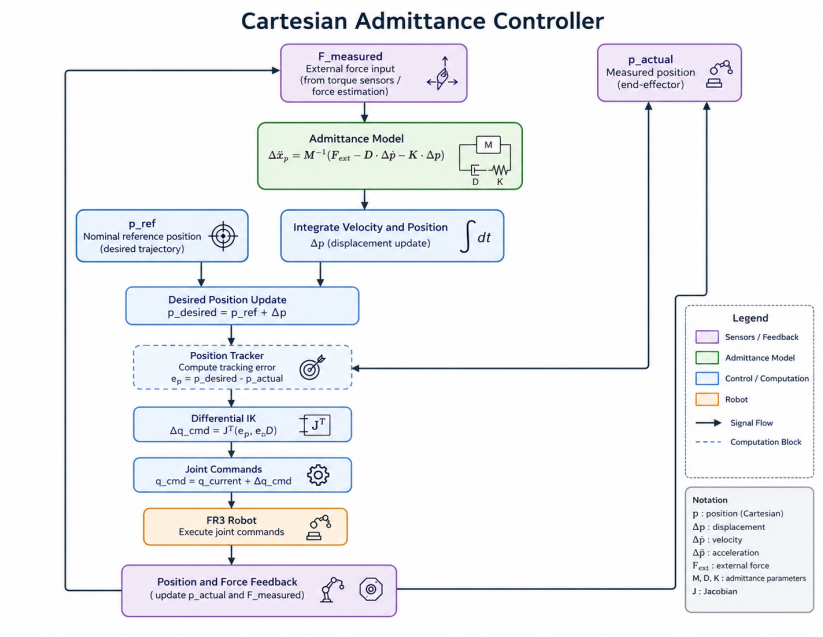

Figure 3.2: Cartesian admittance controller block diagram.

The above block diagram represents a Cartesian admittance controller. While impedance control receives a motion error and produces a force-like response, admittance control follows the opposite logic: it receives an external force measurement and produces a desired motion.

The controller uses a virtual mass-spring-damper model to convert the measured external force into a displacement command. The general admittance equation is:

$$M\Delta\ddot{p} + D\Delta\dot{p} + K\Delta p = F_{ext}$$

Solving for acceleration gives:

$$\Delta\ddot{p} = M^{-1}\left(F_{ext} - D\Delta\dot{p} - K\Delta p\right)$$

This acceleration is integrated once to obtain velocity:

$$\Delta\dot{p}(t + \Delta t) = \Delta\dot{p}(t) + \Delta\ddot{p}\Delta t$$

and integrated again to obtain position displacement:

$$\Delta p(t + \Delta t) = \Delta p(t) + \Delta\dot{p}(t + \Delta t)\Delta t$$

The final desired Cartesian position is then updated as:

$$p_{desired} = p_{ref} + \Delta p$$

This desired position is passed to a position tracker, which generates the required

Cartesian or joint commands. In the diagram, the desired Cartesian motion is converted into joint commands through differential inverse kinematics:

$$\Delta q = J^{\#}(q)\Delta p_{desired}$$

and then integrated to produce the final joint command:

$$q_{cmd} = q_{actual} + \Delta q$$

The FR3 executes the command and returns new position and force feedback to the controller.

The main advantage of admittance control is that the robot can respond smoothly to external forces. For example, if a human pushes the end-effector, the measured external force becomes the input of the admittance model and the robot generates a corresponding compliant displacement. The magnitude of this displacement depends on the virtual parameters M, D, and K. A smaller virtual stiffness allows the robot to move more easily, while a larger virtual stiffness makes the robot resist motion more strongly.

Admittance control is particularly useful for collaborative tasks, soft contact, hand-guiding, and safe human-robot interaction. It is often used when the robot is controlled mainly through position commands but must still display compliant behavior. In the case of the FR3, admittance control can also be implemented by reading the estimated external wrench from the robot state and updating the commanded Cartesian target in real time.

3.6 Hybrid Force/Position Control

Hybrid force/position control is based on the idea that not all directions of motion should be controlled in the same way during contact. The description of an interaction task through physical and artificial constraints requires a controller that assigns desired values only to variables that are not physically constrained by the environment. The control action must not conflict with variables that are constrained by contact, otherwise contradictions between control and environment interaction may occur.

In hybrid control, the task directions are separated into position-controlled directions and force-controlled directions. For example, during a surface-contact task, the robot may apply a desired normal force, such as 10 N, while moving along the tangential directions of the surface. This separation allows the robot to maintain contact force while simultaneously following a desired path.

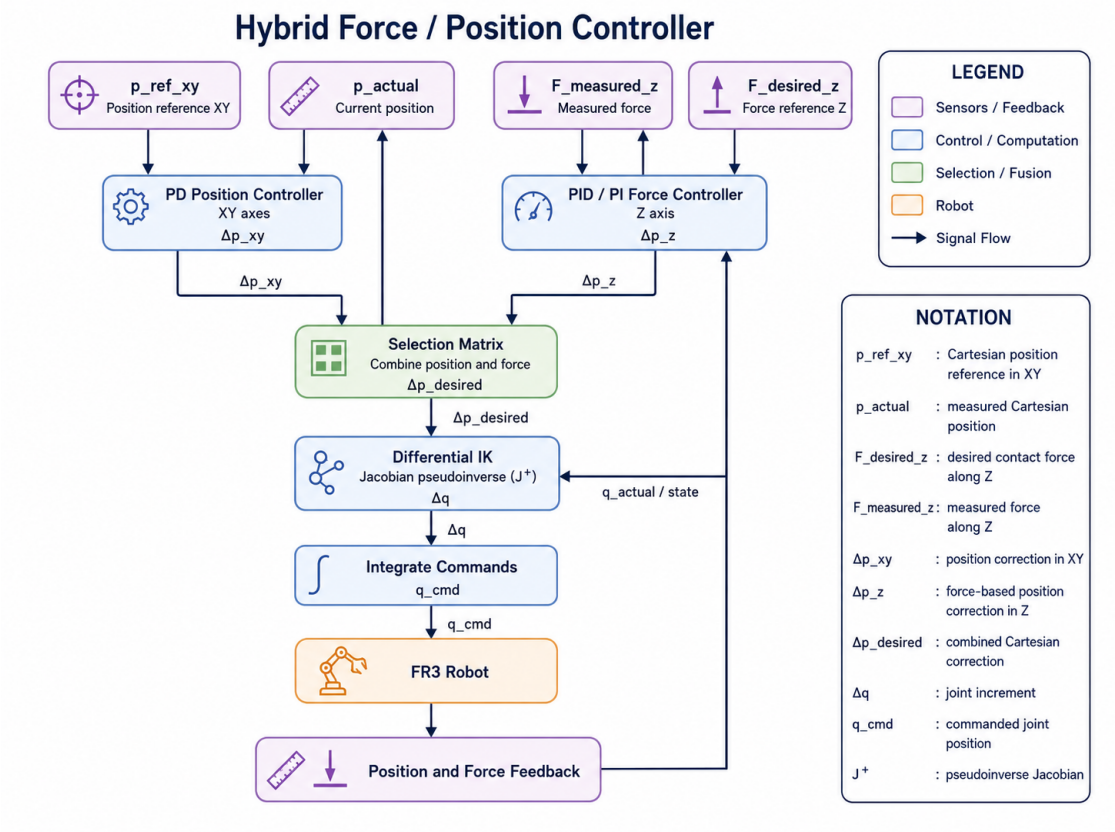

Figure 3.3: Hybrid force/position controller block diagram.

The above block diagram describes a hybrid force/position controller. This controller is designed for tasks where the robot must control motion in some directions and force in other directions at the same time. This is especially useful when the end-effector is in contact with a surface.

A typical example is a surface manipulation task. The robot may need to move along the x and y directions while maintaining a desired normal force along the z direction. In this case, the horizontal motion is controlled by position control, while the vertical contact interaction is controlled by force control.

The controller receives two different references:

$$p_{ref,xy}$$

for the desired Cartesian position in the horizontal plane, and:

$$F_{desired,z}$$

for the desired contact force along the vertical axis. In the block diagram, the desired vertical force is shown as a reference force, for example:

$$F_{desired,z} = 5N$$

The actual Cartesian position is measured as:

$$p_{actual}$$

and the measured vertical contact force is:

$$F_{measured,z}$$

The position-control part computes the Cartesian position error in the controlled position directions:

$$\Delta p_{xy} = p_{ref,xy} - p_{actual,xy}$$

A PD position controller then generates a desired correction in the x and y directions. At the same time, the force-control part compares the desired vertical force with the measured vertical force:

$$e_F = F_{desired,z} - F_{measured,z}$$

A PID or PI force controller can then be used to produce a desired correction along the z-axis. In simplified form, this force-control contribution can be written as:

$$\Delta p_z = K_F e_F + K_I \int e_F dt + K_D \dot{e}_F$$

The selection matrix is the key element of hybrid force/position control. It determines which task-space directions are position-controlled and which are force-controlled. For example, for a task where x and y are position-controlled and z is force-controlled, the selection matrices can be conceptually written as:

$$S_p = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and:

$$S_f = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The final desired Cartesian correction is obtained by combining the position and force contributions:

$$\Delta p_{desired} = S_p \Delta p_{pos} + S_f \Delta p_{force}$$

This desired Cartesian correction is then converted into a joint-space correction using differential inverse kinematics. The relationship between Cartesian velocity and joint velocity is:

$$\dot{x} = J(q)\dot{q}$$

Therefore, a desired Cartesian correction can be mapped to a joint correction using the pseudoinverse of the Jacobian:

$$\Delta q = J^{\#}(q)\Delta p_{desired}$$

where $J^{\#}(q)$ is the pseudoinverse of the Jacobian. The joint command is then produced by integrating the joint correction:

$$q_{cmd}(t + \Delta t) = q_{actual}(t) + \Delta q$$

The command is sent to the FR3 robot, and the robot returns updated position and force feedback. In this way, the controller continuously adjusts the robot motion so that it follows the desired path while maintaining the desired contact force.

The hybrid force/position controller is particularly suitable for assembly tasks, polishing, wiping, pushing, sliding, and surface-following operations. Its main advantage is that it explicitly separates motion control and force control according to the physical constraints of the task.

3.7 Pseudo-Algorithms of the Implemented Controllers

This section presents the pseudo-algorithms of the three main control strategies implemented and studied in this thesis: Cartesian impedance control, Cartesian admittance control, and hybrid force/position control. The algorithms are written in a general form, but they follow the practical implementation logic used for the FR3 experiments. Each controller runs in a cyclic real-time loop, where the robot state is read, the control law is evaluated, safety limits are applied, and the final command is sent to the robot.

3.7.1 Pseudo-Algorithm of the Cartesian Impedance Controller

The impedance controller defines a virtual mechanical relation between the Cartesian pose error of the end-effector and the force or torque response of the robot. In this controller, the desired Cartesian pose is treated as an equilibrium pose. If the robot

is displaced from this pose because of contact or external disturbance, the controller generates a restoring wrench according to a virtual spring-damper model.

The simplified Cartesian impedance law is:

$$F_{cmd} = K_x(x_d - x) + D_x(\dot{x}_d - \dot{x}) \tag{3.5}$$

where K_x is the Cartesian stiffness matrix and D_x is the Cartesian damping matrix. The Cartesian wrench is then mapped to joint torques through:

$$\tau_{cmd} = J(q)^T F_{cmd} + C(q, \dot{q}) \tag{3.6}$$

where $J(q)$ is the Jacobian matrix and $C(q, \dot{q})$ represents Coriolis compensation.

The implemented logic of the Cartesian impedance controller can be described as follows:

Step 1: Define the desired Cartesian pose x_d and desired Cartesian velocity $\dot{x}_d$.

Step 2: Define the Cartesian stiffness matrix K_x and damping matrix D_x. These matrices determine how rigid or compliant the robot behaves during motion and contact.

Step 3: Initialize the previous torque command τ_{prev}, which is used for torque-rate saturation.

Step 4: Read the current robot state from the Franka Control Interface. The most important measured quantities are:

$$q, \quad \dot{q}, \quad x, \quad \dot{x}, \quad \tau_J, \quad \hat{\tau}_{ext} \tag{3.7}$$

Step 5: Compute the current Jacobian matrix:

$$J(q) \tag{3.8}$$

Step 6: Compute the model compensation terms, mainly the Coriolis compensation:

$$C(q, \dot{q}) \tag{3.9}$$

Step 7: Compute the Cartesian position error:

$$e_x = x_d - x \tag{3.10}$$

Step 8: Compute the Cartesian velocity error:

$$e_{\dot{x}} = \dot{x}_d - \dot{x} \tag{3.11}$$

Step 9: Compute the desired Cartesian wrench using the virtual spring-damper law:

$$F_{cmd} = K_x e_x + D_x e_{\dot{x}} \tag{3.12}$$

Step 10: Convert the Cartesian wrench into joint torques:

$$\tau_{raw} = J(q)^T F_{cmd} + C(q, \dot{q}) \tag{3.13}$$

Step 11: Apply torque-rate saturation in order to avoid sudden torque jumps:

$$\Delta\tau = \text{clip}\left(\tau_{raw} - \tau_{prev}, -\Delta\tau_{\max}, \Delta\tau_{\max}\right) \tag{3.14}$$

$$\tau_{cmd} = \tau_{prev} + \Delta\tau \tag{3.15}$$

Step 12: Send the final torque command τ_{cmd} to the FR3 through the Franka Control Interface.

Step 13: Save the relevant robot state, force, torque, and error data to the log file.

Step 14: Update the previous torque command:

$$\tau_{prev} \leftarrow \tau_{cmd} \tag{3.16}$$

Step 15: Repeat the loop until the experiment finishes or a safety stop is triggered.

The main advantage of the impedance controller is that it allows the FR3 to behave like a programmable mechanical system. By changing the stiffness matrix K_x, the robot can behave either rigidly or compliantly. A high stiffness value produces accurate pose tracking, while a low stiffness value allows safer physical interaction with humans or the environment. The damping matrix D_x is used to reduce oscillations and improve stability during contact.

3.7.2 Pseudo-Algorithm of the Cartesian Admittance Controller

The admittance controller uses the opposite logic of impedance control. Instead of converting position error into a force response, it converts measured external force into

a desired motion. This is useful for hand-guiding and collaborative interaction, because the robot moves according to the force applied by the human or the environment.

The admittance model is written as:

$$M_a \Delta\ddot{x} + D_a \Delta\dot{x} + K_a \Delta x = F_{ext} \tag{3.17}$$

where M_a, D_a, and K_a are the virtual mass, damping, and stiffness of the admittance model. Solving for the virtual acceleration gives:

$$\Delta\ddot{x} = M_a^{-1} \left(F_{ext} - D_a \Delta\dot{x} - K_a \Delta x \right) \tag{3.18}$$

The virtual acceleration is integrated to obtain velocity and position corrections. The final desired pose becomes:

$$x_{cmd} = x_{ref} + \Delta x \tag{3.19}$$

The implemented logic of the Cartesian admittance controller can be described as follows:

Step 1: Define the nominal reference pose x_{ref}.

Step 2: Define the admittance parameters: virtual mass M_a, virtual damping D_a, and virtual stiffness K_a.

Step 3: Define the sampling time Δt, which corresponds to the controller loop period.

Step 4: Initialize the virtual displacement:

$$\Delta x \leftarrow 0 \tag{3.20}$$

Step 5: Initialize the virtual velocity:

$$\Delta\dot{x} \leftarrow 0 \tag{3.21}$$

Step 6: Estimate the initial force bias:

$$F_{bias} \tag{3.22}$$

Step 7: Read the current robot state:

$$q, \quad \dot{q}, \quad x, \quad \dot{x}, \quad F_{measured} \tag{3.23}$$

Step 8: Compute the compensated external force:

$$F_{ext} = F_{measured} - F_{bias} \tag{3.24}$$

Step 9: Apply filtering or a small deadband in order to reduce the effect of sensor noise.

Step 10: Compute the virtual acceleration from the admittance model:

$$\Delta\ddot{x} = M_a^{-1}\left(F_{ext} - D_a\Delta\dot{x} - K_a\Delta x\right) \tag{3.25}$$

Step 11: Integrate the virtual acceleration to update the virtual velocity:

$$\Delta\dot{x} \leftarrow \Delta\dot{x} + \Delta\ddot{x}\Delta t \tag{3.26}$$

Step 12: Integrate the virtual velocity to update the virtual displacement:

$$\Delta x \leftarrow \Delta x + \Delta\dot{x}\Delta t \tag{3.27}$$

Step 13: Compute the new desired Cartesian command:

$$x_{cmd} = x_{ref} + \Delta x \tag{3.28}$$

Step 14: Convert the Cartesian command into a joint command using inverse kinematics:

$$q_{cmd} = IK(x_{cmd}) \tag{3.29}$$

Step 15: Send the command q_{cmd} to the robot controller.

Step 16: Save the relevant force, position, and command data to the log file.

Step 17: Repeat the loop until the experiment finishes or a safety stop is triggered.

In this controller, the external force is the input and the motion is the output. If a human pushes the robot, the measured force increases and the admittance model produces a displacement in the direction of the applied force. The parameters M_a, D_a, and K_a determine how easily the robot moves. A low stiffness and low virtual mass make the robot easier to guide, while higher values make the robot more resistant to external forces.

3.7.3 Pseudo-Algorithm of the Hybrid Force/Position Controller

The hybrid force/position controller separates the task into position-controlled directions and force-controlled directions. This is especially useful for surface-contact tasks. For

example, when the robot slides an object on a table, the x and y directions can be controlled by position, while the z direction can be controlled by force.

The Cartesian wrench command is constructed from two parts. The first part comes from position control in the horizontal directions:

$$F_x = K_{xy}(x_d - x) - D_{xy}\dot{x} \tag{3.30}$$

$$F_y = K_{xy}(y_d - y) - D_{xy}\dot{y} \tag{3.31}$$

The second part comes from force control in the vertical direction:

$$e_F = F_{z,target} - F_{z,contact} \tag{3.32}$$

$$F_z = K_f e_F - D_f \dot{z} \tag{3.33}$$

The resulting Cartesian wrench is:

$$W_{cmd} = \begin{bmatrix} F_x \\ F_y \\ F_z \\ M_x \\ M_y \\ M_z \end{bmatrix} \tag{3.34}$$

and the torque command is:

$$\tau_{cmd} = J(q)^T W_{cmd} + C(q, \dot{q}) \tag{3.35}$$

The implemented logic of the hybrid force/position controller can be described as follows:

Step 1: Define the desired planar trajectory $x_d(t)$ and $y_d(t)$.

Step 2: Define the desired normal contact force:

$$F_{z,target} \tag{3.36}$$

Step 3: Define the planar position-control gains K_{xy} and D_{xy}.

Step 4: Define the force-control gain K_f and vertical damping gain D_f.

Step 5: Define the contact-force sign coefficient s_c, which is selected experimentally according to the sign convention of the measured vertical force.

Step 6: Move the robot safely to the initial contact or pre-contact pose.

Step 7: Estimate the vertical force bias:

$$F_{z,bias} = \frac{1}{N} \sum_{i=1}^{N} F_{z,raw,i} \tag{3.37}$$

Step 8: Read the current robot state:

$$q, \quad \dot{q}, \quad x, \quad y, \quad z, \quad \dot{x}, \quad \dot{y}, \quad \dot{z}, \quad F_{z,raw} \tag{3.38}$$

Step 9: Compute the robot Jacobian:

$$J(q) \tag{3.39}$$

Step 10: Compute Coriolis compensation:

$$C(q, \dot{q}) \tag{3.40}$$

Step 11: Compute the compensated vertical force:

$$F_{z,\Delta} = F_{z,raw} - F_{z,bias} \tag{3.41}$$

Step 12: Compute the interpreted contact force:

$$F_{z,contact} = s_c F_{z,\Delta} \tag{3.42}$$

Step 13: Compute the planar position errors:

$$e_x = x_d - x \tag{3.43}$$

$$e_y = y_d - y \tag{3.44}$$

Step 14: Compute the position-control force components:

$$F_x = K_{xy} e_x - D_{xy} \dot{x} \tag{3.45}$$

$$F_y = K_{xy} e_y - D_{xy} \dot{y} \tag{3.46}$$

Step 15: Compute the force error in the vertical direction:

$$e_F = F_{z,target} - F_{z,contact} \tag{3.47}$$

Step 16: Compute the vertical force command:

$$F_z = K_f e_F - D_f \dot{z} \tag{3.48}$$

Step 17: Apply a vertical safety wall if needed:

$$\text{if } z < z_{min}, \qquad F_z \leftarrow F_z + K_{wall}(z_{min} - z) \tag{3.49}$$

Step 18: Construct the commanded Cartesian wrench:

$$W_{cmd} = \begin{bmatrix} F_x \\ F_y \\ F_z \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{3.50}$$

Step 19: Convert the Cartesian wrench into joint torques:

$$\tau_{raw} = J(q)^T W_{cmd} + C(q, \dot{q}) \tag{3.51}$$

Step 20: Apply torque-rate saturation:

$$\Delta\tau = \text{clip}\left(\tau_{raw} - \tau_{prev}, -\Delta\tau_{\max}, \Delta\tau_{\max}\right) \tag{3.52}$$

$$\tau_{cmd} = \tau_{prev} + \Delta\tau \tag{3.53}$$

Step 21: Send the final torque command τ_{cmd} to the FR3 through the Franka Control Interface.

Step 22: Save position, force, and torque data to the log file.

Step 23: Update the previous torque command:

$$\tau_{prev} \leftarrow \tau_{cmd} \tag{3.54}$$

Step 24: Repeat the loop until the surface-transfer task finishes or a safety stop is triggered.

The hybrid controller is suitable for contact tasks because it allows the robot to control different task-space directions in different ways. In the surface-transfer experiment, the horizontal directions were used for trajectory tracking, while the vertical direction was used for force regulation. This allowed the robot to slide the object along the table while maintaining contact force.

3.7.4 Comparison of the Three Algorithms

The three algorithms differ in the way they relate motion and force.

The impedance controller receives a desired pose and generates a force or torque response when the robot deviates from this pose. Therefore, it can be interpreted as:

$$\text{motion error} \rightarrow \text{force response} \tag{3.55}$$

The admittance controller receives an external force and generates a motion response. Therefore, it can be interpreted as:

$$\text{external force} \rightarrow \text{motion command} \tag{3.56}$$

The hybrid force/position controller separates the task directions. Some directions are controlled by position, while other directions are controlled by force. Therefore, it can be interpreted as:

$$\text{position control in selected axes} + \text{force control in selected axes} \tag{3.57}$$

In this thesis, the impedance controller was mainly used to achieve compliant behavior, the admittance controller was used to study force-driven motion and hand-guiding behavior, and the hybrid controller was used for surface-contact tasks where the robot had to move along a surface while regulating the normal force.

Chapter 4

Experimental Results

4.1 Franka Control Interface

The Franka Control Interface (FCI) allows custom controllers to run on an external real-time capable Linux computer. It provides access to robot-state data and allows torque commands to be sent at high frequency. This interface is essential for the work of this thesis because impedance control requires fast feedback from measured joint states, estimated external torques, Cartesian poses, and force-related quantities.

The real-time communication loop is sensitive to latency. If the loop misses deadlines, the robot may stop for safety reasons. For this reason, the control computer was configured with a real-time Linux kernel and a direct Ethernet connection to the robot controller. The controller node was designed to avoid heavy computations inside the hard real-time loop, ensuring stable communication and safe execution of torque commands.

The FCI was used as the main interface for the real-robot experiments of this thesis. Through this interface, the implemented controllers could read the robot state, estimate interaction forces, send joint torque commands, and log experimental data for post-processing. The experiments were designed to evaluate the behavior of the FR3 during free-space motion, contact with the environment, external disturbances caused by human interaction, compliant hand-guiding, and object manipulation.

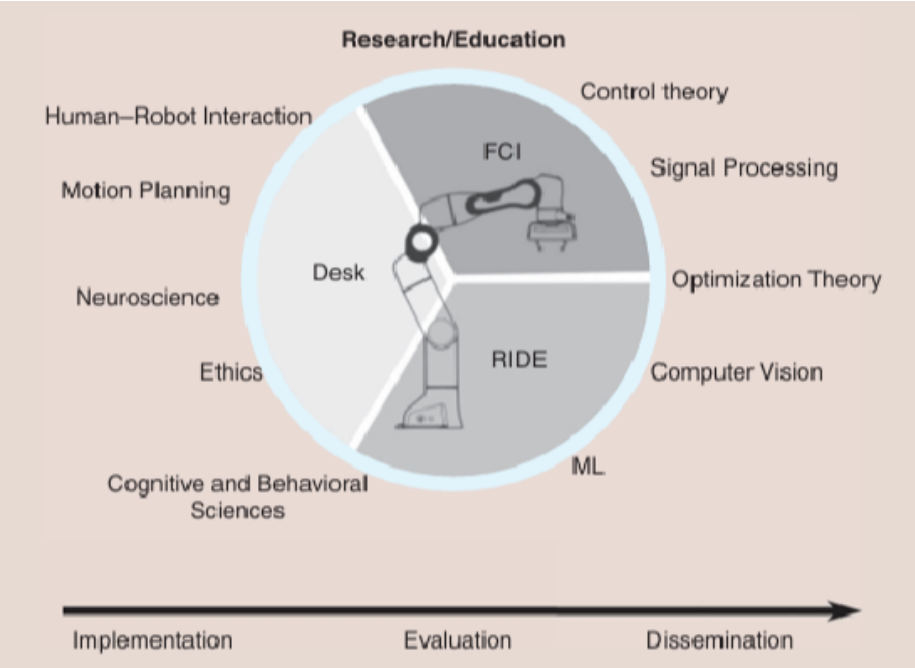

Figure 4.1: Franka Research 3 as a research and education platform. The figure illustrates the role of the FR3 ecosystem, including Desk, FCI, and RIDE, and shows how the platform supports several research fields such as human–robot interaction, control theory, motion planning, computer vision, machine learning, signal processing, and optimization.

The figure above presents the Franka Research 3 as a multidisciplinary research and education platform. The central part of the figure shows the FR3 robot together with the main software and control components of the Franka ecosystem. Desk provides the user interface for robot monitoring, safety configuration, unlocking, recovery, and activation of the robot. The Franka Control Interface (FCI) provides the low-level communication layer that allows external real-time controllers to access robot-state data and send commands to the robot. RIDE represents the higher-level development and execution environment provided by the Franka ecosystem.

The surrounding terms in the figure show that the FR3 can be used in many different research areas. In the context of this thesis, the most relevant areas are human–robot interaction, control theory, motion planning, signal processing, optimization, computer vision, and machine learning. Human–robot interaction is directly related to the goal of safe physical contact between the robot and the user. Control theory is necessary for the design of impedance, admittance, and hybrid force/position controllers. Signal processing is useful for filtering torque and force estimates. Optimization theory is related to redundancy resolution, null-space control, and trajectory generation. Computer vision and machine learning can be used in future extensions for object detection, human intention prediction, and adaptive control.

The horizontal axis at the bottom of the figure indicates the typical research workflow

from implementation to evaluation and dissemination. This is also the structure followed in this thesis. First, the control algorithms were implemented using libfranka. Then, the methods were evaluated through real-robot experiments on the FR3. Finally, the results were analyzed and documented in order to assess the suitability of the FR3 for force-control and human-interaction tasks.

4.1.1 Experimental Campaign

The experimental campaign was organized in a progressive way. The first experiments focused on simple and safe free-space motion, while later experiments introduced contact, force regulation and human interaction. This gradual structure was necessary because force-control experiments are more sensitive than pure position-control experiments. Before testing contact-rich tasks, the robot motion, network communication, controller stability, and safety limits had to be validated.

For every experiment, the following quantities were considered important for evaluation:

- joint positions and joint velocities,
- end-effector Cartesian position,
- desired and actual trajectory,
- position tracking error,
- measured or estimated external wrench,
- external joint torques,
- commanded joint torques,
- contact force, when contact was present,
- controller stability and safety behavior.

The experiments were executed using conservative limits at the beginning. The robot was always monitored through Desk, and the emergency stop was kept accessible. The purpose of the experiments was not only to demonstrate that the robot could move, but also to evaluate how the implemented controllers behaved when the robot interacted with its environment or with a human operator.

4.1.2 Experiment 1: Baseline Free-Space Joint Trajectory

The first experiment was a baseline free-space joint trajectory. The purpose of this experiment was to verify that the robot could execute a smooth joint-space motion

without contact with the environment. This experiment was used as the reference case for the rest of the experimental campaign.

In this experiment, a desired joint trajectory was generated for one or more joints of the FR3. The commanded trajectory was intentionally small and smooth in order to avoid sudden accelerations or velocity discontinuities. The robot moved in free space, without touching any object or surface.

The controller objective was to minimize the joint tracking error:

$$e_q(t) = q_{ref}(t) - q(t)$$

where $q_{ref}(t)$ is the desired joint trajectory and $q(t)$ is the measured joint position. A simple joint-space control law can be written as:

$$\tau_{cmd} = K_p\,(q_{ref} - q) + K_d\,(\dot{q}_{ref} - \dot{q})$$

where K_p and K_d are the proportional and derivative gain matrices.

The main goal of this experiment was to confirm that the FCI communication, controller loop, and torque command pipeline were working correctly. Since there was no intentional contact, the measured external forces and external torques were expected to remain small. Any large external torque estimate during this experiment would indicate a possible modeling error, incorrect payload configuration, or unexpected contact.

The logged data from this experiment were used to evaluate the quality of the trajectory tracking. Important plots include actual versus desired joint position, joint tracking error, commanded torque, and external torque norm. This experiment provided the baseline behavior of the FR3 before introducing contact forces.

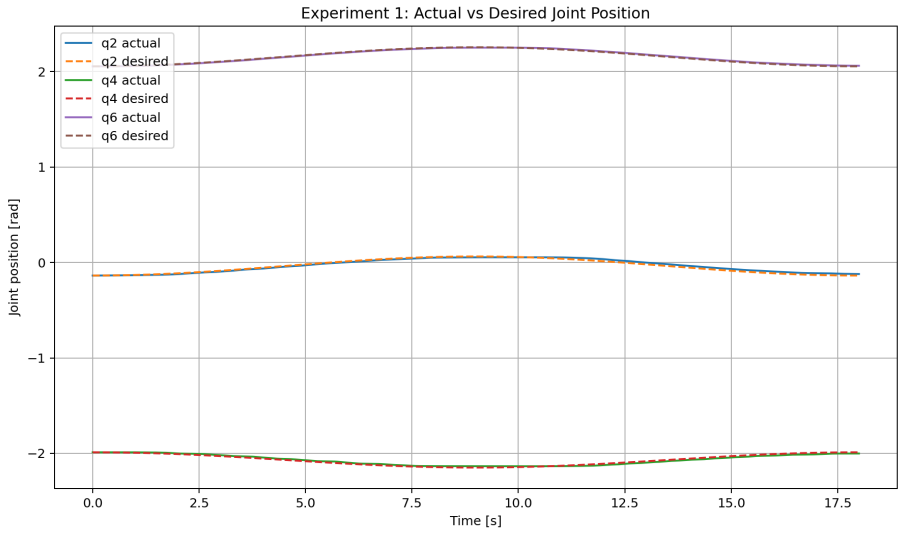

Figure 4.2: The solid and dashed curves almost overlap, so the robot followed the reference trajectory well.

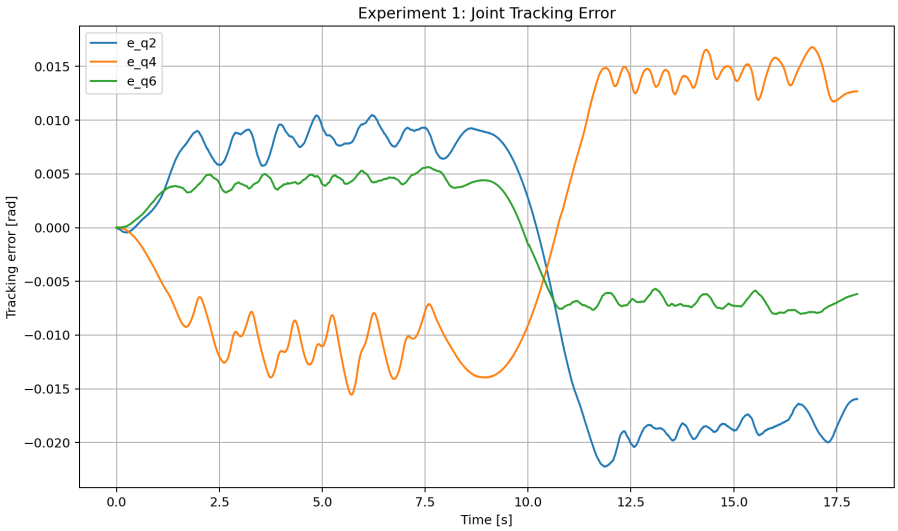

Figure 4.3: The error stays small, roughly within about ±0.02 rad, which is acceptable for a smooth torque-controlled baseline.

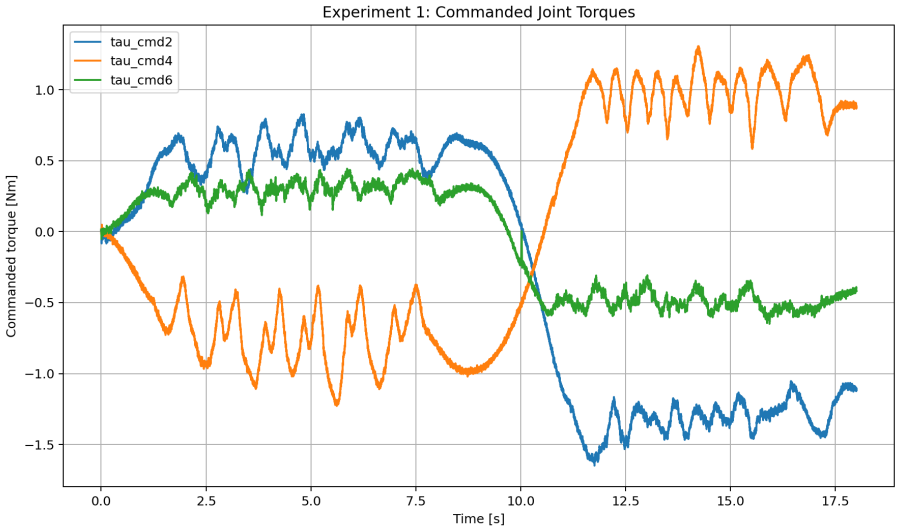

Figure 4.4: the torques are smooth and remain low, around less than ±1.7 Nm, so the controller is not producing aggressive commands.

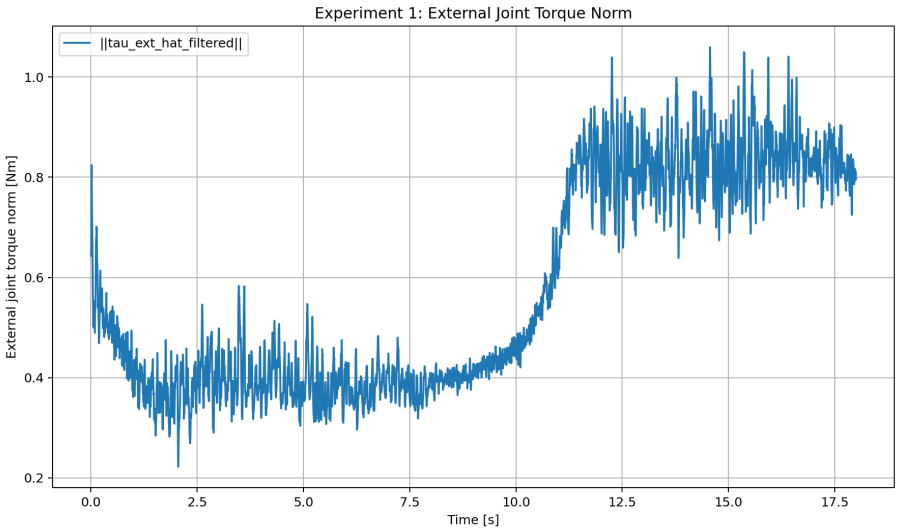

Figure 4.5: it remains low, approximately below 1.1 Nm.

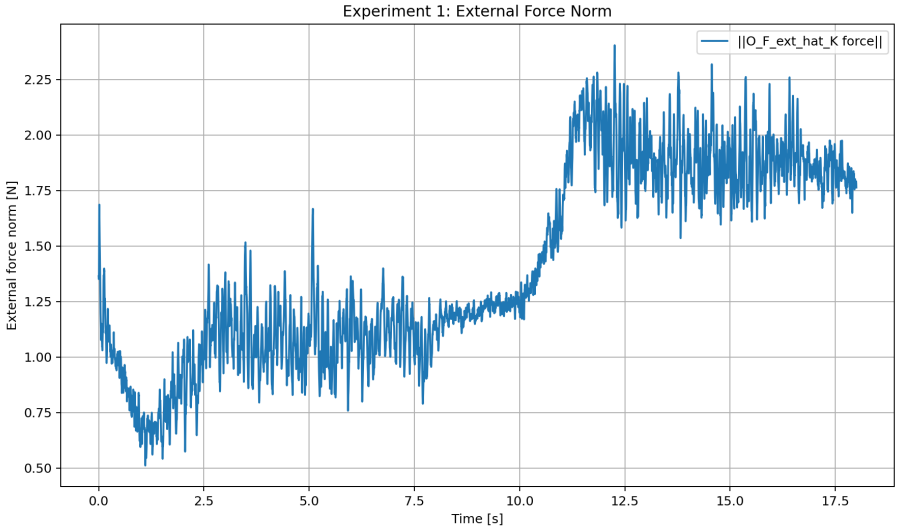

Figure 4.6: it stays low, below about 2.5 N, so there was no significant contact with the environment.

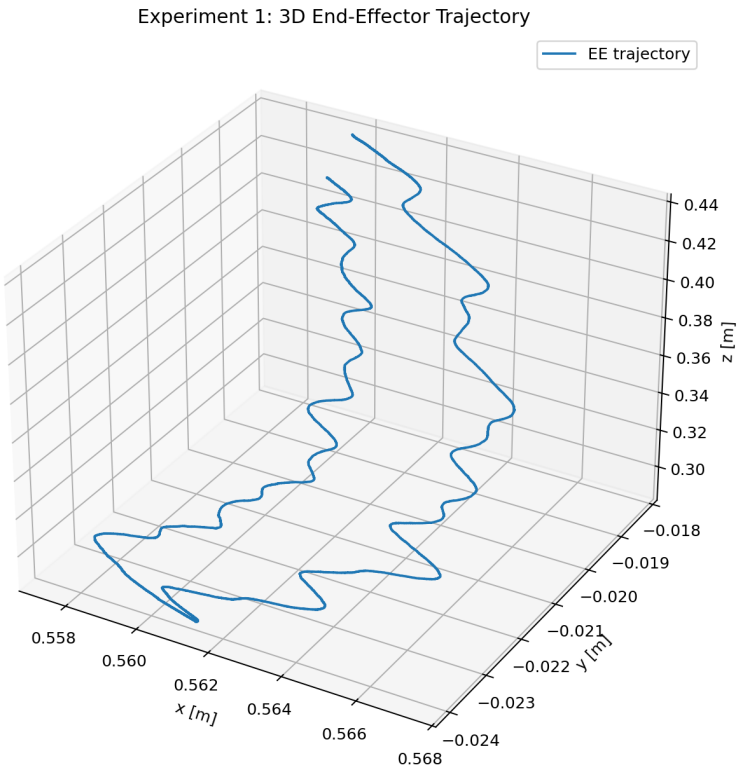

Figure 4.7: the end-effector follows a small but visible free-space path, which is exactly what this experiment should show.

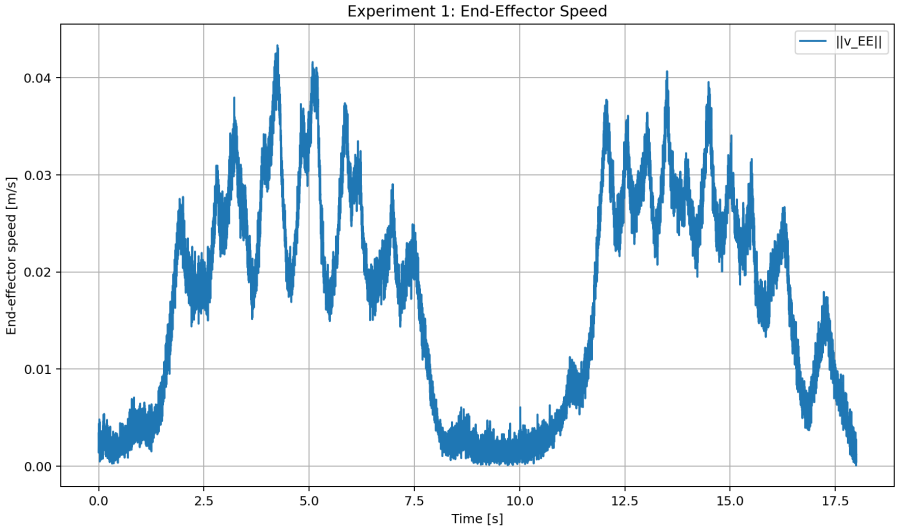

Figure 4.8: the speed stays low, below about 0.045 m/s, so the motion is safe and smooth.

The only thing to mention carefully is that the external force/torque estimates are not exactly zero. That is normal because the FR3 estimates external forces from the dynamic model, gravity compensation, payload assumptions, sensor noise, and motion effects. Since the values are small, this does not indicate problematic contact.

Conclusion

Overall, this experiment provides a reliable reference case for the following force-control experiments. It verifies that the robot can execute controlled joint-space motion without contact and establishes the expected baseline behavior of the FR3 in terms of tracking accuracy, commanded torques, external force estimates, and end-effector motion.

4.1.3 Experiment 2: Pick and Place Demonstration with a dice

The second experiment evaluated a pick-and-place task with a dice. The robot first moved to a pregrasp pose above the object, then descended toward the grasping position, grasped the object, lifted it, transferred it to a new location, and finally placed it back on the table. During the whole task, Cartesian position, tracking error, external force estimates, joint torques, end-effector speed, and phase information were recorded.

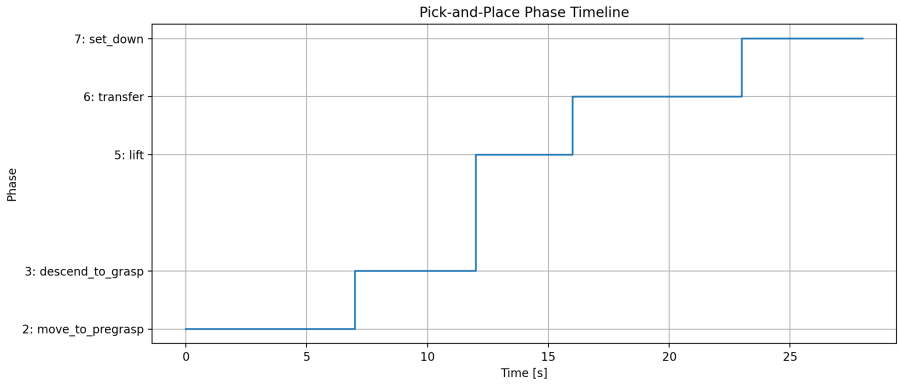

Figure 4.9: The main motion phases are shown: `move_to_pregrasp`, `descend_to_grasp`, `lift`, `transfer`, and `set_down`. The missing phase 4 is not an error; it corresponds to the gripper grasp action, which is not logged inside `moveToPose()`, so it does not appear in the phase timeline.

The task sequence was:

1. move to a pre-grasp pose,
2. open the gripper,
3. approach the object,
4. close the gripper and grasp the object,
5. lift the object,
6. move to the target pose,
7. release the object,

8. return to a safe pose.

The pick and place experiment was important because it tested the integration of motion generation and gripper control. Although this task does not necessarily require continuous force control, it provides the foundation for more advanced manipulation tasks. In particular, it validates whether the robot can approach an object safely, grasp it, and move it without losing stability.

For the FR3, this experiment also helped calibrate practical parameters such as grasp height, gripper width, grasping force, approach speed, and safe lifting distance. If the gripper did not confirm the grasp, the object width, grasp height, or grasping force had to be adjusted.

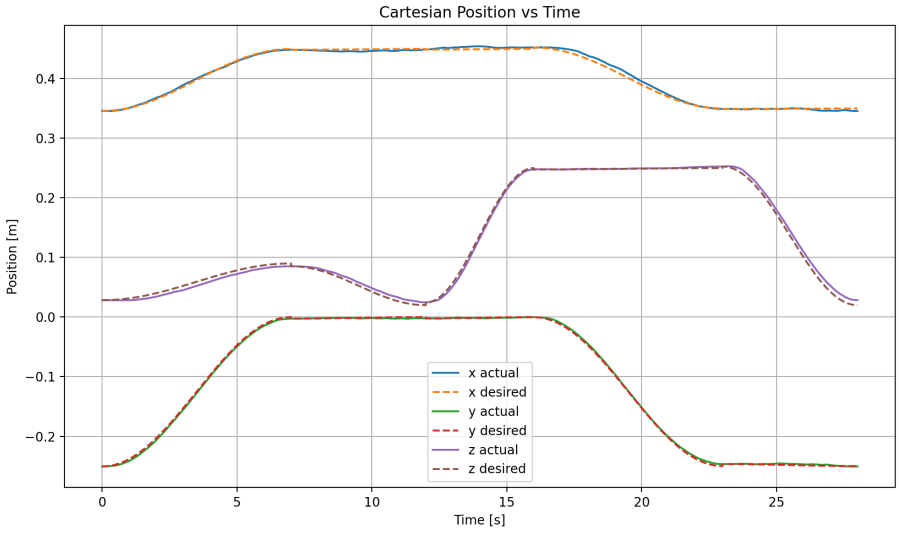

Figure 4.10: The actual and desired x, y, and z curves almost overlap. This shows that the robot followed the commanded Cartesian trajectory correctly during the pick-and-place motion.

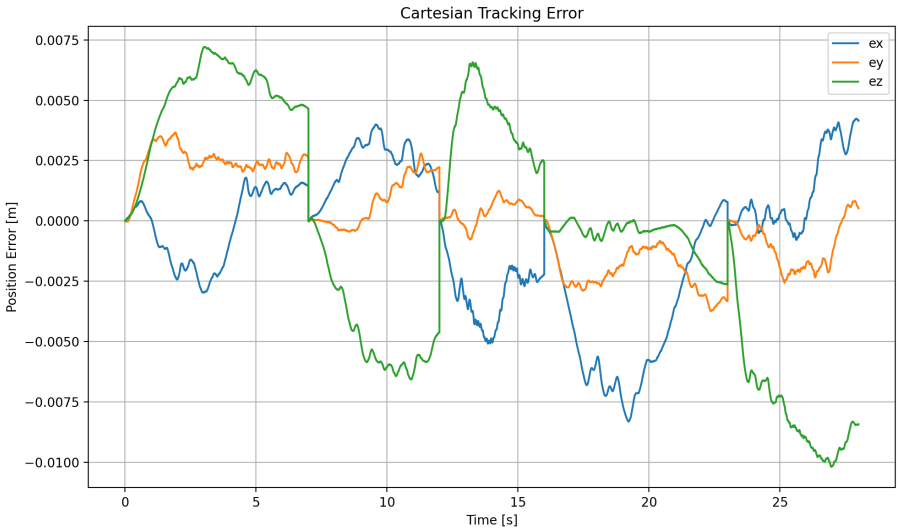

Figure 4.11: The tracking error remained small throughout the task, with deviations on the order of a few millimeters and a maximum error of approximately one centimeter.

This indicates that the Cartesian motion controller was able to execute the pick-and-place trajectory accurately, even during the phases involving object interaction.

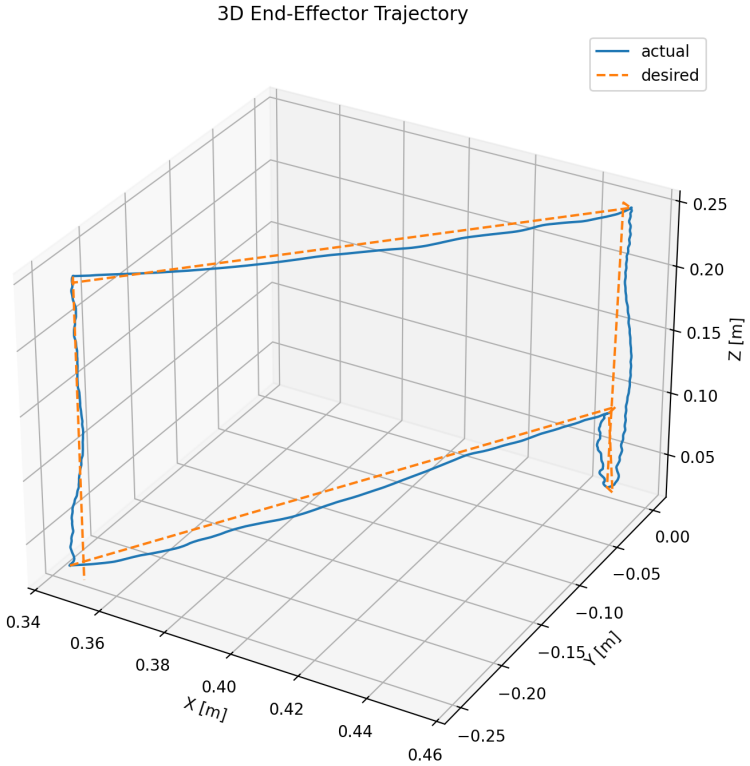

Figure 4.12: It clearly shows the robot moving from the pick position, lifting the object, transferring it to the place position, and lowering it again. The actual trajectory follows the desired trajectory closely.

The external force measurements remained within a safe range during the experiment.

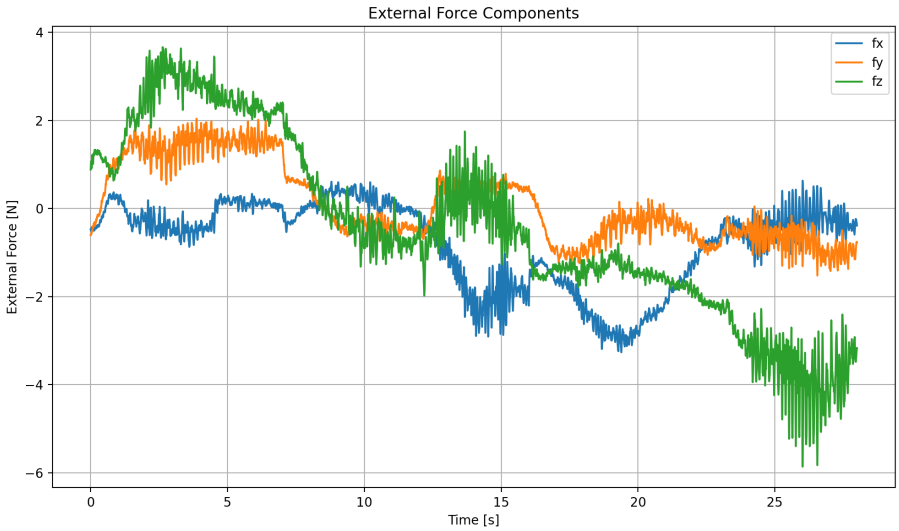

Figure 4.13: The maximum estimated external force was approximately 6 N, which indicates that the interaction with the soft object did not generate large contact forces.

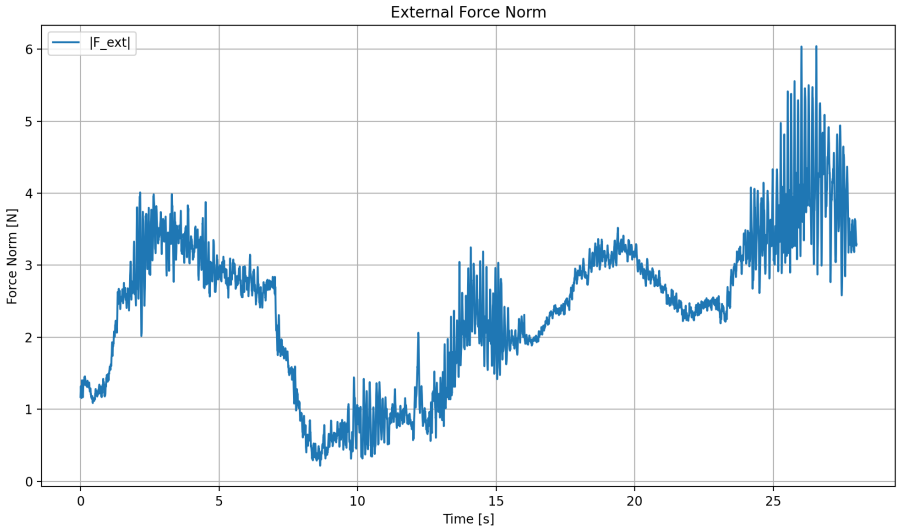

Figure 4.14: The force norm increased during phases where physical interaction was expected, especially around grasping, lifting, transfer, and set-down.

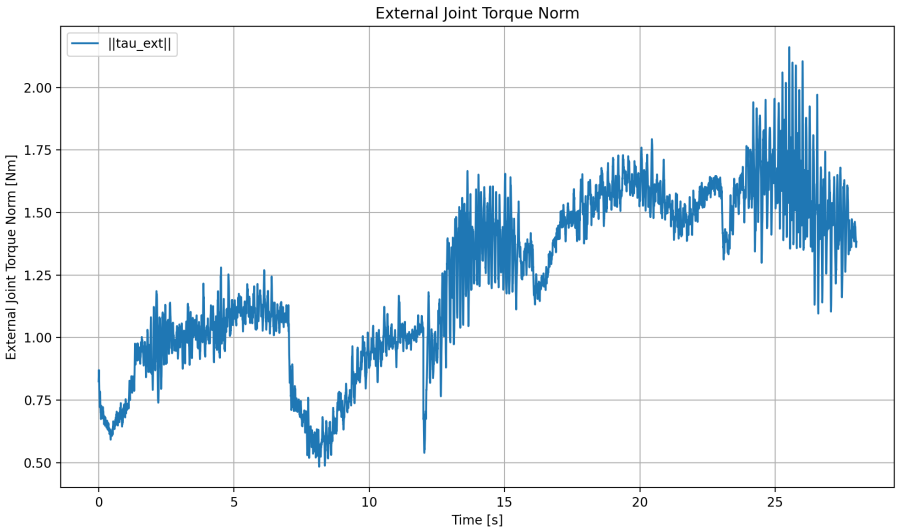

Figure 4.15: The external joint torque norm also remained moderate, suggesting that no unsafe collision or excessive external disturbance occurred.

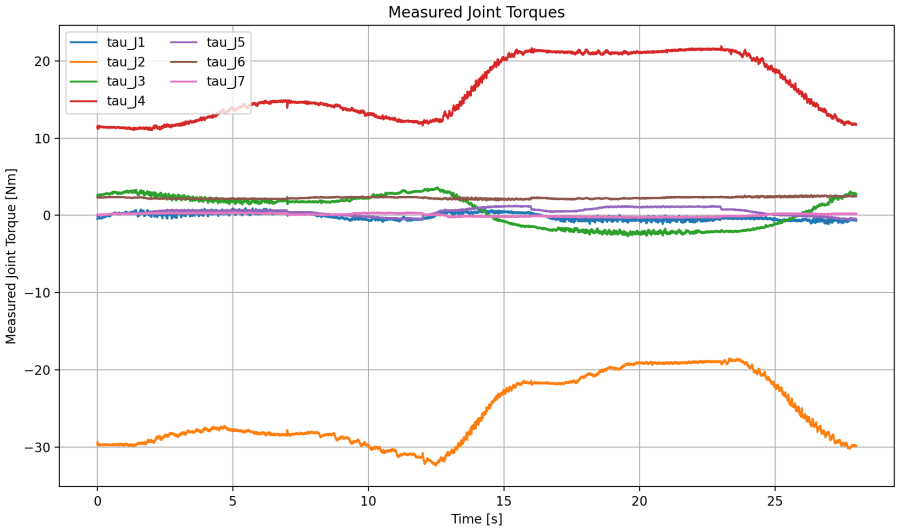

Figure 4.16: Joints 2 and 4 have the largest torque values, which is normal because these joints carry much of the robot arm weight and are strongly affected by gravity and configuration changes. These are measured joint torques, not necessarily the contact torques.

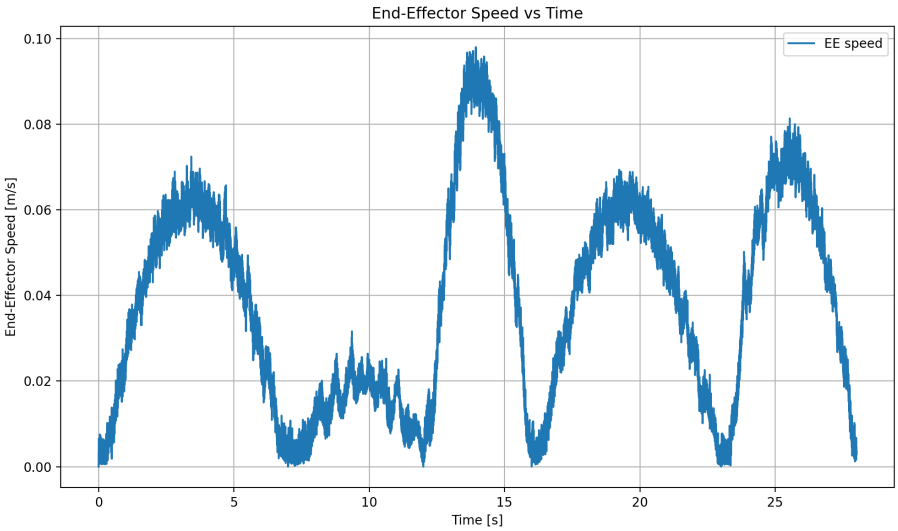

Figure 4.17: The maximum speed is around 0.10 m/s. This is still safe for a controlled pick-and-place experiment.

Conclusion

Overall, the experiment was successful. It demonstrated that the FR3 manipulator can perform a complete soft-object pick-and-place task while maintaining accurate Cartesian tracking and safe external force levels. The results also confirm that the external wrench and joint torque logging pipeline works correctly, providing useful data for the analysis of contact-aware manipulation and future force-control experiments.

4.1.4 Experiment 3: Hybrid Position/Force Interaction with a Soft Ball

This experiment was designed to evaluate the ability of the FR3 manipulator to maintain physical interaction with a compliant object while performing a tangential motion. A soft foam ball was placed on the table. The purpose of the experiment was to examine how the robot behaves during a contact-rich task where both position tracking and force interaction are present.

The experimental sequence consisted of five main phases. First, the gripper was closed to a predefined width in order to create a stable contact interface. Then, the end-effector moved slowly downward along the vertical direction until contact with the soft ball was detected through the estimated external force. After contact detection, the robot continued moving slightly downward in order to compress the ball and generate a visible deformation. Then, while maintaining the compressed configuration, the end-effector performed a horizontal motion along the table surface, dragging the ball in the commanded direction. Finally, the robot retracted upward and the experiment was completed.

During the experiment, the measured Cartesian position shows that the end-effector moved mainly along the desired horizontal direction. In the successful run, the displacement along the y-axis was approximately 7 cm, which is close to the commanded 8 cm displacement. Small deviations from the ideal trajectory are expected because the robot was interacting with a deformable object and because friction between the ball and the table affected the motion. The z-position decreased during the approach and touchdown phases and then remained approximately constant during the dragging phase, indicating that the robot maintained contact with the ball instead of lifting away from it.

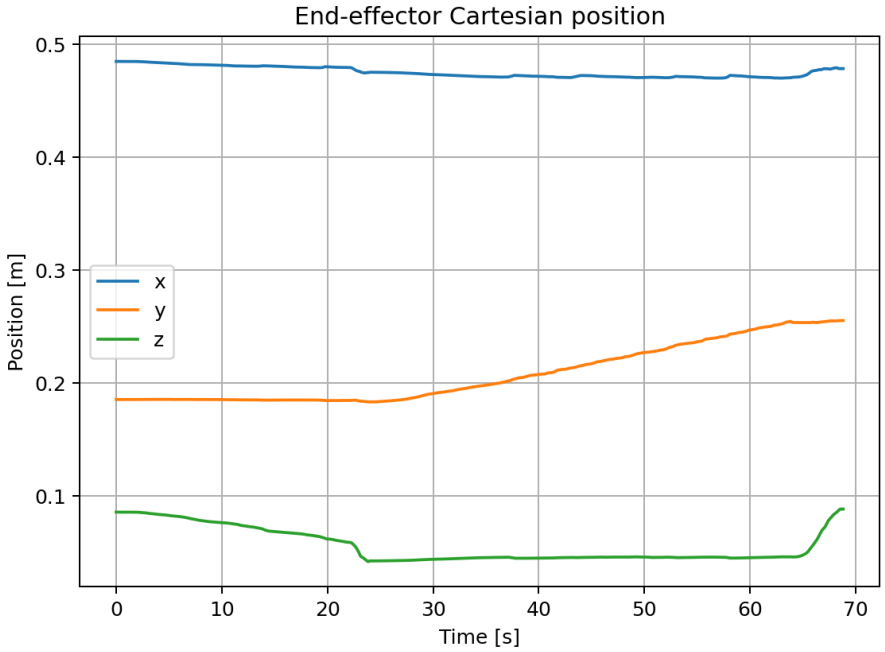

Figure 4.18: End-effector Cartesian position during the soft ball dragging experiment. The vertical motion corresponds to the approach and touchdown phases, while the horizontal displacement along the y-axis corresponds to the dragging phase.

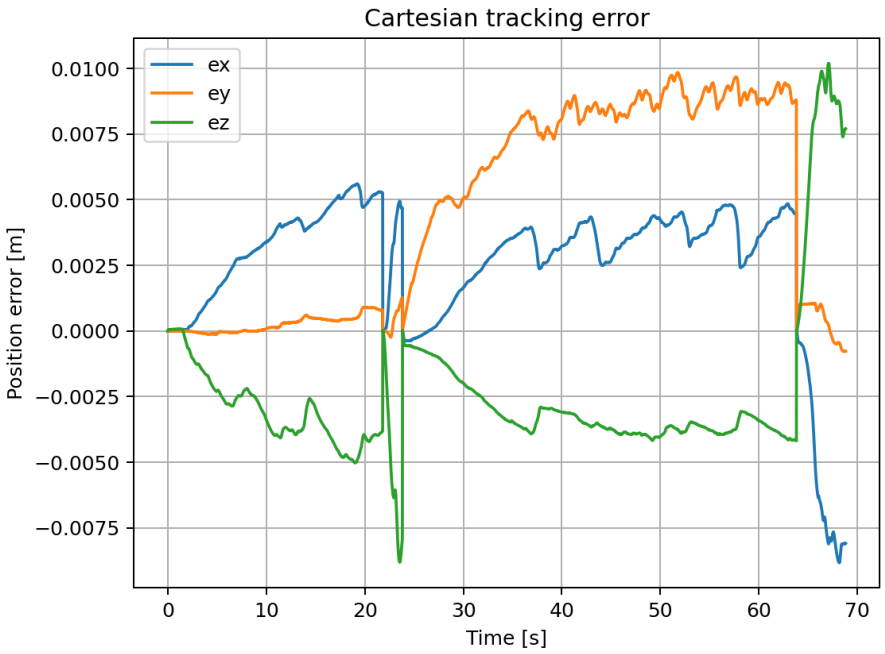

Figure 4.19: Cartesian tracking error during the soft ball dragging experiment. Small deviations appear during contact because the robot interacts with a compliant object and must overcome friction between the ball and the table.

The estimated normal contact force also confirms the presence of physical interaction. During the touchdown phase, the contact force increased and reached a peak of approximately 4.5 N, corresponding to the compression of the foam ball. During the dragging phase, the normal contact force remained lower but clearly positive, mainly around 1.5–2 N. This indicates that the robot maintained contact with the soft object while moving tangentially across the table. The bias-compensated external force norm remained below approximately 5 N, which shows that the interaction remained within a safe range.

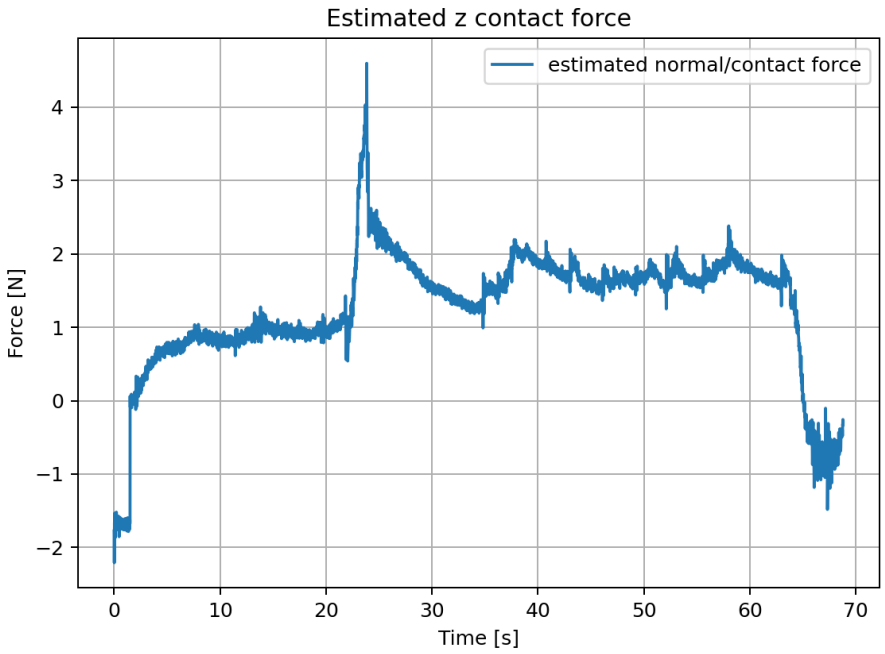

Figure 4.20: Estimated normal contact force during the interaction with the soft ball. The force increases during touchdown and remains positive during the dragging phase, indicating maintained contact.

The external force components provide additional information about the contact behavior. The vertical force component increased during compression, while the horizontal force components changed during the dragging phase due to friction and resistance from the ball. This behavior is expected in a sliding interaction, since the robot must overcome tangential friction while also maintaining vertical contact. The measured joint torques remained smooth throughout the experiment, with the largest torques appearing mainly in the joints that carry most of the manipulator weight. The external joint torque norm remained low, below approximately 1.3 Nm, indicating that the task did not generate excessive disturbances at the joint level.

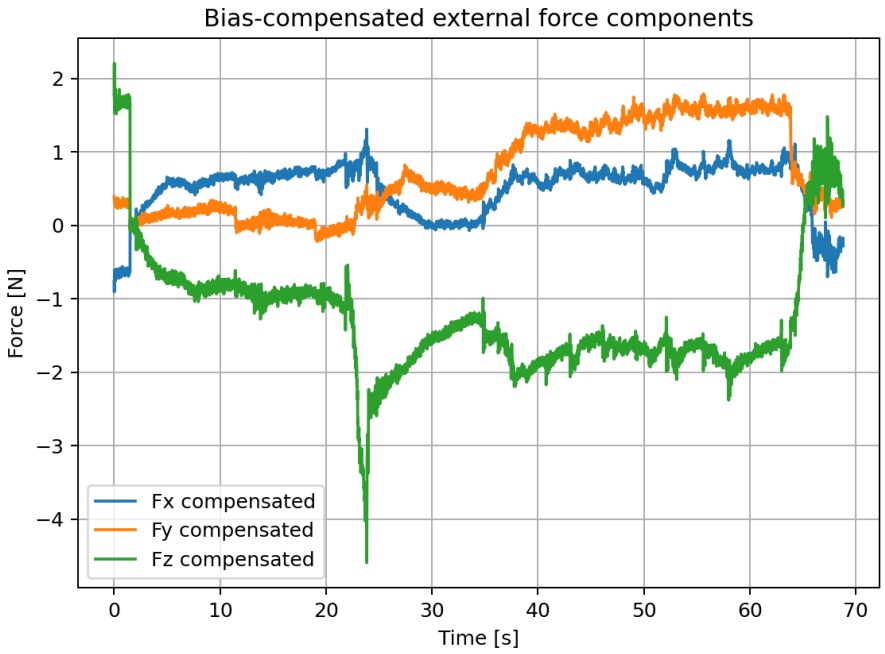

Figure 4.21: Bias-compensated external force components during the soft ball dragging experiment. The vertical component corresponds mainly to compression, while the horizontal components are affected by tangential friction during dragging.

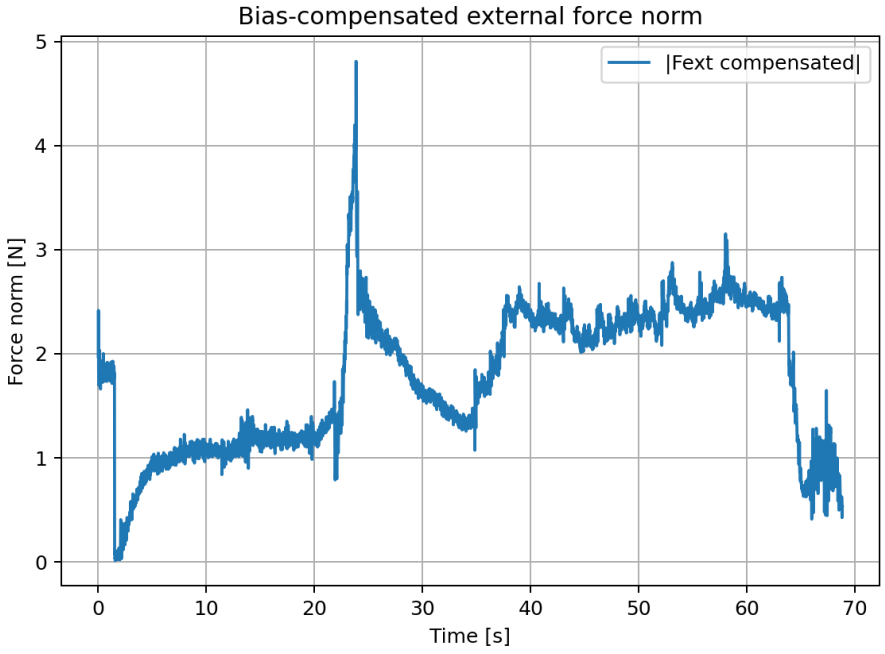

Figure 4.22: Bias-compensated external force norm during the soft ball dragging experiment. The force norm increases during contact and remains within the predefined safety limits throughout the task.

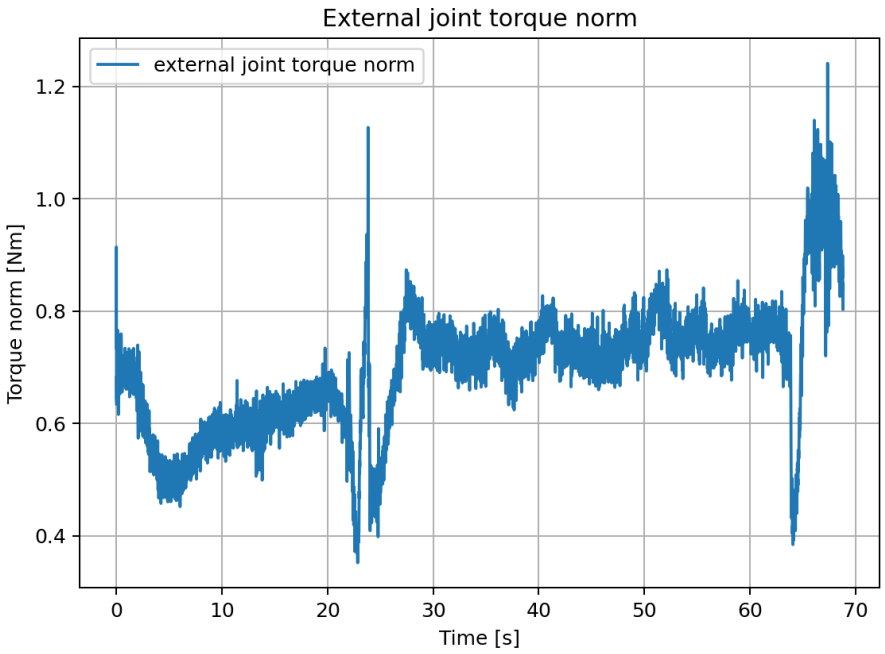

Figure 4.23: External joint torque norm during the experiment. The low magnitude indicates that the contact interaction remained within safe limits.

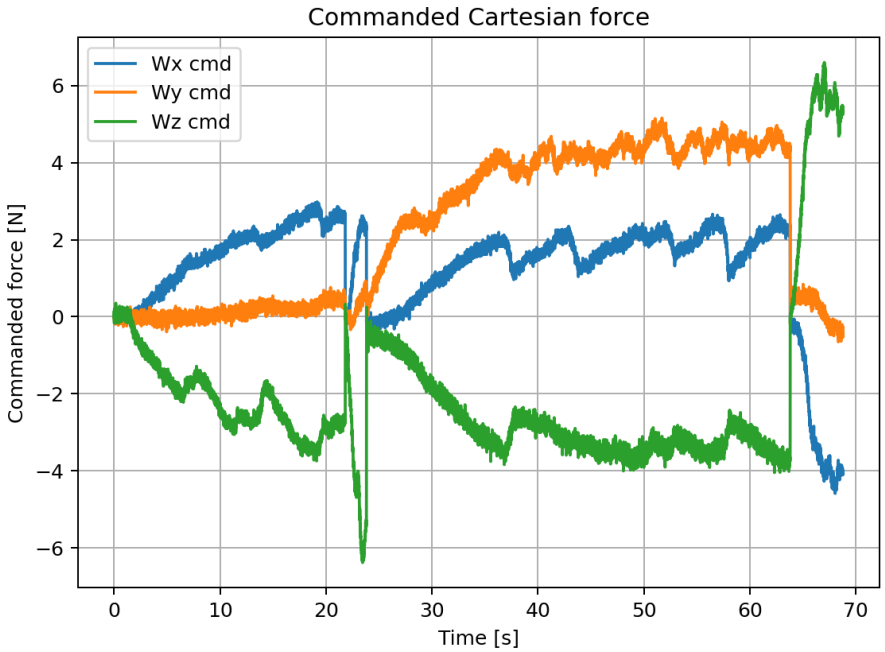

Figure 4.24: Commanded Cartesian force during the soft ball dragging experiment. The commanded force changes according to the approach, touchdown, dragging, and retract phases of the task.

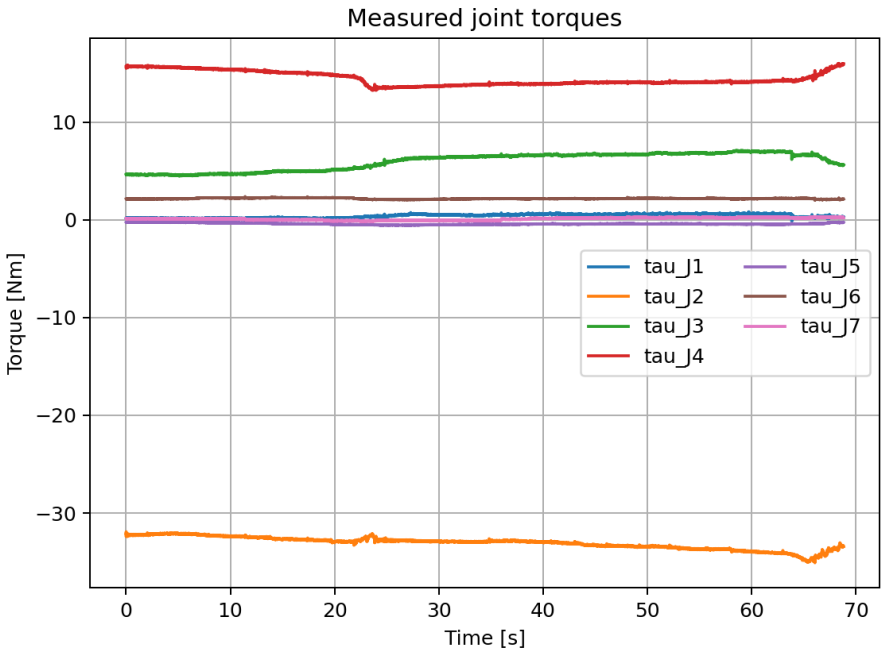

Figure 4.25: Measured joint torques during the soft ball dragging experiment. The torques remain smooth and within reasonable values throughout the contact-rich task.

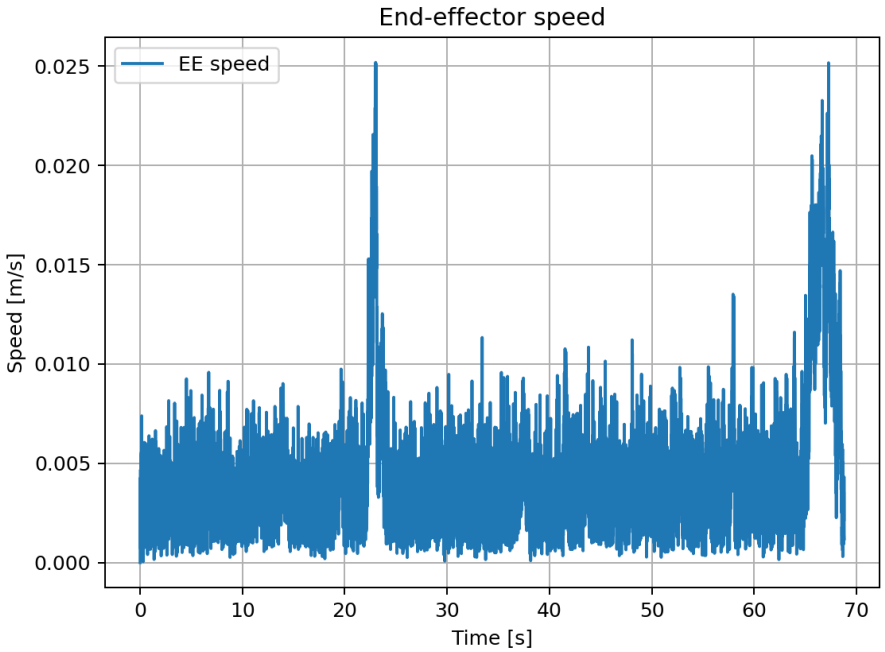

Figure 4.26: End-effector speed during the soft ball dragging experiment. The velocity remains low, confirming that the task was executed in a slow and safe manner.

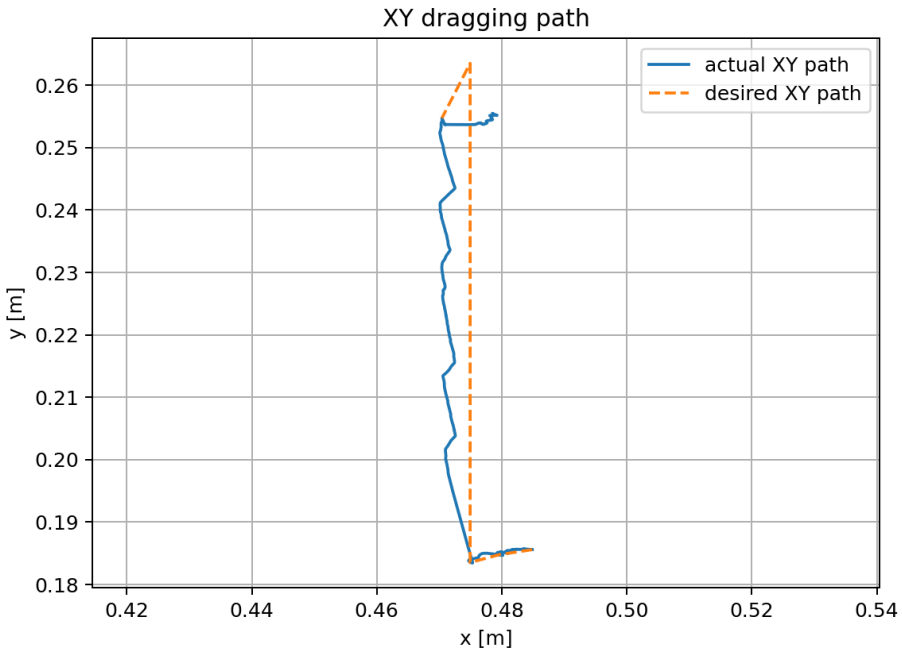

Figure 4.27: End-effector path in the x-y plane during the soft ball dragging experiment. The trajectory shows the horizontal motion used to drag the ball along the table surface.

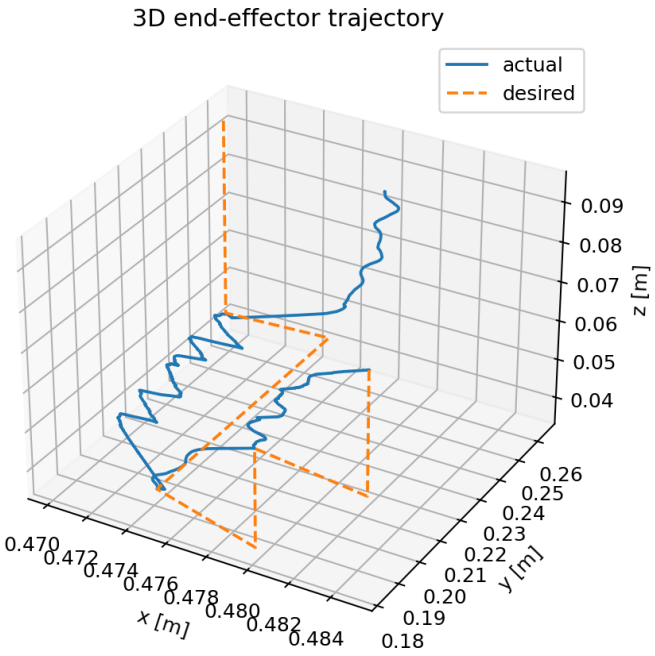

Figure 4.28: Three-dimensional end-effector trajectory during the soft ball dragging experiment. The trajectory includes the approach, touchdown, horizontal dragging, and retract phases.

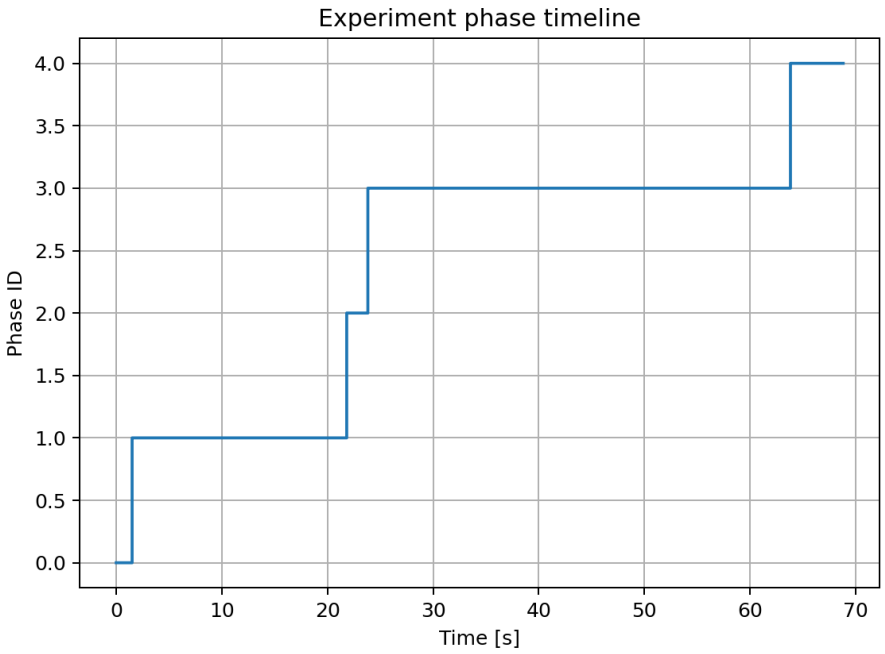

Figure 4.29: Phase timeline of the soft ball dragging experiment. The task consists of bias estimation, approach, touchdown, dragging, and retract phases.

Conclusion

Overall, this experiment demonstrates that the FR3 manipulator can perform a stable contact-maintaining manipulation task with a compliant object. The robot successfully approached the ball, compressed it, dragged it along the table, and retracted without exceeding the predefined safety limits. The results show that the controller was able to combine Cartesian position tracking with force monitoring during physical interaction. Although the force was not regulated to an exactly constant value, the experiment validates the ability of the robot to maintain safe contact while executing a useful tangential motion.

4.1.5 Experiment 4: Human Push and Disturbance Rejection

The fourth experiment evaluated the response of the FR3 manipulator to external human disturbances. The purpose was to examine whether the robot can remain stable, safe, and compliant when a human applies a push to the arm or to the end-effector. This experiment is important for physical human–robot interaction, because a collaborative robot should not behave as a perfectly rigid system during contact with a human operator.

During the experiment, the robot was commanded either to hold a desired Cartesian pose or to follow a slow reference trajectory. While the controller was running, a human applied external pushes to the robot. The response was evaluated using the end-effector

displacement, Cartesian position error, estimated external wrench, external joint torque norm, commanded Cartesian force, measured joint torques, and end-effector speed.

For the pose-holding case, the Cartesian impedance controller was used. The desired pose was treated as an equilibrium point, and the robot generated a restoring force when it was displaced from this reference:

$$F_{\text{cmd}} = K_x(x_{\text{ref}} - x) + D_x(\dot{x}_{\text{ref}} - \dot{x}), \tag{4.1}$$

where K_x is the Cartesian stiffness matrix, D_x is the Cartesian damping matrix, x_{ref} is the reference pose, and x is the measured end-effector pose. The commanded Cartesian force was mapped to joint torques through the Jacobian transpose:

$$\tau_{\text{cmd}} = J^T(q)F_{\text{cmd}} + C(q, \dot{q}), \tag{4.2}$$

where $J(q)$ is the robot Jacobian and $C(q, \dot{q})$ represents the Coriolis compensation term.

Two impedance configurations were tested: a soft/compliant configuration and a stiff configuration. The soft configuration was expected to allow larger displacement under human contact, while the stiff configuration was expected to reject disturbances more strongly. Therefore, this experiment demonstrates the fundamental trade-off between physical compliance and disturbance rejection.

Soft/Compliant Push Experiment

In the soft configuration, the robot behaved compliantly during the human pushes. The desired pose remained approximately constant, while the actual end-effector position moved away from the reference during the disturbance intervals. This confirms that the robot did not behave rigidly, but allowed safe displacement under external human force.

The main quantitative results of the soft push experiment are summarized in Table 4.1.

Table 4.1: Main metrics of the soft/compliant human push experiment.

Metric	**Value**
Maximum position error	$0.026\,\text{m} \approx 2.6\,\text{cm}$
Mean position error	$0.010\,\text{m} \approx 1.0\,\text{cm}$
Maximum external force norm	$6.18\,\text{N}$
Mean force during disturbance	$4.58\,\text{N}$
Maximum external joint torque norm	$3.85\,\text{Nm}$
Maximum end-effector speed	$0.031\,\text{m/s}$

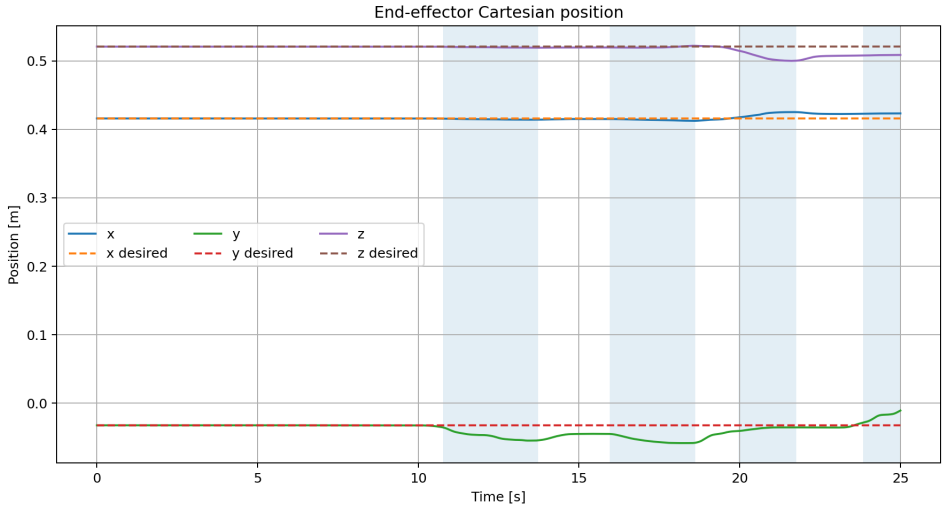

Figure 4.30: Cartesian position during the soft/compliant human push experiment. The desired pose remains approximately constant, while the actual end-effector position changes during the human disturbances.

The position error components show that the largest displacement occurred mainly in the y- and z-directions. The position error norm increased during the human push intervals and reached approximately 2.6 cm. This is expected for a low-stiffness impedance controller, because the robot intentionally allows displacement instead of resisting the human push aggressively.

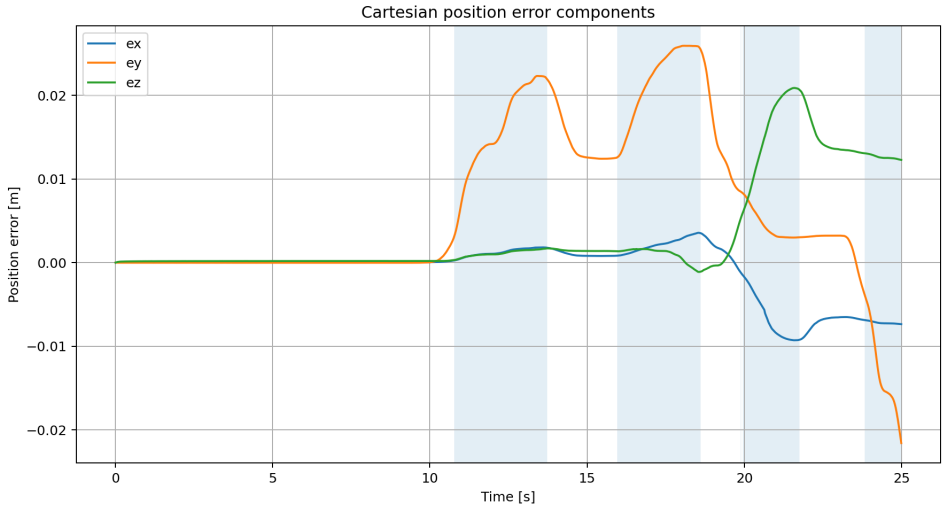

Figure 4.31: Cartesian position error components during the soft/compliant push experiment. The error increases mainly during the disturbance intervals, especially in the directions affected by the human push.

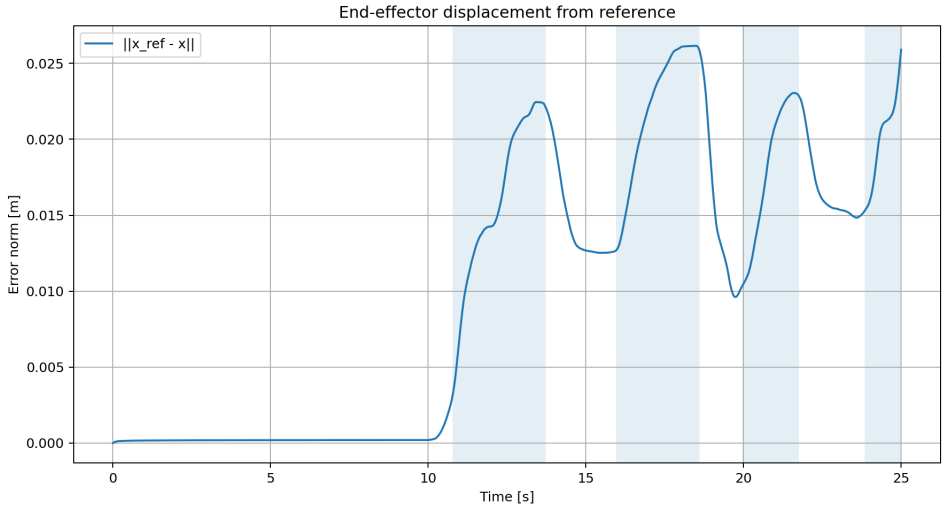

Figure 4.32: Position error norm during the soft/compliant push experiment. The error increases when the robot is pushed and remains bounded, showing compliant displacement under external human force.

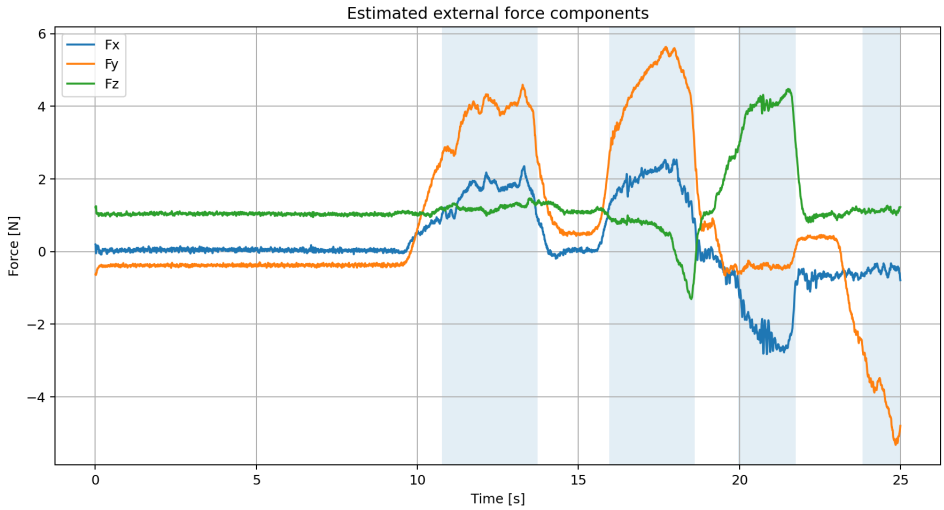

Figure 4.33: Estimated external force components during the soft/compliant push experiment. The force components increase during human contact, confirming that the FR3 detected the external disturbance.

The estimated external force norm increased during the disturbance intervals and reached approximately 6.18 N. This confirms that the FR3 detected the physical interaction through its external force estimation. The external joint torque norm also increased during the pushes, but remained below 4 Nm, indicating that the interaction stayed within safe limits.

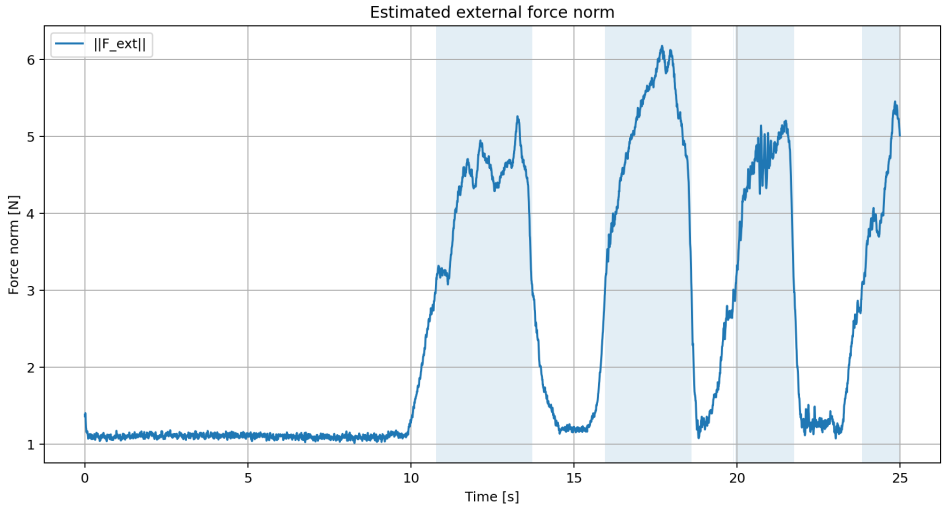

Figure 4.34: Estimated external force norm during the soft/compliant push experiment. The force norm remains moderate, indicating safe physical interaction with the human operator.

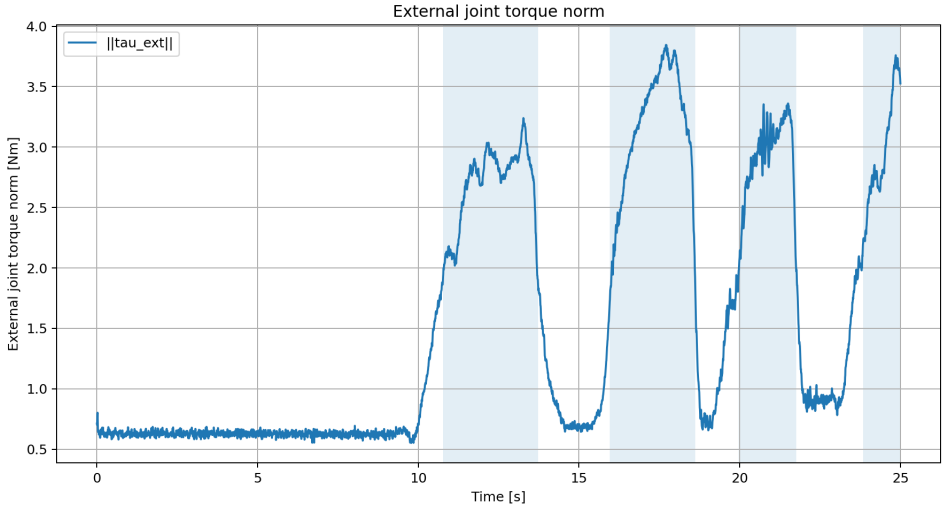

Figure 4.35: External joint torque norm during the soft/compliant push experiment. The increase during the pushes shows that the disturbance was also detected at the joint-torque level.

The end-effector speed remained low, with a maximum value of approximately 0.031 m/s. Therefore, even when the robot was pushed, the resulting motion was slow and safe. The measured joint torques remained smooth, with larger torque values appearing mainly in joints that carry more of the robot weight, such as joints 2 and 4. No dangerous torque spikes were observed.

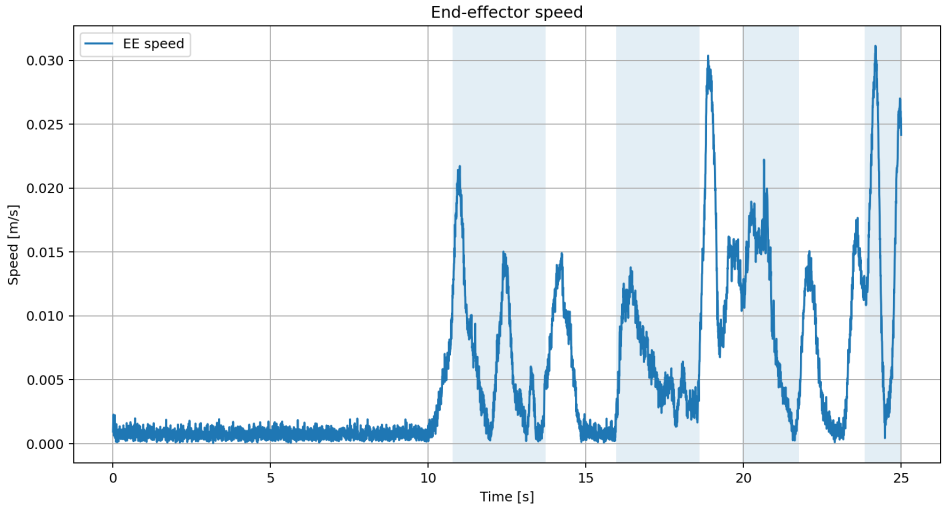

Figure 4.36: End-effector speed during the soft/compliant push experiment. The speed remains low throughout the interaction, confirming slow and safe robot motion.

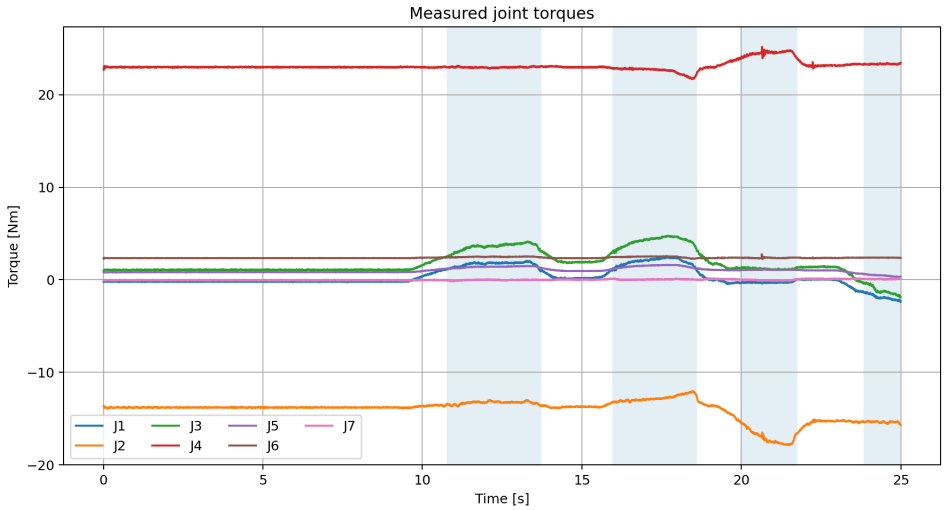

Figure 4.37: Measured joint torques during the soft/compliant push experiment. The torques remain smooth, with larger offsets mainly caused by gravity and the robot configuration.

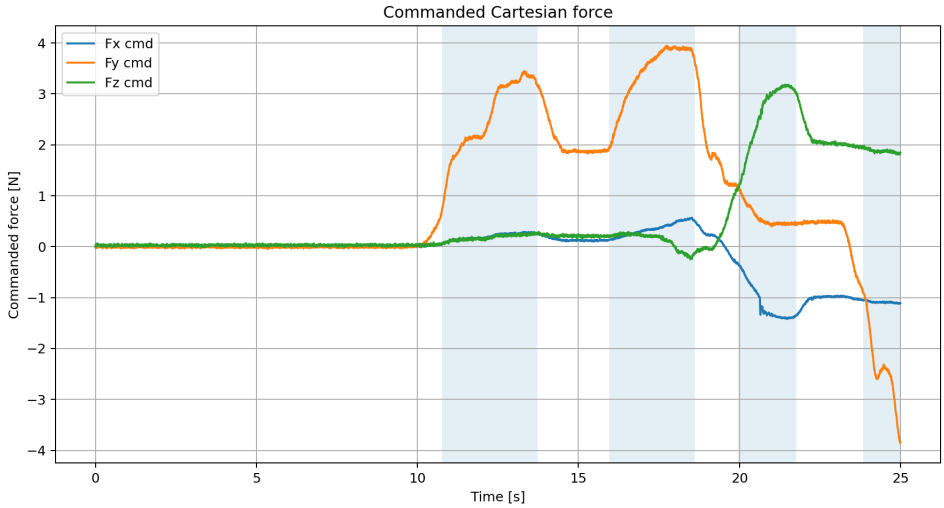

Figure 4.38: Commanded Cartesian force during the soft/compliant push experiment. The controller generates a restoring force when the end-effector is displaced from the reference pose.

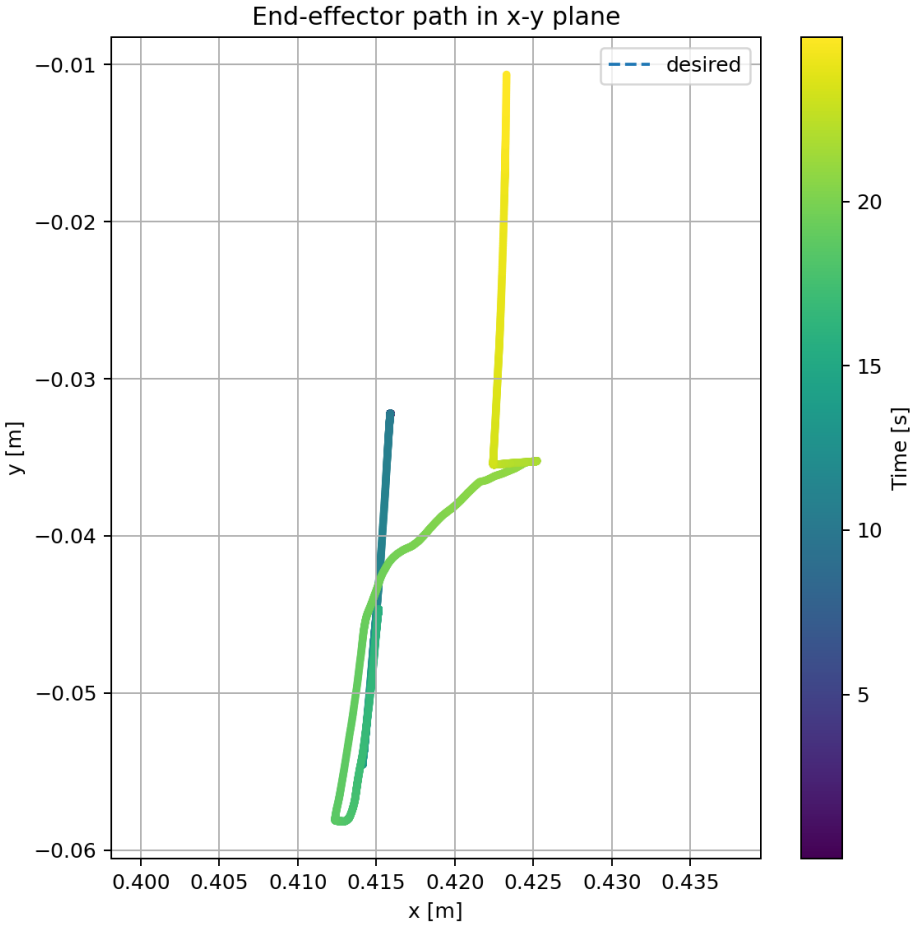

Figure 4.39: End-effector trajectory in the x-y plane during the soft/compliant push experiment. Since the robot was holding a pose, this plot represents the displacement caused by the human pushes rather than a planned path.

Overall, the soft push experiment confirms stable and safe compliant behavior. The robot allowed a moderate displacement under human force, detected the external disturbance, and remained slow and controlled throughout the interaction.

Stiff Push Experiment

In the stiff configuration, the robot rejected the human disturbance more strongly. Compared with the soft case, the end-effector displacement was smaller, but the measured interaction forces and commanded restoring forces were larger. This behavior is expected because a higher stiffness value makes the robot resist displacement more aggressively.

The main quantitative results of the stiff push experiment are summarized in Table 4.2.

Table 4.2: Main metrics of the stiff human push experiment.

Metric	**Value**
Maximum position error	$0.019\,\mathrm{m} \approx 1.9\,\mathrm{cm}$
Maximum external force norm	approximately $11\,\mathrm{N}$
Maximum external joint torque norm	approximately $7\,\mathrm{Nm}$
Maximum end-effector speed	approximately 0.040–$0.045\,\mathrm{m/s}$

The Cartesian position plots show that the desired pose remained almost constant, while the actual position moved only slightly during the human pushes. The position error norm reached approximately $1.9\,\mathrm{cm}$, which is smaller than the $2.6\,\mathrm{cm}$ observed in the soft configuration. This confirms that the stiff controller improved disturbance rejection.

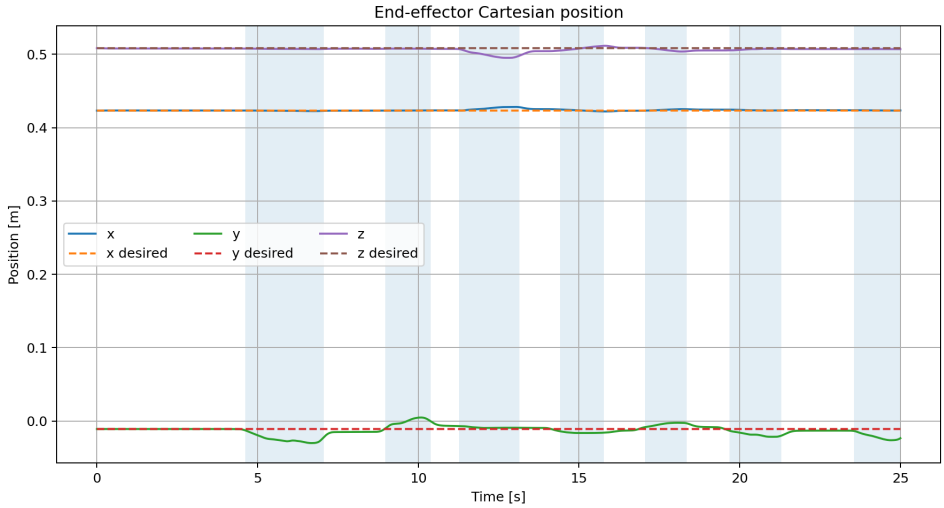

Figure 4.40: Cartesian position during the stiff human push experiment. The desired pose remains approximately constant, while the actual end-effector position changes only slightly during the human disturbances, showing stronger disturbance rejection than in the soft configuration.

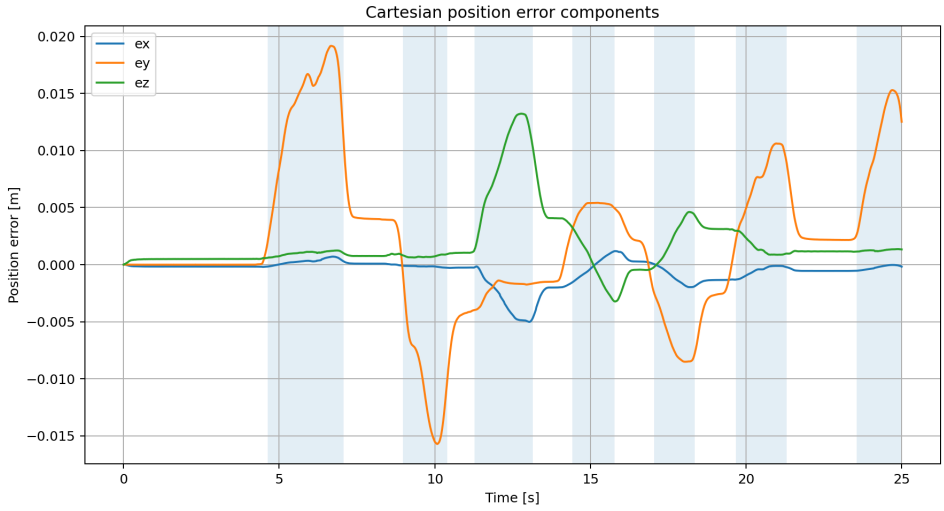

Figure 4.41: Cartesian position error components during the stiff push experiment. The error increases during the disturbance intervals, mainly in the directions affected by the human push, but remains smaller than in the soft configuration.

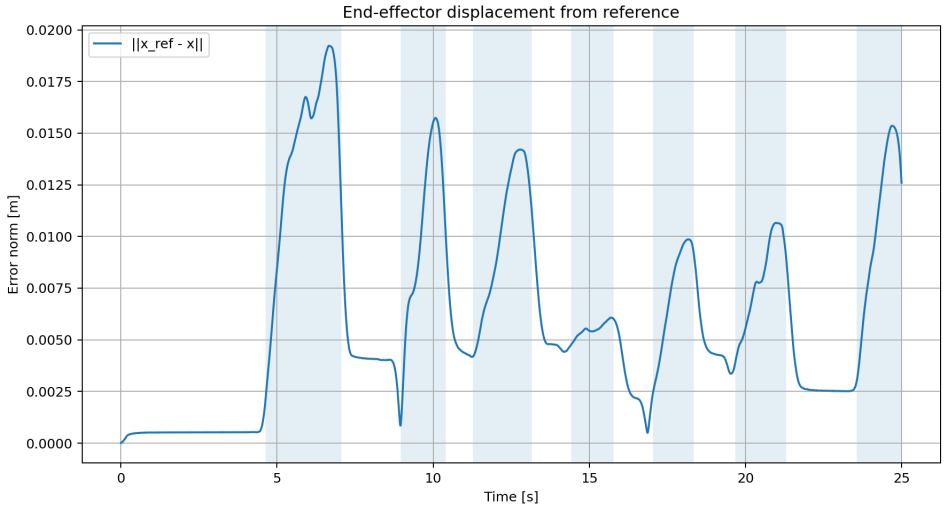

Figure 4.42: Position error norm during the stiff push experiment. The error increases when the robot is pushed and decreases after the disturbance is removed, demonstrating recovery toward the reference pose.

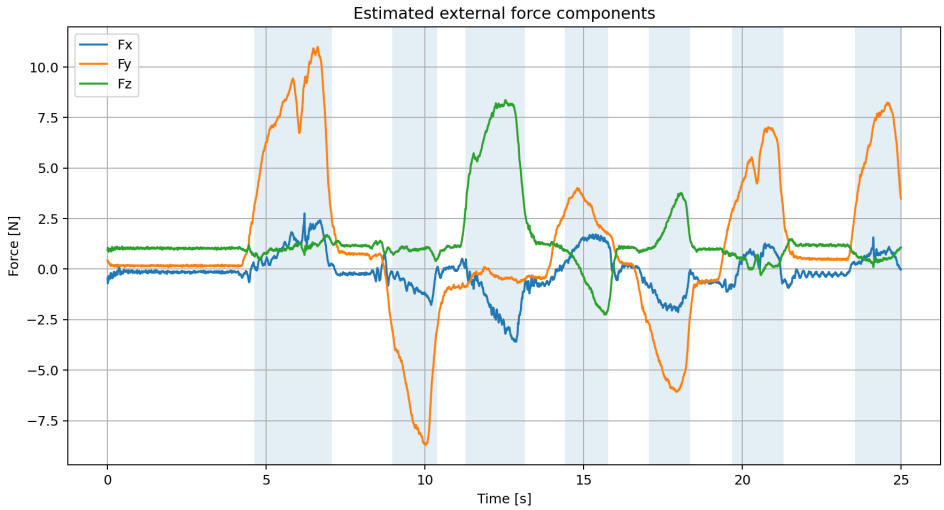

Figure 4.43: Estimated external force components during the stiff push experiment. The force components increase during human contact, confirming that the external disturbance was detected by the FR3 force-estimation system.

However, the force norm reached approximately 11 N, which is higher than in the soft experiment. This shows that the human had to apply a larger force in order to displace the robot. The external joint torque norm also increased, reaching approximately 7 Nm, but it remained within safe limits. The end-effector speed stayed low, with peaks around 0.040–0.045 m/s, confirming that the motion was still controlled and safe.

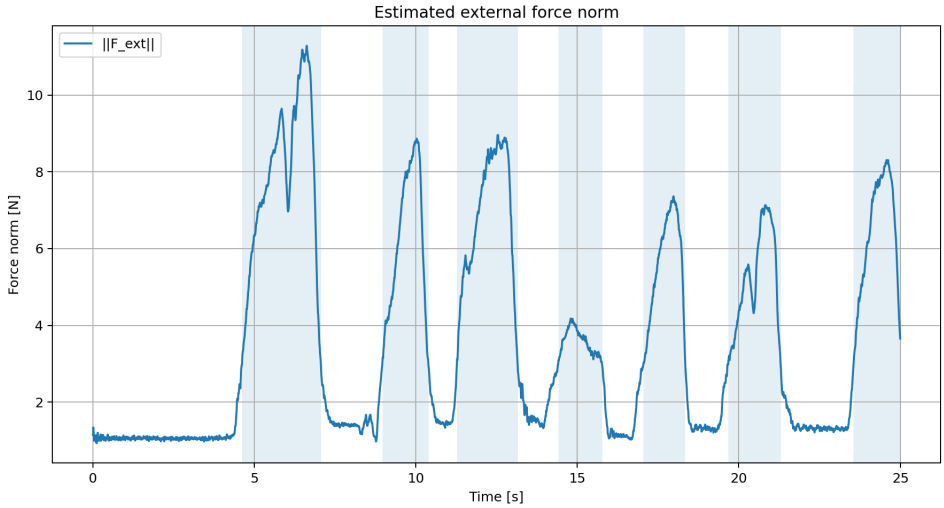

Figure 4.44: Estimated external force norm during the stiff push experiment. The force peaks are higher than in the soft configuration, showing that the robot resisted the human push more strongly.

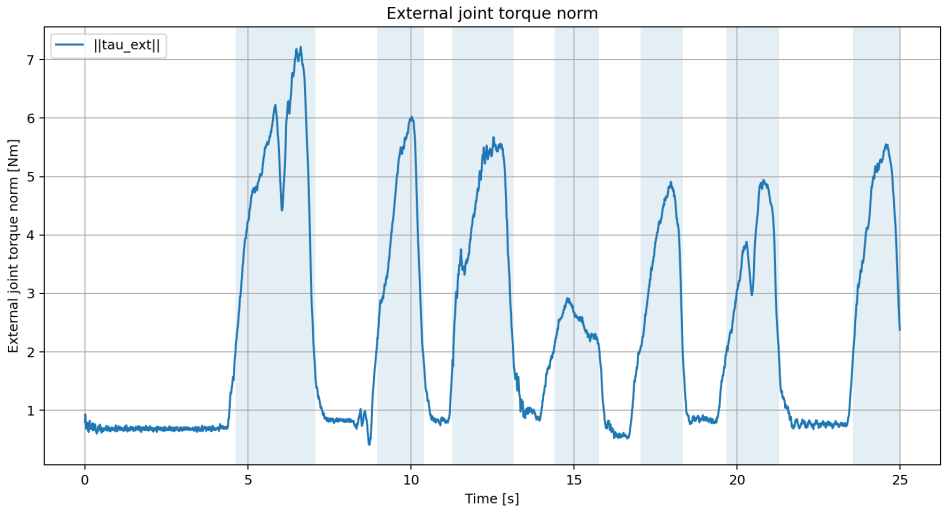

Figure 4.45: External joint torque norm during the stiff push experiment. The torque norm increases during the human disturbances but remains within safe limits, indicating stable physical interaction.

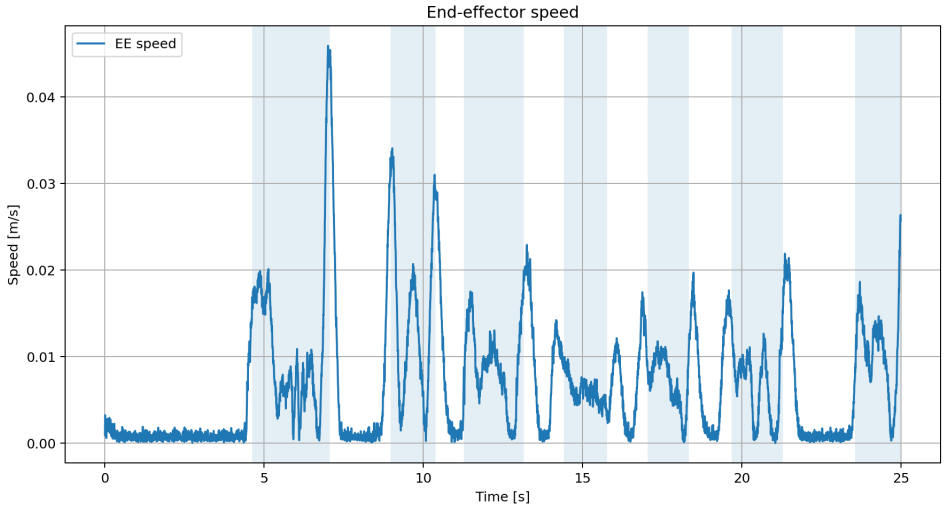

Figure 4.46: End-effector speed during the stiff push experiment. The speed remains low during the interaction, confirming that the robot motion was controlled and safe despite the external pushes.

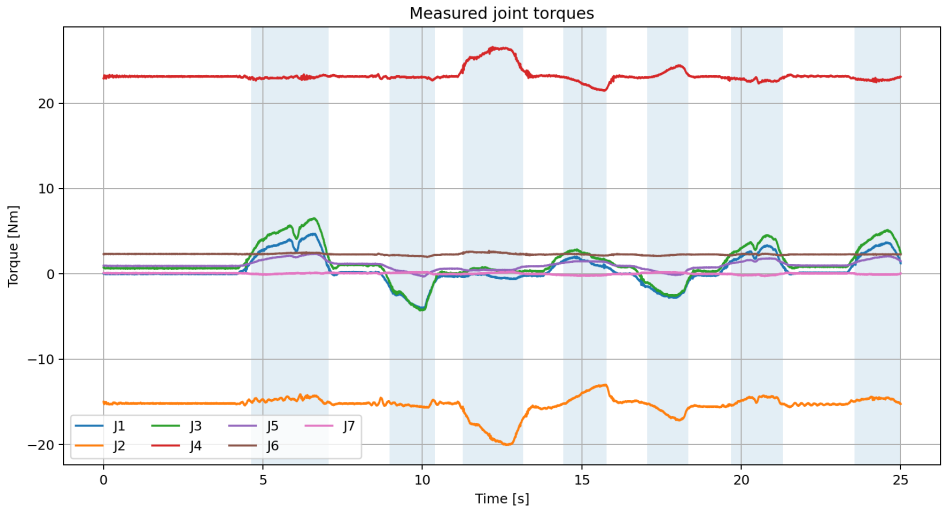

Figure 4.47: Measured joint torques during the stiff push experiment. The torques remain smooth and physically reasonable, with larger offsets mainly caused by gravity compensation and the robot configuration.

The commanded Cartesian force increased during the disturbances because the controller generated a stronger restoring action to return the end-effector toward the reference pose. The measured joint torques remained smooth and physically reasonable, with no discontinuities or unsafe peaks.

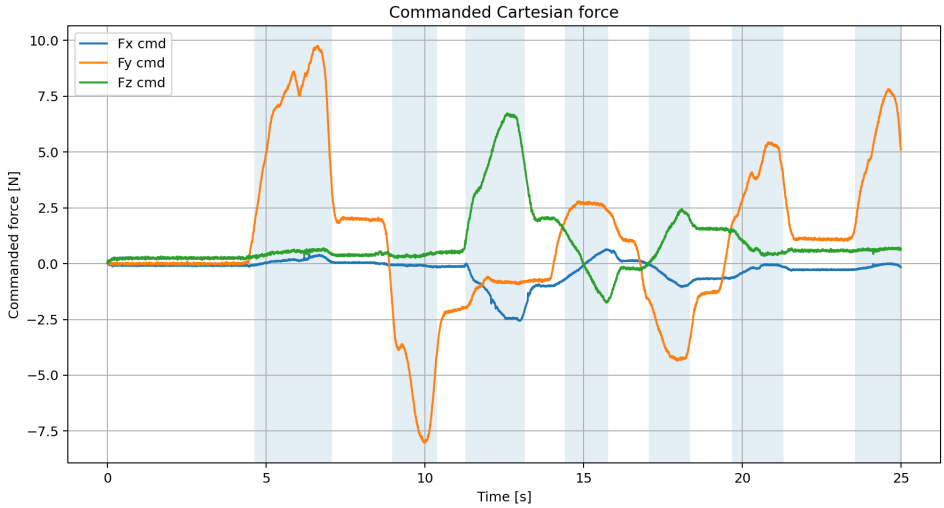

Figure 4.48: Commanded Cartesian force during the stiff push experiment. The controller generates a stronger restoring force when the end-effector is displaced from the reference pose, which explains the improved disturbance rejection.

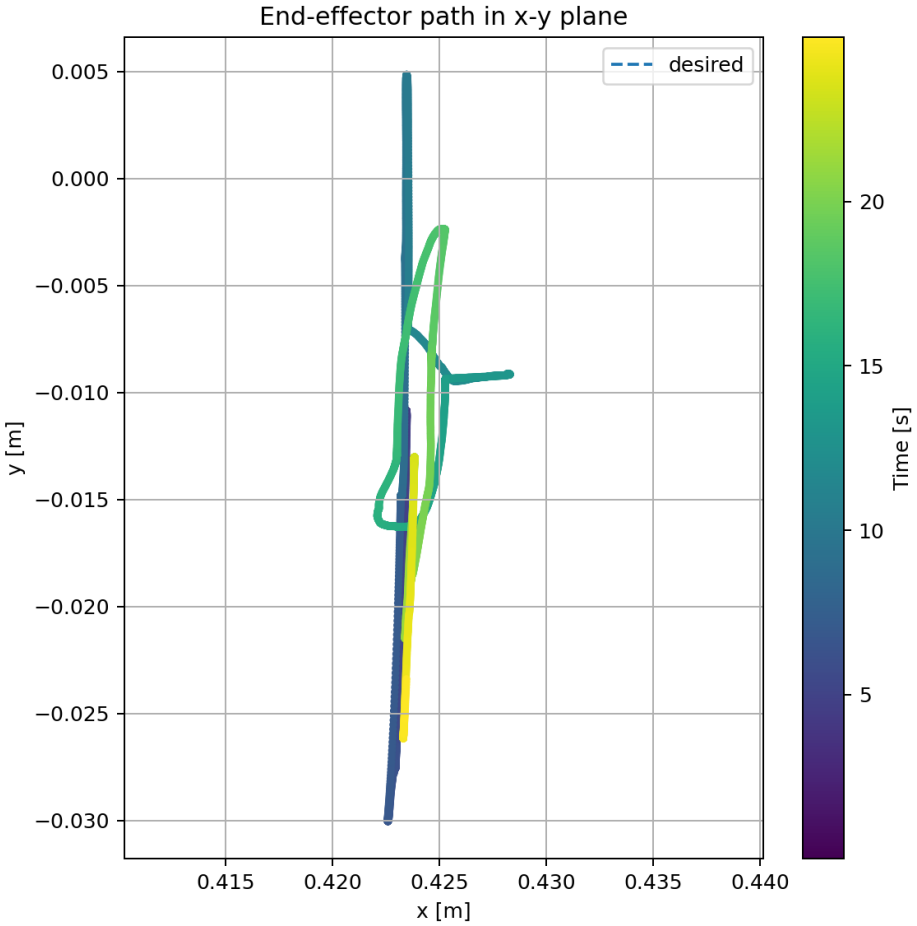

Figure 4.49: End-effector trajectory in the x-y plane during the stiff push experiment. Since the robot was holding a pose, this plot represents the displacement caused by the human pushes rather than a planned geometric trajectory.

Comparison Between Soft and Stiff Behavior

The comparison between the two configurations shows the expected stiffness–compliance trade-off. The soft controller allowed larger end-effector displacement but produced lower interaction forces. The stiff controller reduced the displacement from the reference pose but produced higher interaction forces during the human push.

Table 4.3: Qualitative comparison between soft and stiff impedance behavior.

Quantity	**Soft Configuration**	**Stiff Configuration**
Maximum displacement	larger, about 2.6 cm	smaller, about 1.9 cm
External force	lower, about 6 N	higher, about 11 N
External joint torque norm	lower, about 3.85 Nm	higher, about 7 Nm
Physical feeling	more compliant	more resistant
Disturbance rejection	weaker	stronger
Human safety/comfort	higher compliance	higher resistance

This comparison should be interpreted qualitatively, because the human pushes were not perfectly identical in both experiments. Nevertheless, the results clearly show the expected trend: increasing stiffness improves disturbance rejection, but reduces physical compliance and increases interaction force.

Human Pushes During Joint-Space Trajectory

In addition to the pose-holding experiment, human disturbances can also be applied while the robot follows a slow joint-space trajectory. In this case, the robot is not fixed at one pose, but moves according to a predefined periodic reference. For example, joints 2, 4, and 6 can be commanded using sinusoidal trajectories:

$$q_{2,d}(t) = q_{2,0} + A_2 \sin(2\pi f t), \tag{4.3}$$

$$q_{4,d}(t) = q_{4,0} + A_4 \sin(2\pi f t + \frac{\pi}{2}), \tag{4.4}$$

$$q_{6,d}(t) = q_{6,0} + A_6 \sin(2\pi f t + \pi), \tag{4.5}$$

where $q_{i,0}$ is the initial joint position, A_i is the motion amplitude, and f is the trajectory frequency. The phase shifts between the joints create a richer motion than moving all joints in phase.

For this trajectory case, a joint-space impedance control law can be used:

$$\tau_{\text{cmd}} = K_q(q_d - q) + D_q(\dot{q}_d - \dot{q}) + C(q, \dot{q}), \tag{4.6}$$

where q_d and $\dot{q}_d$ are the desired joint position and velocity vectors, K_q is the joint stiffness matrix, D_q is the joint damping matrix, and $C(q, \dot{q})$ is the Coriolis compensation term.

This version of the experiment is useful because it evaluates disturbance rejection during motion, not only during static pose holding. The expected behavior is that the robot continues following the reference trajectory after the push is removed, while the external force and torque estimates increase during the disturbance. This provides a more realistic validation of the controller under dynamic human–robot interaction.

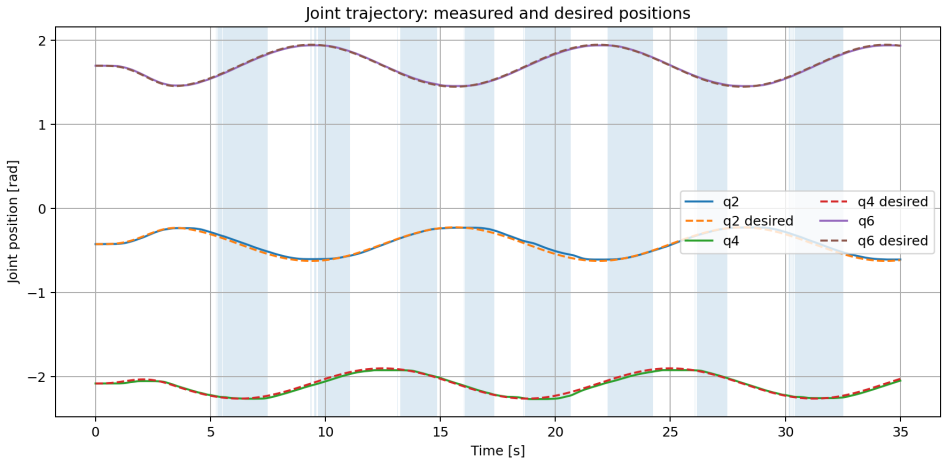

Figure 4.50: Desired and measured joint positions for joints 2, 4, and 6 during the joint-space trajectory with human pushes. The measured joint positions follow the sinusoidal references while external disturbances are applied.

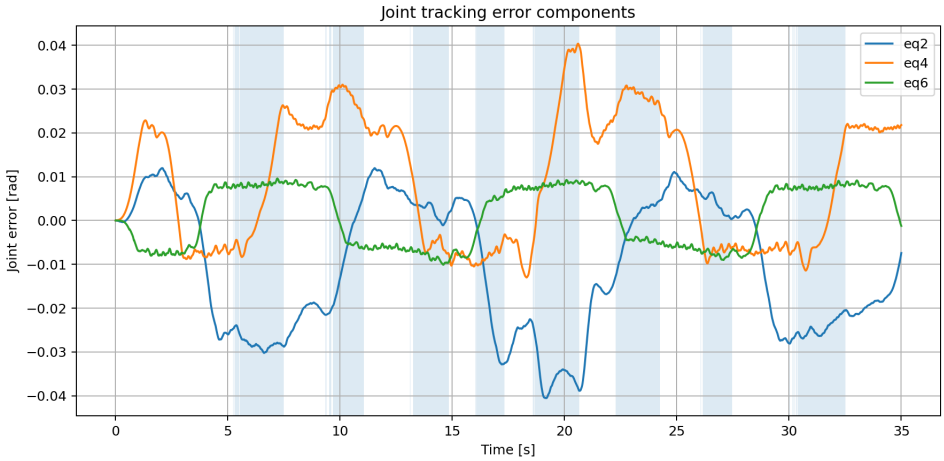

Figure 4.51: Joint tracking errors for joints 2, 4, and 6 during the trajectory experiment. The errors remain bounded, showing that the controller maintains trajectory tracking despite the external human disturbances.

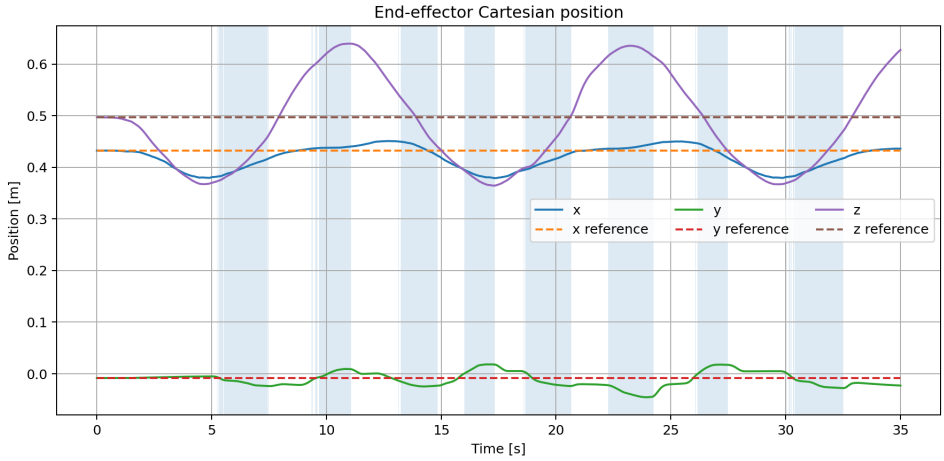

Figure 4.52: Cartesian end-effector position during the joint-space trajectory with human pushes. Although the commanded motion is defined in joint space, the resulting Cartesian motion shows the effect of the coordinated motion of joints 2, 4, and 6.

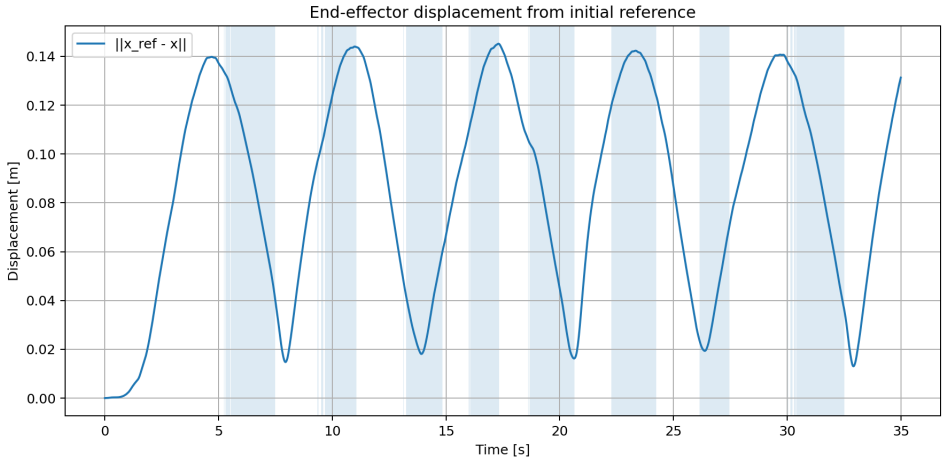

Figure 4.53: Cartesian displacement norm during the joint-space trajectory experiment. The plot shows the overall end-effector displacement produced by the joint-space motion and the additional deviations caused by human pushes.

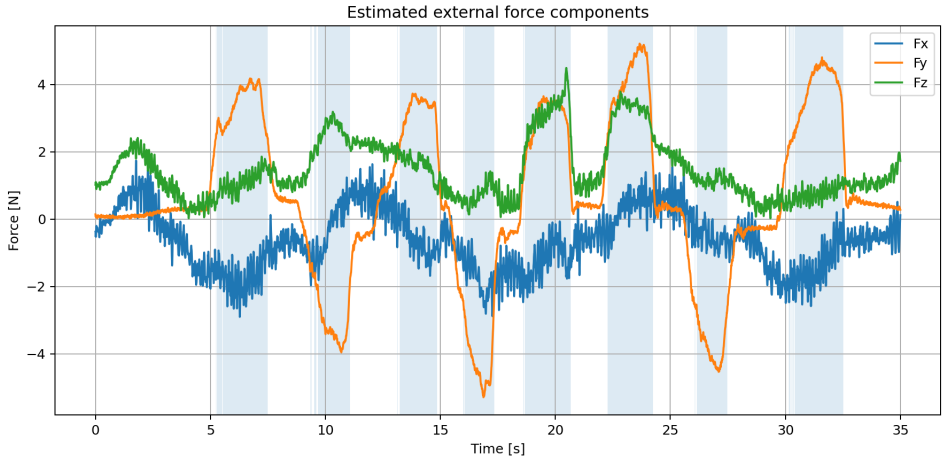

Figure 4.54: Estimated external force components during the joint-space trajectory with human pushes. The force components increase during the disturbance intervals, confirming that the robot detected the physical interaction.

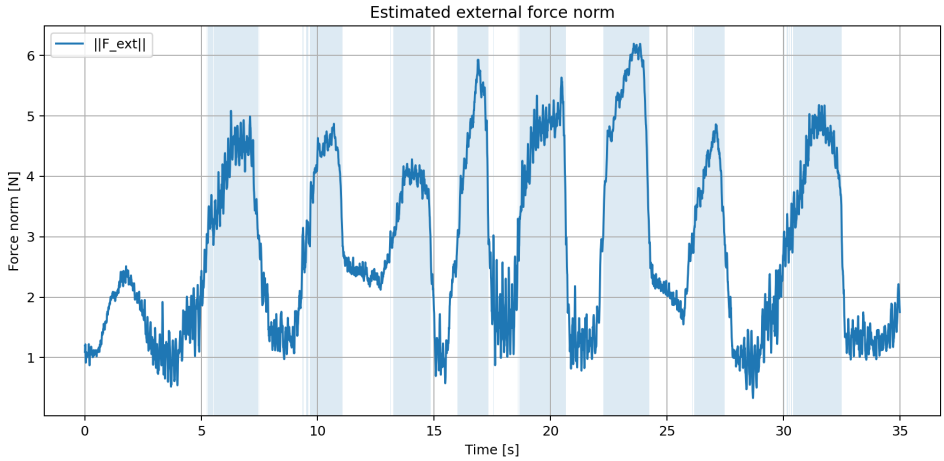

Figure 4.55: Estimated external force norm during the joint-space trajectory experiment. The force norm increases when the human pushes the robot and remains within safe interaction limits.

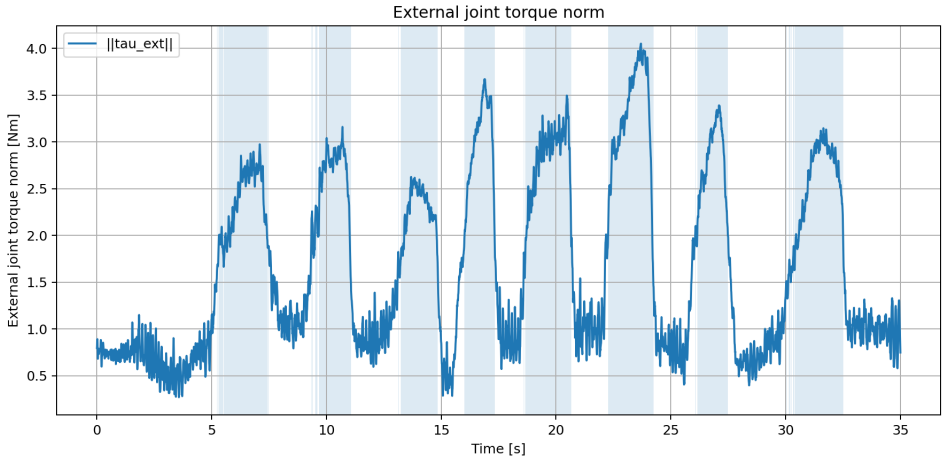

Figure 4.56: External joint torque norm during the joint-space trajectory with human pushes. The increases in the norm indicate that the external disturbances were detected at the joint-torque level.

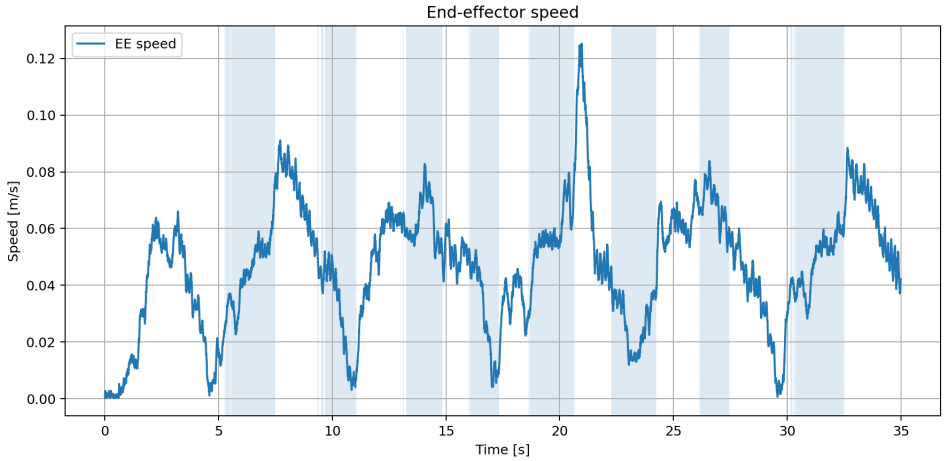

Figure 4.57: End-effector speed during the joint-space trajectory experiment. The speed remains low, confirming that the robot motion was smooth and safe during the human interaction.

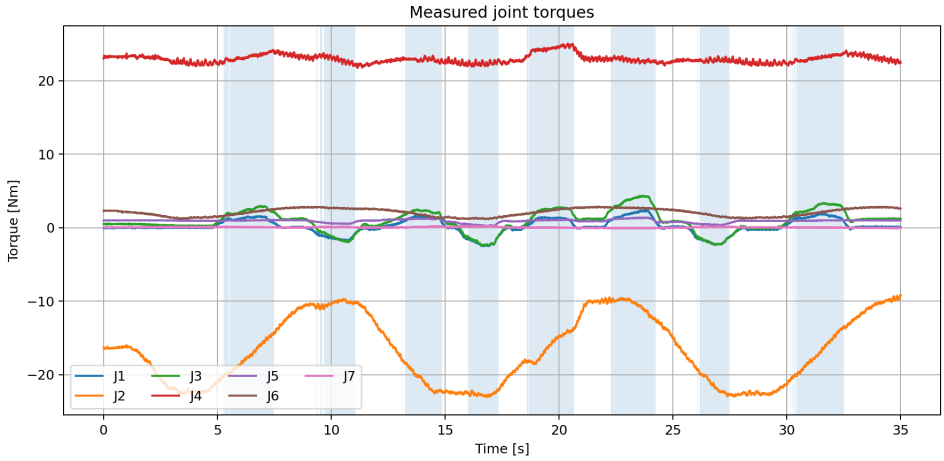

Figure 4.58: Measured joint torques during the joint-space trajectory with human pushes. The torques remain smooth and physically reasonable, with larger values mainly related to gravity, robot configuration, and the motion of the selected joints.

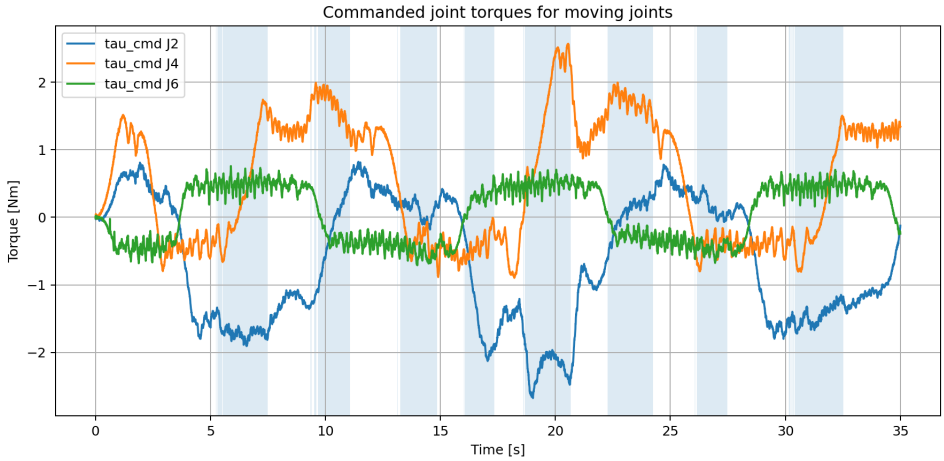

Figure 4.59: Commanded joint torques for joints 2, 4, and 6 during the joint-space trajectory experiment. The controller generates torque commands to track the sinusoidal references and recover from external disturbances.

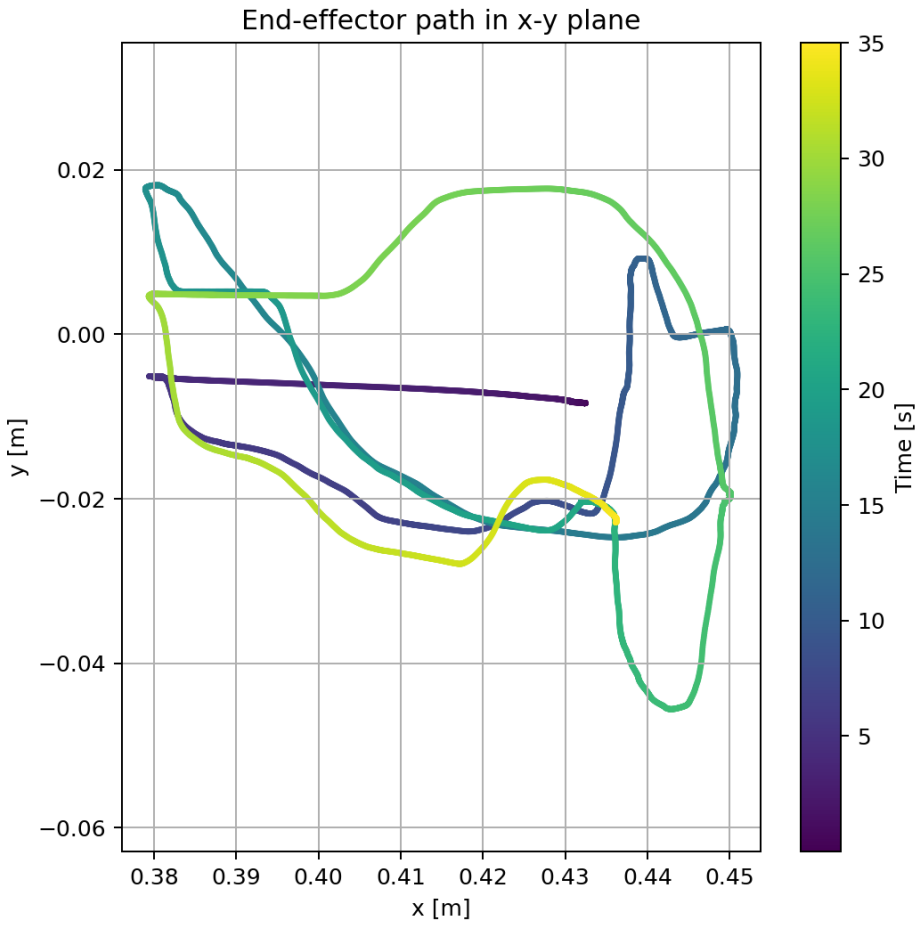

Figure 4.60: End-effector trajectory in the x-y plane during the joint-space trajectory with human pushes. The curve represents the Cartesian motion produced by the joint-space trajectory and the deviations caused by human interaction.

Conclusion

The human push and disturbance rejection experiment was successfully performed with both soft and stiff impedance configurations. In the soft case, the FR3 allowed larger displacement and lower interaction forces, demonstrating compliant and comfortable behavior. In the stiff case, the robot rejected the disturbance more strongly and returned closer to the reference pose, but this required higher interaction forces.

The results confirm the expected behavior of impedance control: stiffness determines how strongly the robot resists displacement, while damping helps maintain stable and smooth recovery. The experiment also verifies that the FR3 can detect human disturbances through its estimated external wrench and external joint torque signals. Since the end-effector speed, force levels, and torque norms remained within safe values, the experiment supports the use of the FR3 as a suitable platform for safe physical human–robot interaction.

4.1.6 Experiment 5: Hand-Guiding and Lead-Through Experiment

The fifth experiment evaluates the ability of the FR3 manipulator to operate in a hand-guiding, or lead-through, mode. The purpose of this experiment is to allow a human operator to physically guide the robot by applying external forces to the end-effector or to the robot body. This type of behavior is important in collaborative robotics, where the robot should not behave as a rigid system, but should instead move compliantly and safely in response to human interaction.

In hand-guiding, the robot is expected to offer low resistance to the human operator. Therefore, the controller should provide low virtual stiffness together with sufficient damping, so that the robot can move smoothly when pushed and stop without oscillation when the applied force is removed. This behavior can be achieved using an admittance-based control approach, where the measured external force is used as the input to a virtual mass-spring-damper system.

The admittance model is expressed as

$$M\Delta\ddot{x} + D\Delta\dot{x} + K\Delta x = F_{ext},$$

where M, D, and K are the virtual mass, damping, and stiffness matrices, respectively. The term F_{ext} represents the measured external force, while Δx, $\Delta\dot{x}$, and $\Delta\ddot{x}$ represent the virtual displacement, velocity, and acceleration generated by the admittance model.

Solving for the virtual acceleration gives

$$\Delta\ddot{x} = M^{-1}\left(F_{ext} - D\Delta\dot{x} - K\Delta x\right).$$

The virtual acceleration is then integrated to obtain a desired velocity and a desired position offset. This position offset is added to the initial reference pose, allowing the robot to move in the direction of the human-applied force. In this experiment, the admittance command was generated mainly in the horizontal plane, while the vertical direction was not intentionally commanded by the admittance model.

The expected behavior is that, when the human pushes the robot, the end-effector moves smoothly in the direction of the applied force. When the human stops pushing, the admittance velocity should decrease and the robot should stop without excessive oscillation. If the virtual damping is too low, the robot may continue moving after the push is removed. On the other hand, if the virtual stiffness or damping is too high, the robot becomes difficult to guide and the interaction feels less natural.

The Cartesian position and trajectory plots further show that the robot was physically guided through space rather than simply holding a fixed pose. The x-y trajectory

illustrates the planar motion produced by the operator, while the three-dimensional trajectory shows the complete end-effector motion during the lead-through task. These results demonstrate that the FR3 can respond compliantly to human guidance and can be moved through physical contact in a controlled manner.

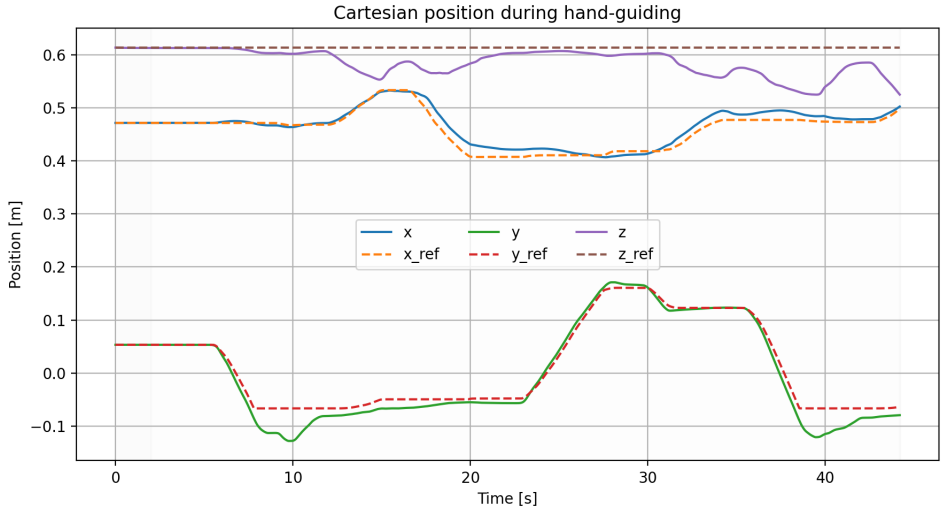

Figure 4.61: Cartesian position during the hand-guiding experiment. The end-effector position changes according to the motion physically applied by the human operator.

The admittance-generated position offset shows that the virtual displacement changes mainly along the x and y directions, while the commanded admittance displacement along z remains approximately zero. This indicates that the admittance controller generated horizontal motion according to the human-applied forces. The corresponding admittance velocity also follows the interaction: velocity increases when the human applies force and decreases when the applied force is removed.

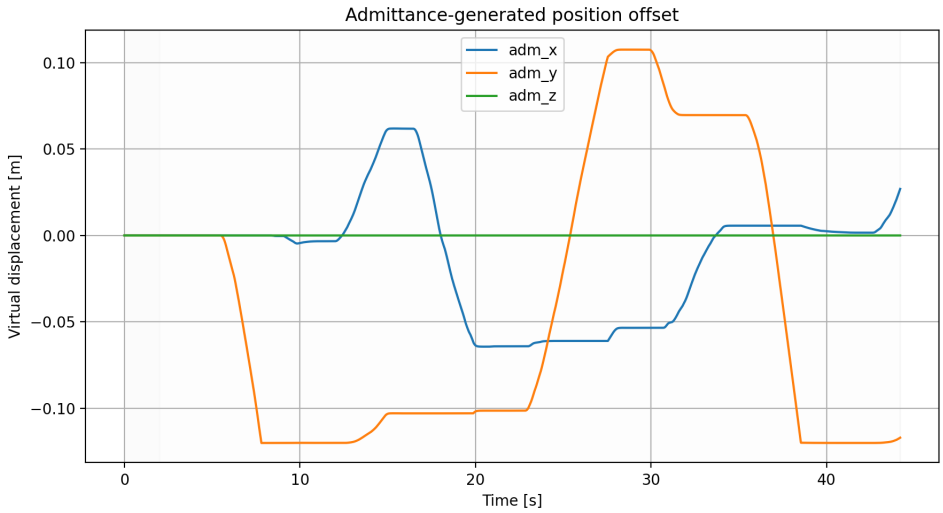

Figure 4.62: Admittance position offset during the hand-guiding experiment. The offset represents the compliant displacement generated by the admittance controller from the measured external force.

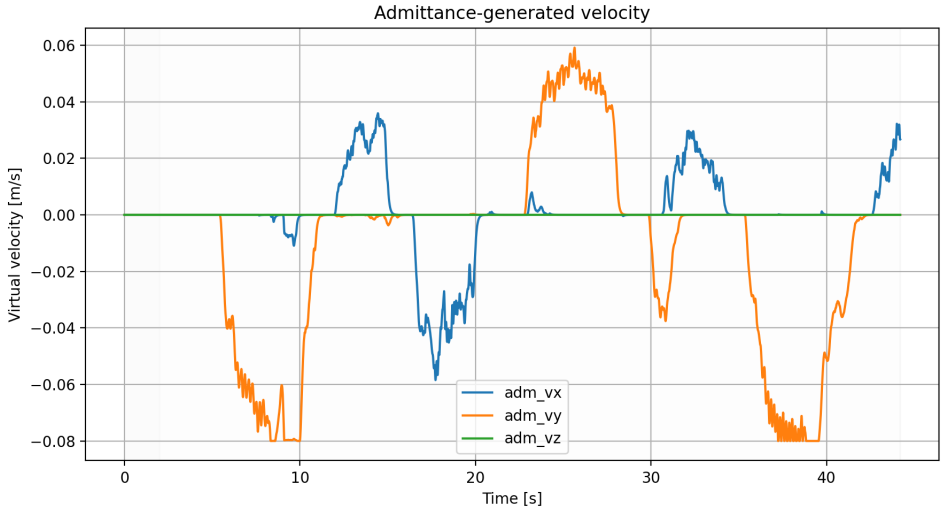

Figure 4.63: Admittance velocity during the hand-guiding experiment. The velocity response shows how the virtual admittance model converts human-applied forces into smooth motion commands.

The Cartesian tracking error is also useful for interpreting the experiment. The tracking error components show the difference between the reference position generated by the admittance controller and the measured end-effector position. Since the robot was intentionally controlled with low Cartesian stiffness, the tracking error is not expected to remain close to zero as in a rigid position-control task. Instead, the error reflects the compliant behavior of the robot during physical interaction. In particular, the x and y errors remain within a reasonable range during the hand-guided motion, while the z-axis error becomes more noticeable. This occurs because the admittance command

was mainly generated in the horizontal plane, but the low-stiffness Cartesian impedance allowed the human operator to also displace the robot vertically.

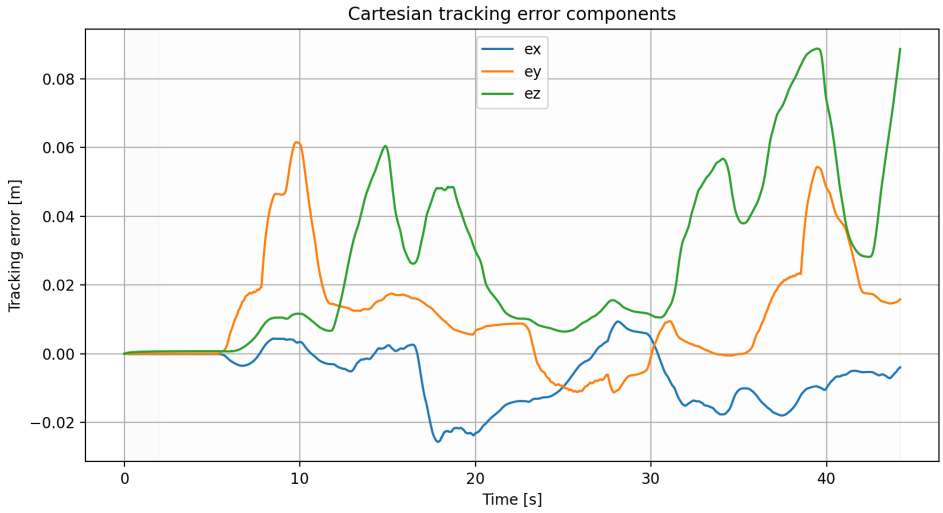

Figure 4.64: the tracking error plot confirms that the robot remained compliant during interaction, but it also highlights the limitation that the motion was not perfectly constrained to the x-y plane.

The external force components and the external force norm demonstrate clear physical interaction between the human and the robot. During the guiding task, the external force norm reaches approximately 12 N. This value indicates that the operator applied noticeable forces to guide the robot, while the robot remained stable throughout the experiment. The end-effector speed stayed below approximately 0.09 m/s, confirming that the motion remained slow and safe for physical human-robot interaction.

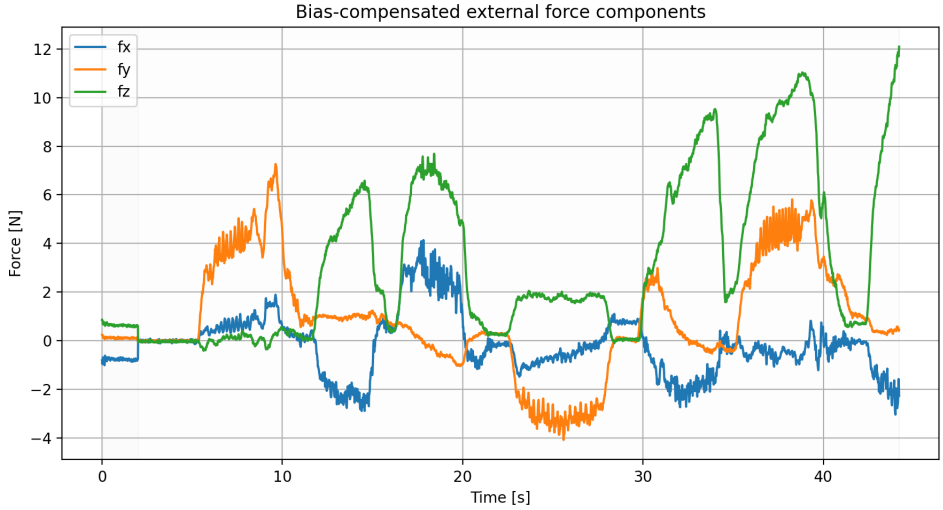

Figure 4.65: Estimated external force components during the hand-guiding experiment. The force components increase when the human pushes or pulls the end-effector, confirming that the robot detected the physical interaction.

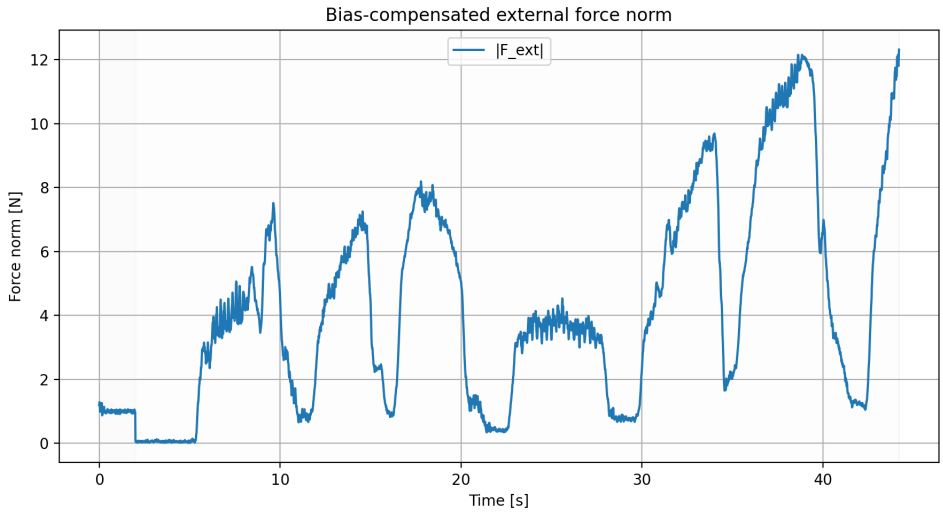

Figure 4.66: Estimated external force norm during the hand-guiding experiment. The force norm reflects the intensity of the human-applied guidance forces throughout the task.

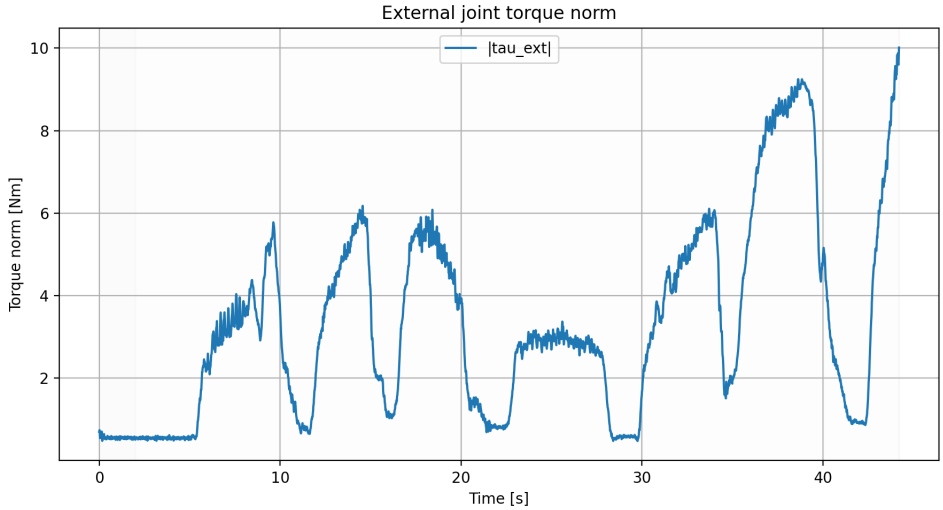

Figure 4.67: External joint torque norm during the hand-guiding experiment. The increase in the norm during interaction shows that the human disturbance was also detected at the joint-torque level.

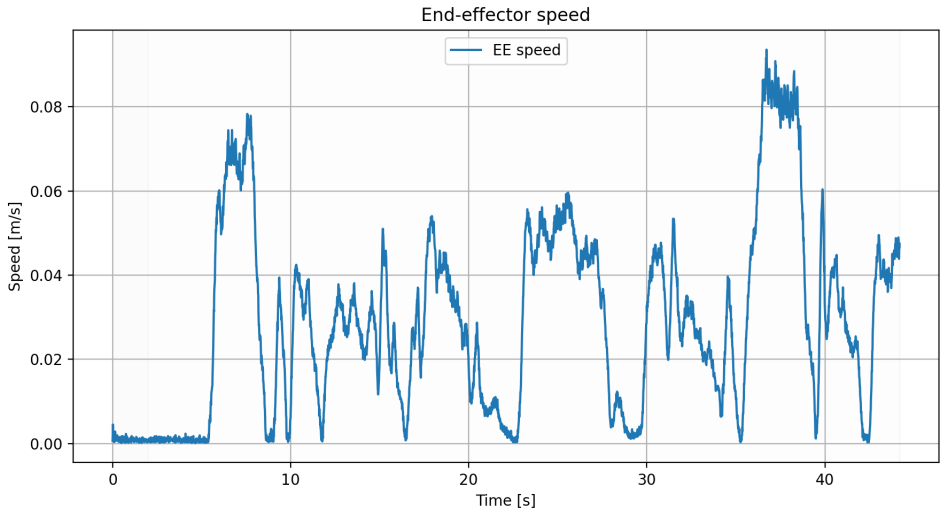

Figure 4.68: End-effector speed during the hand-guiding experiment. The speed remains low, indicating that the robot motion was smooth and safe during physical guidance.

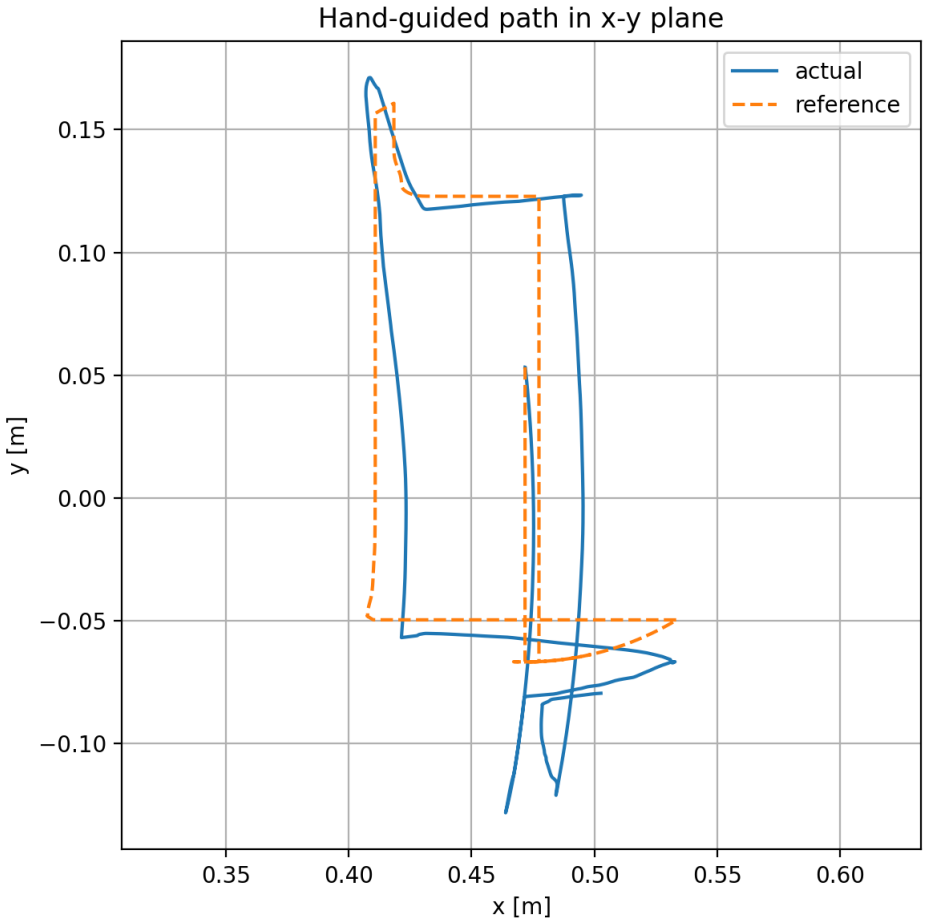

Figure 4.69: Hand-guided end-effector path in the x-y plane. The trajectory shows the motion produced by the human operator through physical interaction with the robot.

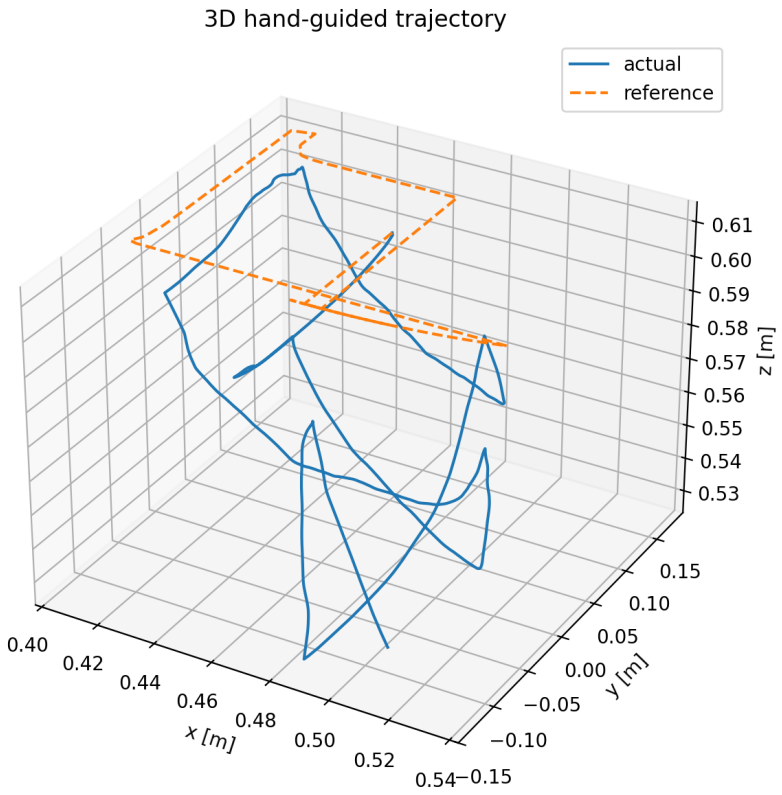

Figure 4.70: Three-dimensional end-effector trajectory during the hand-guiding experiment. The plot shows the complete spatial motion of the robot while it was physically guided by the human operator.

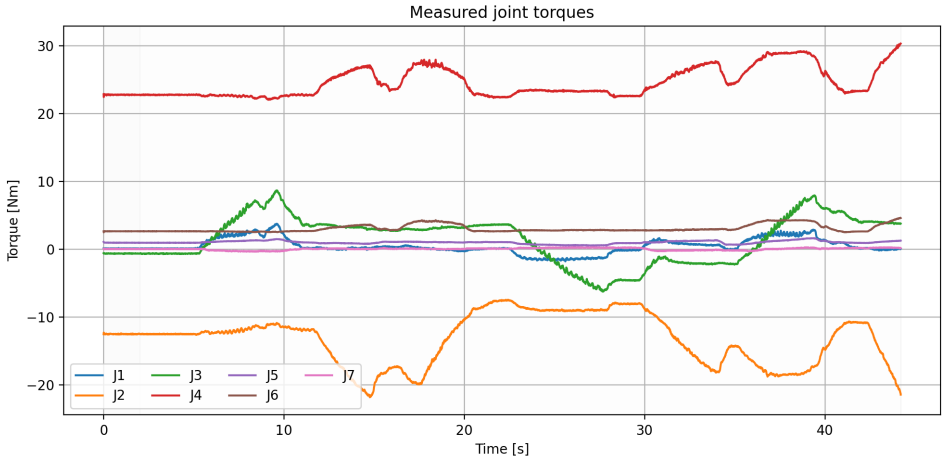

Figure 4.71: Measured joint torques during the hand-guiding experiment. The torques remain smooth and physically reasonable, with larger values mainly caused by gravity compensation and the robot configuration.

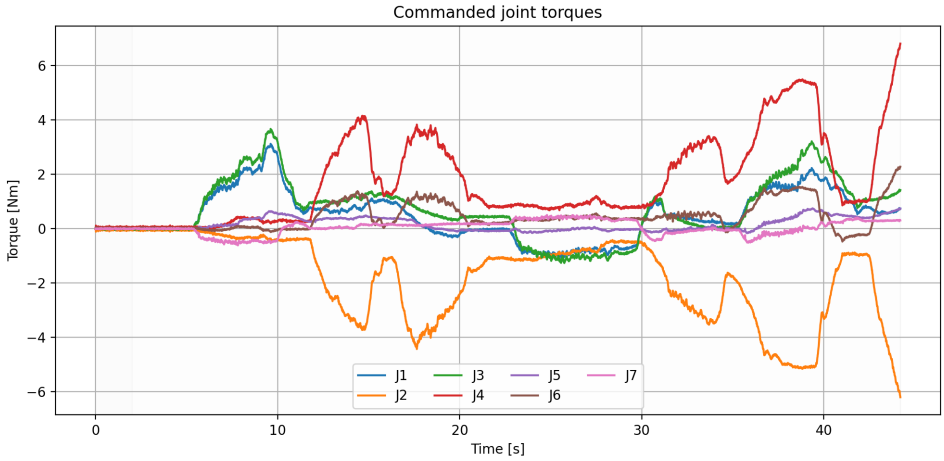

Figure 4.72: Commanded joint torques during the hand-guiding experiment. The torque commands remain smooth while the controller responds to the human-applied forces.

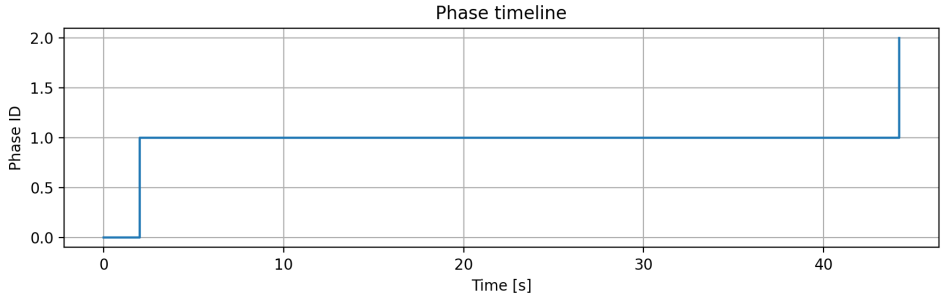

Figure 4.73: Phase timeline of the hand-guiding experiment. The plot shows the main phases of the experiment, including initialization, admittance-based hand-guiding, and completion.

One important observation is that the measured z-position changes during the experiment, even though the admittance command was generated mainly in the horizontal plane. This is also visible in the Cartesian tracking error plot, where the z-axis error becomes larger than the horizontal tracking errors. This vertical drift occurs because the Cartesian impedance gains were intentionally kept low to make the robot compliant. As a result, the robot could still be displaced vertically by the human operator. Therefore, this experiment should not be interpreted as a perfectly constrained planar x-y motion. Instead, it is better described as a safe and compliant hand-guiding experiment with low Cartesian stiffness, where the admittance command is primarily horizontal but the robot remains physically compliant in all Cartesian directions.

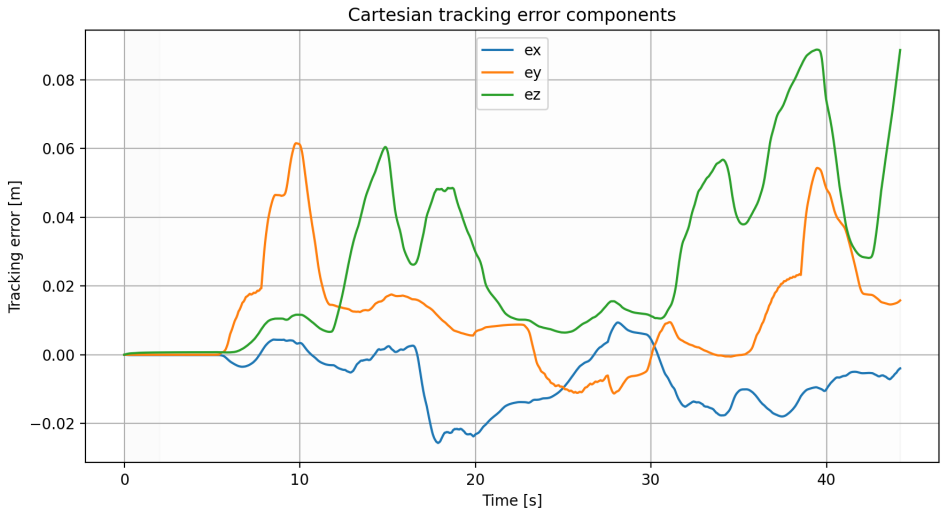

Figure 4.74: Cartesian tracking error components during the hand-guiding experiment. The horizontal errors correspond to the difference between the admittance-generated reference and the measured end-effector motion, while the larger z-axis error indicates vertical compliance caused by the low-stiffness controller.

The tracking error is not interpreted as poor control performance, but as a consequence of the intentionally compliant behavior required for hand-guiding.

Overall, the experiment demonstrates successful lead-through behavior. The robot detects human-applied forces, generates admittance-based motion, follows the physical guidance of the operator, and keeps the end-effector velocity within safe limits. The joint torque measurements remain smooth, indicating stable interaction without abrupt torque changes. These results show that the FR3 can be used in collaborative tasks where a human operator physically guides or teaches the robot through direct interaction.

Conclusion

The hand-guiding experiment successfully demonstrated compliant physical interaction between the human operator and the FR3 robot. The admittance controller generated position and velocity commands based on the measured external forces, allowing the robot to move in response to human guidance. The robot behaved compliantly rather than rigidly, which is desirable for safe human-robot interaction.

4.1.7 Experiment 6: Resistance-Aware Assist-as-Needed Shoulder Rehabilitation Using the Franka Research 3

Objective and Scope

The purpose of this experiment was to evaluate whether the Franka Research 3 (FR3) can execute a repetitive shoulder-elevation exercise while adapting its motion to the

physical resistance applied by a human participant. The experiment was designed as an engineering validation of physical human–robot interaction and not as a clinical trial or as validation of a certified rehabilitation device.

The participant maintained the arm in an approximately extended configuration and held an ergonomic handle grasped by the Franka Hand. Starting from the current robot pose, the end-effector moved along an arc in the Cartesian X-Z plane, representing an approximate shoulder flexion–extension movement. Five complete repetitions were executed, corresponding to ten half-cycles. The controller combined Cartesian impedance control, a virtual movement channel, resistance-dependent trajectory progression, and bounded assist-as-needed force.

Shoulder-Arc Trajectory

Let

$$\boldsymbol{p}_s \in \mathbb{R}^3$$

denote the estimated shoulder-centre position expressed in the robot base frame, and let

$$\boldsymbol{p}_0 \in \mathbb{R}^3$$

denote the initial end-effector position. The initial shoulder-to-handle vector is

$$\boldsymbol{r}_0 = \boldsymbol{p}_0 - \boldsymbol{p}_s, \qquad L = \|\boldsymbol{r}_0\|_2,$$

where L is the estimated shoulder-to-handle distance.

The desired end-effector position is generated by rotating $\boldsymbol{r}_0$ around the unit shoulder-rotation axis

$$\hat{\boldsymbol{a}} \in \mathbb{R}^3, \qquad \|\hat{\boldsymbol{a}}\|_2 = 1.$$

The desired Cartesian trajectory is therefore

$$\boldsymbol{p}_d(t) = \boldsymbol{p}_s + \boldsymbol{R}(\hat{\boldsymbol{a}}, \phi(t))\, \boldsymbol{r}_0, \tag{4.7}$$

where $\phi(t)$ is the desired shoulder rotation angle and $\boldsymbol{R}(\hat{\boldsymbol{a}}, \phi) \in SO(3)$ is obtained from Rodrigues' formula:

$$\boldsymbol{R}(\hat{\boldsymbol{a}}, \phi) = \boldsymbol{I}_3 + \sin\phi\, [\hat{\boldsymbol{a}}]_\times + (1 - \cos\phi)[\hat{\boldsymbol{a}}]_\times^2. \tag{4.8}$$

The skew-symmetric matrix associated with $\hat{\boldsymbol{a}} = [a_x, a_y, a_z]^T$ is

$$[\hat{\boldsymbol{a}}]_\times = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}. \tag{4.9}$$

Each upward or downward movement is treated as one half-cycle. Let

$$\sigma_h(t) \in [0, 1]$$

be the normalized progression variable of half-cycle h, and let $\phi_{h,0}$ and $\phi_{h,f}$ be its initial and final shoulder angles. A fifth-order minimum-jerk scaling function is used:

$$s(\sigma) = 10\sigma^3 - 15\sigma^4 + 6\sigma^5. \tag{4.10}$$

The desired angle during half-cycle h is

$$\phi_h(\sigma_h) = \phi_{h,0} + (\phi_{h,f} - \phi_{h,0})\, s(\sigma_h). \tag{4.11}$$

Its derivative with respect to σ_h is

$$\frac{d\phi_h}{d\sigma_h} = (\phi_{h,f} - \phi_{h,0}) \left(30\sigma_h^2 - 60\sigma_h^3 + 30\sigma_h^4\right). \tag{4.12}$$

Consequently, the desired angular velocity is

$$\dot{\phi}_h = \frac{d\phi_h}{d\sigma_h}\dot{\sigma}_h, \tag{4.13}$$

and the desired Cartesian velocity of the handle is

$$\dot{\boldsymbol{p}}_d = \dot{\phi}_h \left[\hat{\boldsymbol{a}} \times \boldsymbol{R}(\hat{\boldsymbol{a}}, \phi_h)\, \boldsymbol{r}_0\right]. \tag{4.14}$$

Tracking-Error Decomposition

The Cartesian position error is defined as

$$\boldsymbol{e}_p = \boldsymbol{p}_d - \boldsymbol{p}, \tag{4.15}$$

where $\boldsymbol{p}$ is the measured end-effector position. Its Euclidean norm is

$$e_{\text{norm}} = \|\boldsymbol{e}_p\|_2 = \sqrt{e_x^2 + e_y^2 + e_z^2}. \tag{4.16}$$

When $\|\dot{\boldsymbol{p}}_d\|_2$ is non-zero, the unit tangent vector of the desired trajectory is

$$\hat{\boldsymbol{t}} = \frac{\dot{\boldsymbol{p}}_d}{\|\dot{\boldsymbol{p}}_d\|_2}. \tag{4.17}$$

The scalar tangential tracking error is

$$e_{\parallel} = \hat{\boldsymbol{t}}^T \boldsymbol{e}_p, \tag{4.18}$$

whereas the perpendicular error vector is

$$\boldsymbol{e}_{\perp} = \left(\boldsymbol{I}_3 - \hat{\boldsymbol{t}}\hat{\boldsymbol{t}}^T\right)\boldsymbol{e}_p. \tag{4.19}$$

Its magnitude is

$$e_{\perp} = \|\boldsymbol{e}_{\perp}\|_2. \tag{4.20}$$

This decomposition distinguishes lag along the desired path from deviation away from the virtual movement channel.

Interaction-Force Estimation

The force estimate recorded during the initial stationary calibration interval is

$$\boldsymbol{F}_{\text{bias}} = \frac{1}{N_b}\sum_{k=1}^{N_b}\boldsymbol{F}_{\text{ext}}(k), \tag{4.21}$$

where N_b is the number of calibration samples. The bias-corrected interaction force is

$$\boldsymbol{F}_h = \boldsymbol{F}_{\text{ext}} - \boldsymbol{F}_{\text{bias}}. \tag{4.22}$$

The absolute external-force norm and the force change from the initial bias are

$$F_{\text{ext,norm}} = \|\boldsymbol{F}_{\text{ext}}\|_2, \tag{4.23}$$

$$\Delta F_{\text{bias}} = \|\boldsymbol{F}_h\|_2. \tag{4.24}$$

The component opposing the desired movement is estimated by projecting the bias-corrected force onto the path tangent:

$$F_{\text{opp}} = \max\left(0, -s_F\hat{\boldsymbol{t}}^T\boldsymbol{F}_h\right), \qquad s_F \in \{-1, +1\}, \tag{4.25}$$

where s_F is selected experimentally to match the force-sign convention of the implementation.

To reduce sensitivity to measurement noise, a first-order low-pass filter is applied:

$$F_{\text{opp},f}[k] = \beta F_{\text{opp},f}[k-1] + (1-\beta)F_{\text{opp}}[k], \qquad 0 < \beta < 1. \tag{4.26}$$

Resistance-Dependent Trajectory Progression

The controller does not advance the reference trajectory solely as a function of time. Instead, the phase rate depends on the filtered opposing force. A commanded speed

scale is defined as

$$\rho_{\mathrm{cmd}}(F) = \begin{cases} 1, & F \le F_{\mathrm{free}}, \\ \dfrac{F_{\mathrm{pause}} - F}{F_{\mathrm{pause}} - F_{\mathrm{free}}}, & F_{\mathrm{free}} < F < F_{\mathrm{pause}}, \\ 0, & F \ge F_{\mathrm{pause}}. \end{cases} \tag{4.27}$$

The filtered progression scale is limited to

$$0 \le \rho(t) \le 1. \tag{4.28}$$

For a nominal half-cycle duration T_h, the phase evolution is

$$\dot{\sigma}_h = \frac{\rho(t)}{T_h}. \tag{4.29}$$

Thus, low resistance gives $\rho \approx 1$, moderate resistance reduces the phase rate, and sufficiently high resistance causes $\rho = 0$, preventing the desired position from continuing to move away from the participant.

A hysteresis condition is used for resuming motion after a compliant hold:

$$F_{\mathrm{opp},f} \le F_{\mathrm{resume}} < F_{\mathrm{pause}}. \tag{4.30}$$

The force must remain below F_{resume} for a specified persistence time before the controller returns to active progression.

Assist-as-Needed Force

Additional assistance is generated only when the participant lags behind the desired trajectory. With a tangential-error deadband e_{db}, the scalar assistive force is

$$F_{\mathrm{assist}} = \mathrm{sat}_{[0,F_{\mathrm{assist,max}}]} \left[k_a \max\left(0, e_{\parallel} - e_{\mathrm{db}}\right) \right], \tag{4.31}$$

where k_a is the assistance gain. The corresponding Cartesian assistive-force vector is

$$\boldsymbol{F}_{\mathrm{assist}} = F_{\mathrm{assist}} \hat{\boldsymbol{t}}. \tag{4.32}$$

For the presented experiment,

$$0 \le F_{\mathrm{assist}} \le 4\,\mathrm{N}. \tag{4.33}$$

During a compliant hold, the phase rate, tangential assistance, and tangential path

stiffness are reduced:

$$\dot{\sigma}_h = 0, \qquad F_{\text{assist}} = 0, \qquad k_{\parallel} = 0. \tag{4.34}$$

This prevents the robot from continuously increasing the force against a participant who is resisting the movement.

Cartesian Impedance and Joint-Torque Command

The virtual-channel position feedback is expressed as

$$\boldsymbol{F}_{\text{vc}} = k_{\parallel} e_{\parallel} \hat{\boldsymbol{t}} + k_{\perp} \boldsymbol{e}_{\perp}, \qquad k_{\perp} > k_{\parallel}, \tag{4.35}$$

where $k_{\perp}$ keeps the handle near the desired arc and $k_{\parallel}$ provides compliant guidance along the path.

The translational damping force is

$$\boldsymbol{F}_d = \boldsymbol{D}_p \left(\dot{\boldsymbol{p}}_d - \dot{\boldsymbol{p}} \right), \tag{4.36}$$

where $\boldsymbol{D}_p \succeq 0$ is the translational damping matrix. If a constant vertical support term is enabled, it is written as

$$\boldsymbol{F}_{\text{support}} = F_z^{\text{support}} \hat{\boldsymbol{z}}. \tag{4.37}$$

The complete commanded Cartesian force is

$$\boldsymbol{F}_{\text{cmd}} = \boldsymbol{F}_{\text{vc}} + \boldsymbol{F}_d + \boldsymbol{F}_{\text{assist}} + \boldsymbol{F}_{\text{support}}. \tag{4.38}$$

Let $\boldsymbol{e}_R \in \mathbb{R}^3$ denote the orientation-error vector and $\boldsymbol{\omega}$, $\boldsymbol{\omega}_d$ the measured and desired angular velocities. The commanded Cartesian moment is

$$\boldsymbol{M}_{\text{cmd}} = \boldsymbol{K}_R \boldsymbol{e}_R + \boldsymbol{D}_R \left(\boldsymbol{\omega}_d - \boldsymbol{\omega} \right), \tag{4.39}$$

where low rotational stiffness is selected to avoid unnecessarily constraining the participant's wrist.

The complete commanded wrench is

$$\boldsymbol{W}_{\text{cmd}} = \begin{bmatrix} \boldsymbol{F}_{\text{cmd}} \\ \boldsymbol{M}_{\text{cmd}} \end{bmatrix} \in \mathbb{R}^6. \tag{4.40}$$

The desired joint-torque command is obtained through the Jacobian-transpose mapping:

$$\boldsymbol{\tau}_{\text{des}} = \boldsymbol{J}(\boldsymbol{q})^T \boldsymbol{W}_{\text{cmd}} + \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}), \tag{4.41}$$

where

$$\boldsymbol{J}(\boldsymbol{q}) \in \mathbb{R}^{6\times 7}$$

is the FR3 geometric Jacobian and $\boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ is the Coriolis compensation term.

To avoid discontinuous torque commands, the applied torque is rate-limited:

$$\tau_{\mathrm{cmd},j}[k] = \tau_{\mathrm{cmd},j}[k-1] + \mathrm{sat}_{[-\Delta\tau_{\max},\,\Delta\tau_{\max}]}\left(\tau_{\mathrm{des},j}[k] - \tau_{\mathrm{cmd},j}[k-1]\right). \tag{4.42}$$

Controller Modes

The supervisory controller uses the following modes:

$$\texttt{CALIBRATION} \rightarrow \texttt{MOVE} \leftrightarrow \texttt{SLOW_ASSIST} \leftrightarrow \texttt{COMPLIANT_HOLD} \rightarrow \texttt{FINAL_HOLD}.$$

Their intended behaviour is:

- `CALIBRATION`: estimates the external-force bias while the handle remains stationary.
- `MOVE`: executes the trajectory at approximately nominal speed.
- `SLOW_ASSIST`: reduces the phase speed and applies bounded tangential assistance.
- `COMPLIANT_HOLD`: freezes the reference progression and removes tangential assistance until the resistance decreases.
- `FINAL_HOLD`: holds the final position compliantly after all half-cycles have been completed.

Safety Conditions

The software safety layer monitors the external force, force change from bias, external joint torque, end-effector speed, and perpendicular path error. A generic stop condition can be expressed as

$$\mathcal{S}_{\mathrm{stop}} = \{F_{\mathrm{ext,norm}} > F_{\mathrm{abs,max}}\} \cup \{\Delta F_{\mathrm{bias}} > F_{\Delta,\max}\} \cup \{\|\boldsymbol{\tau}_{\mathrm{ext}}\|_2 > \tau_{\mathrm{ext,max}}\} \cup \{\|\dot{\boldsymbol{p}}\|_2 > v_{\max}\} \cup \{e_{\perp} > e_{\perp,\max}$$

(4.43)

Where persistence is used, the corresponding threshold must remain exceeded for a specified duration T_{persist} before the stop is triggered.

Performance Metrics

For N logged samples, the Cartesian tracking root-mean-square error is

$$\mathrm{RMSE}_p = \sqrt{\frac{1}{N}\sum_{k=1}^{N} \|\boldsymbol{p}_d[k] - \boldsymbol{p}[k]\|_2^2}. \tag{4.44}$$

The maximum tracking error is

$$e_{\max} = \max_k \|\boldsymbol{p}_d[k] - \boldsymbol{p}[k]\|_2. \tag{4.45}$$

The maximum measured end-effector speed is

$$v_{\max,\text{meas}} = \max_k \|\dot{\boldsymbol{p}}[k]\|_2. \tag{4.46}$$

For repetition n, occupying the time interval $[t_n, t_n + T_n]$, the average assistive force is

$$\overline{F}^{(n)}_{\text{assist}} = \frac{1}{T_n} \int_{t_n}^{t_n+T_n} F_{\text{assist}}(t)\, dt. \tag{4.47}$$

The normalized required-assistance metric is

$$A^{(n)}_{\text{required}} = \frac{\overline{F}^{(n)}_{\text{assist}}}{F_{\text{assist,max}}}, \qquad 0 \le A^{(n)}_{\text{required}} \le 1. \tag{4.48}$$

A decrease under identical task conditions,

$$A^{(n+1)}_{\text{required}} < A^{(n)}_{\text{required}}, \tag{4.49}$$

indicates that less robotic assistance was required in the later repetition. However, this inequality should not be interpreted as evidence of clinical recovery unless it is observed consistently across repeated sessions under controlled experimental conditions.

The running normalized assistance shown during the experiment is

$$A_{\text{running}}(t) = \frac{\int_{t_0}^{t} F_{\text{assist}}(\lambda)\, d\lambda}{F_{\text{assist,max}}(t - t_0)}, \qquad t > t_0. \tag{4.50}$$

Experimental Procedure

The experimental sequence was:

1. The robot was placed at the lower endpoint of the shoulder exercise.
2. A handle with a diameter of approximately 2.5 cm was positioned between the gripper fingers.
3. The gripper closed around the handle using a low grasping force.
4. The participant held the handle with the arm approximately extended.
5. The shoulder centre, shoulder-to-handle distance, movement axis, and angular amplitude were defined.
6. The force bias was estimated while the participant remained relaxed.

7. The robot executed ten half-cycles, corresponding to five complete repetitions.

8. Position, velocity, force, joint torque, controller mode, speed scale, phase, and assistance data were recorded.

9. The handle was supported and the participant released it before the gripper opened.

Experimental Results

Figure 4.75 compares the measured and desired Cartesian end-effector positions. The periodic variations in the (x)- and (z)-coordinates are consistent with the intended shoulder arc, whereas the desired (y)-coordinate remains approximately constant.

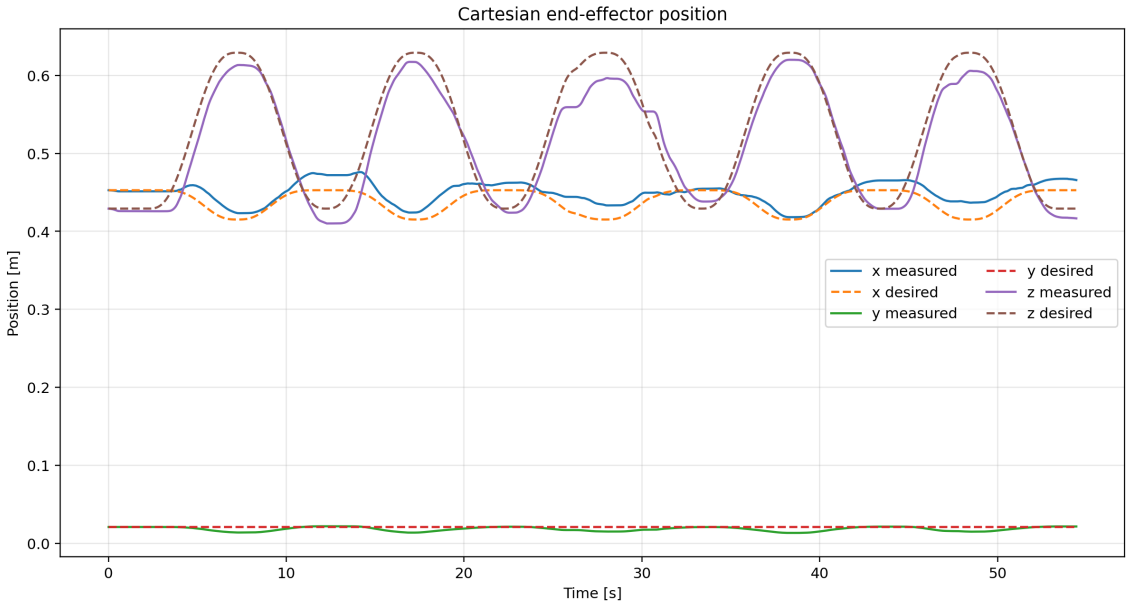

Figure 4.75: Measured and desired Cartesian end-effector positions during the resistance-aware shoulder exercise. Five complete repetitions are visible in the periodic (x)- and (z)-coordinate profiles.

Figure 4.76 presents the Cartesian tracking-error components. The largest variations occur in the x- and z-directions, which define the primary plane of motion, whereas the y-axis error remains comparatively small.

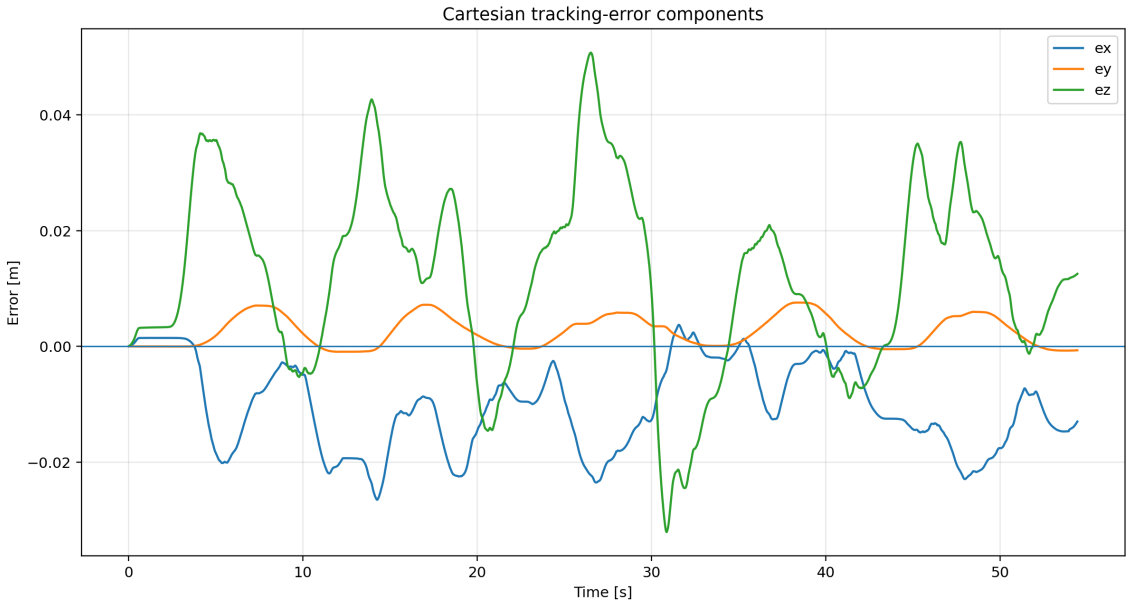

Figure 4.76: Cartesian tracking-error components e_x, e_y, and e_z. The relatively small y-axis error indicates limited out-of-plane deviation.

Figure 4.77 shows that the largest component of the tracking deviation is tangential to the desired path, whereas the perpendicular error remains lower. This indicates that the participant and the robot primarily lagged along the intended arc rather than deviating substantially from the virtual movement channel.

Figure 4.77: Tracking-error norm, absolute tangential error, and perpendicular error. The tangential error is dominant, whereas the perpendicular deviation remains lower.

Figure 4.78 presents the bias-corrected interaction-force components. The (x)- and (z)-components are dominant because the exercise is performed primarily in the (X)-(Z) plane, whereas the (y)-component mainly represents out-of-plane interaction.

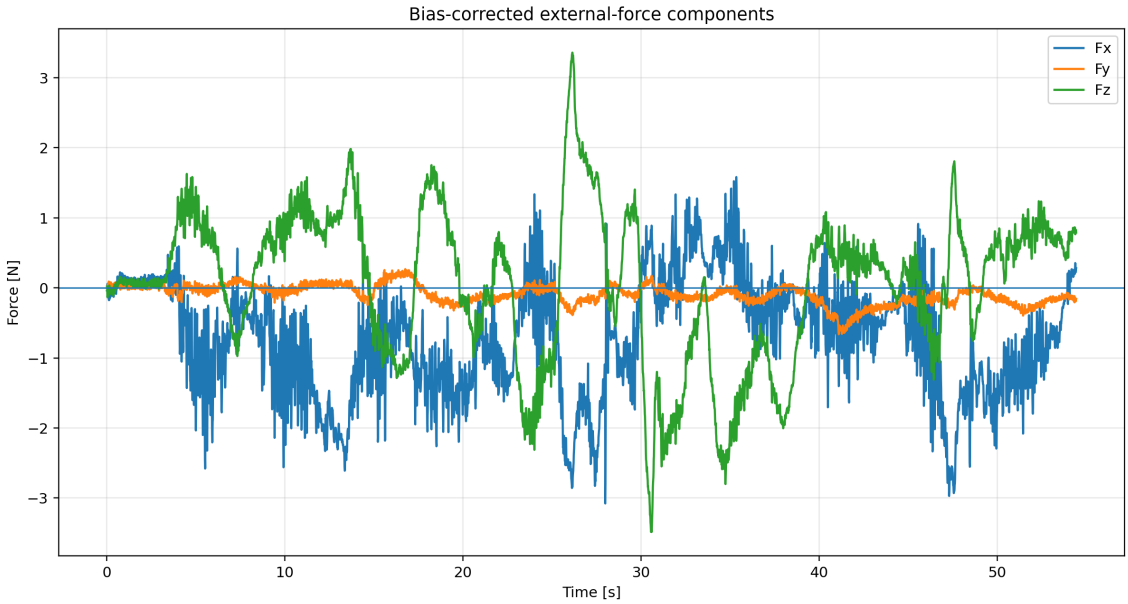

Figure 4.78: Bias-corrected Cartesian external-force components during physical human–robot interaction.

Figure 4.79 shows the absolute external-force norm and the force change relative to the initial calibration bias. Both quantities remain bounded throughout the exercise, indicating that the interaction forces remained within the configured safety limits.

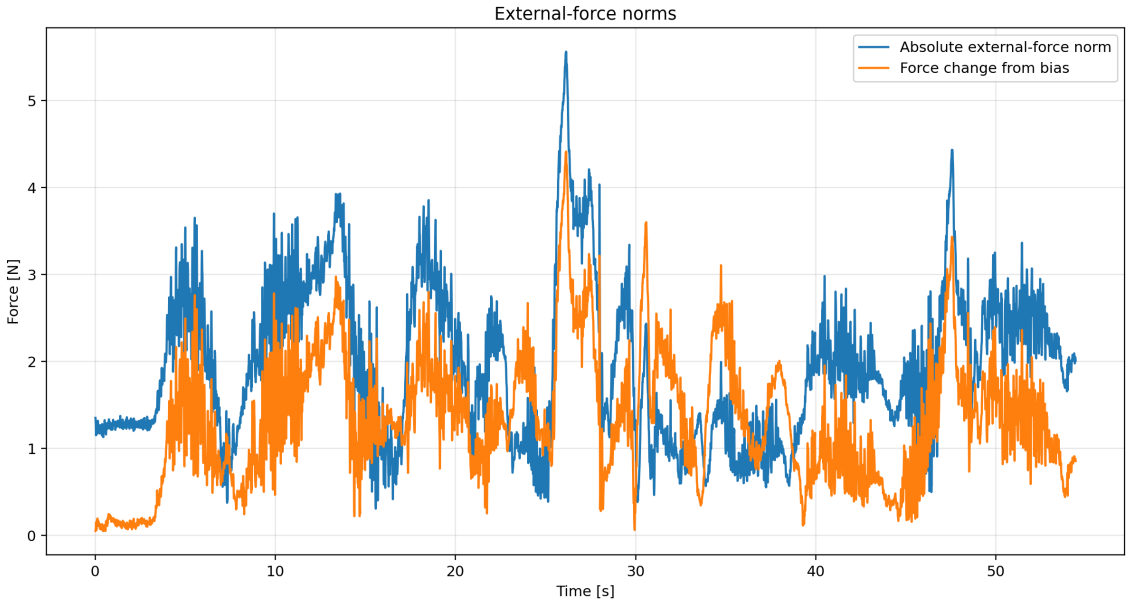

Figure 4.79: Absolute external-force norm and force change relative to the initial calibration bias.

Figure 4.80 presents the estimated opposing force together with the controller thresholds. The filtered signal follows the overall trend of the raw opposing-force estimate while attenuating high-frequency fluctuations. During several intervals, the filtered opposing force exceeds the free-motion threshold, causing the controller to

reduce the trajectory progression speed. Although the signal does not generally remain above the pause threshold for a prolonged interval, brief threshold crossings may still result in temporary resistance-hold behaviour. Overall, the plot is primarily consistent with `SLOW_ASSIST` operation.

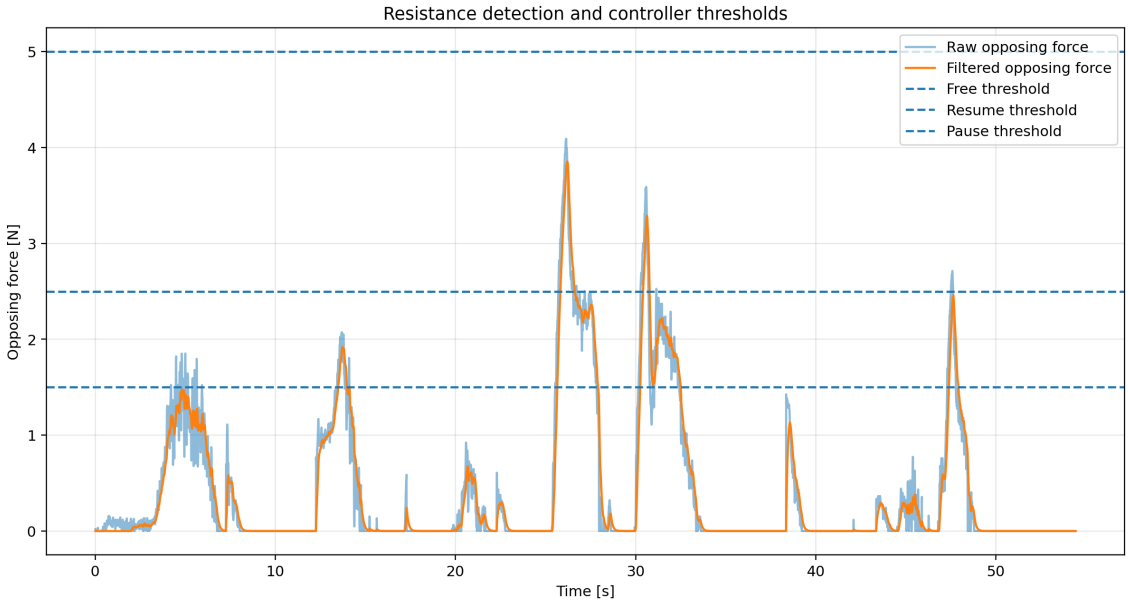

Figure 4.80: Raw and filtered opposing-force estimates together with the free-motion, resume, and pause thresholds.

Figure 4.81 shows the bounded assistive force and the running normalized required-assistance metric. The assistive force is activated when tangential lag is detected and is saturated at the configured maximum value of 4 N. The normalized assistance metric indicates the relative robotic contribution required during the exercise.

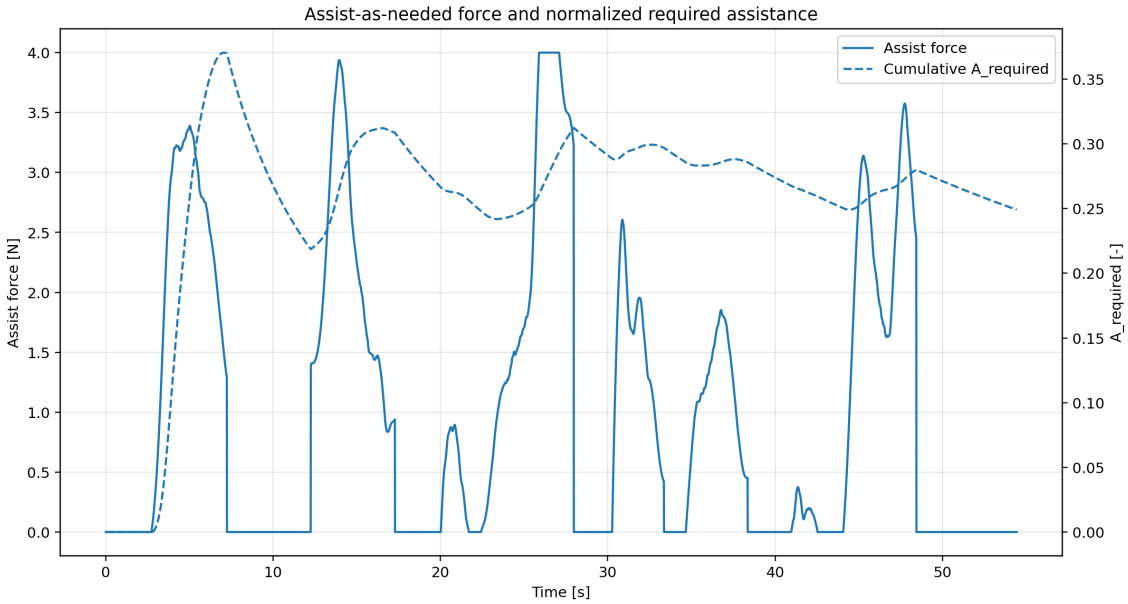

Figure 4.81: Bounded assist-as-needed force and running normalized required assistance.

Figure 4.82 demonstrates the adaptive trajectory-progression mechanism. Under low resistance, the speed scale remains close to one, and the reference trajectory progresses at approximately its nominal rate. During periods of increased resistance, the speed scale decreases, and the controller enters `SLOW_ASSIST`. A temporary resistance-hold interval may also occur when the corresponding force condition is satisfied. After the tenth half-cycle is completed, the controller transitions to `FINAL_HOLD`.

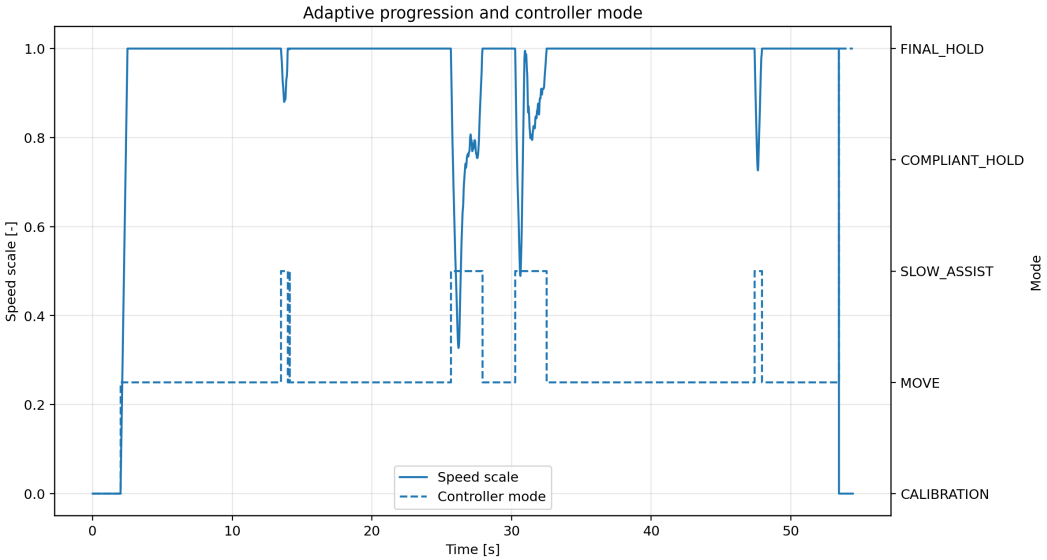

Figure 4.82: Adaptive trajectory speed scale and controller operating mode.

Figure 4.83 compares the measured and desired end-effector speeds. The desired speed follows the selected smooth trajectory profile and approaches zero near each

reversal of the shoulder motion. The measured speed follows the same periodic structure but contains additional variations caused by physical human interaction, controller compliance, and adaptive trajectory slowing.

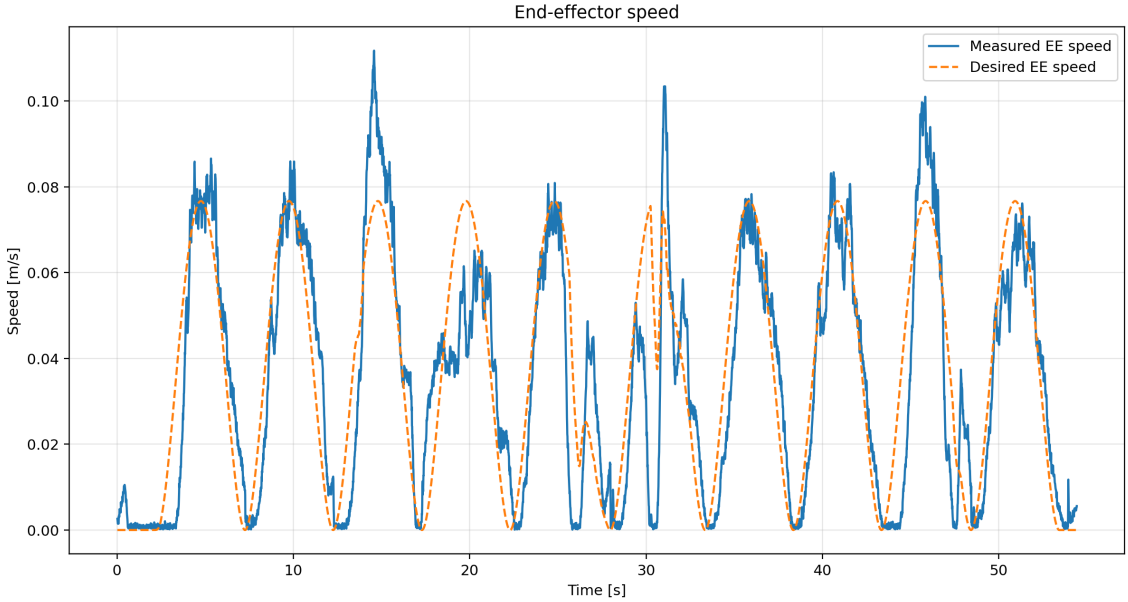

Figure 4.83: Measured and desired end-effector speeds during the shoulder exercise.

Figure 4.84 presents the desired and measured end-effector paths in the (X)-(Z) plane. The desired trajectory forms an arc around the estimated shoulder centre. The measured trajectory follows the general arc geometry but exhibits repetition-to-repetition variation caused by participant resistance, controller compliance, and small changes in the participant's posture.

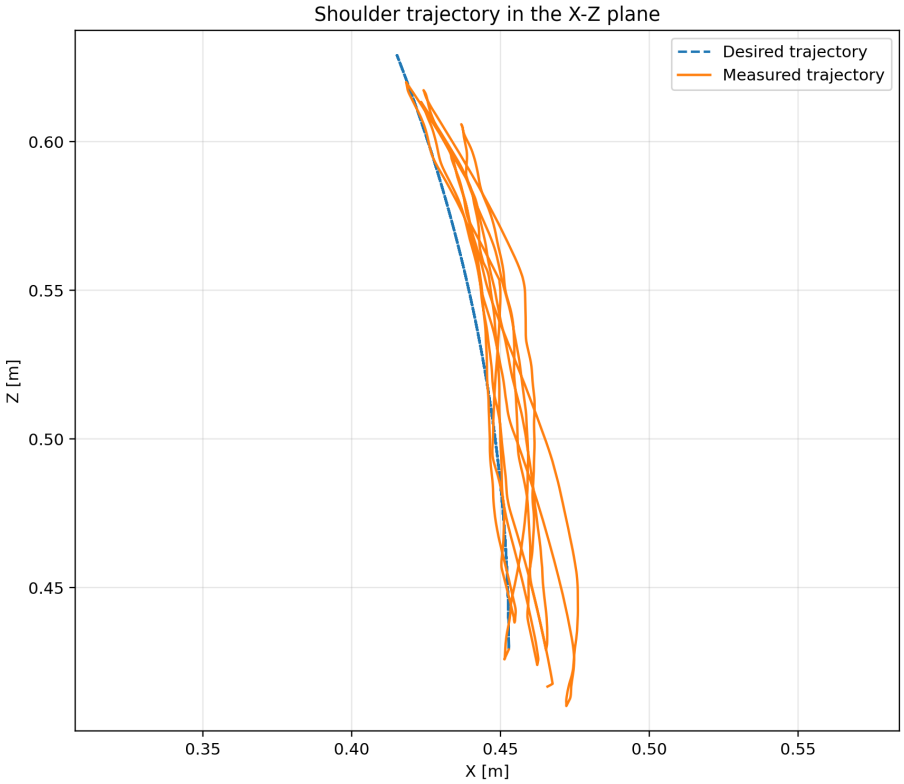

Figure 4.84: Desired and measured shoulder trajectories in the Cartesian (X)-(Z) plane.

Figure 4.85 presents the measured torques of the seven FR3 joints. The static offsets observed in some joints are primarily associated with the robot configuration, gravitational loading, and the effort required to support the arm and end-effector during the exercise. The time-varying components reflect the periodic motion and the physical interaction between the participant and the robot.

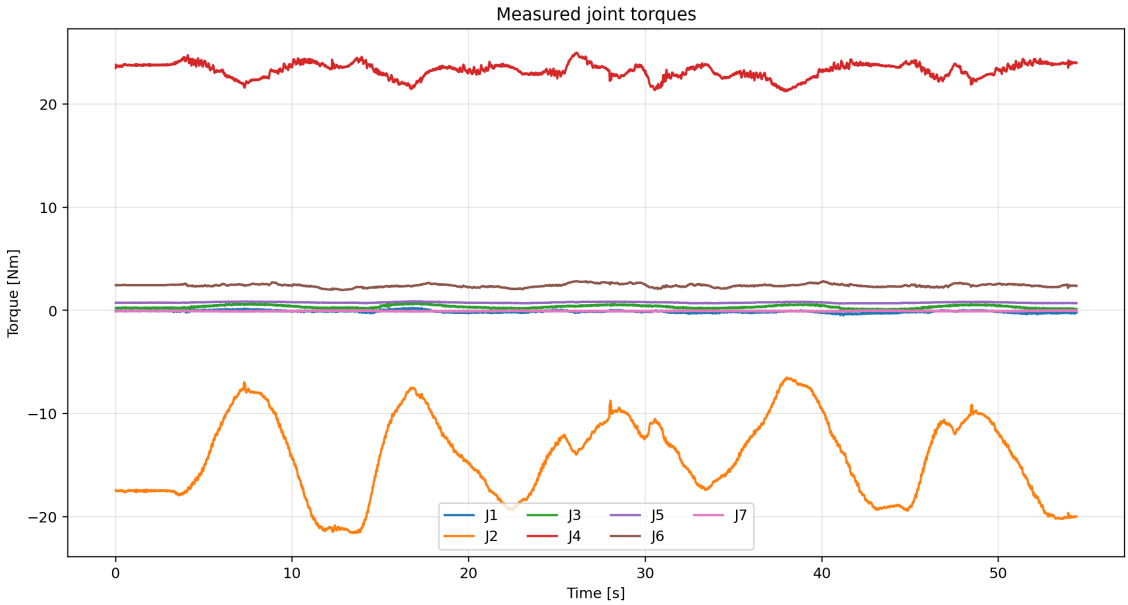

Figure 4.85: Measured torques of the seven FR3 joints during the shoulder exercise.

Figure 4.86 shows the commanded joint torques generated by the resistance-aware Cartesian impedance controller. These torques combine the Cartesian tracking contribution, compliant behaviour, dynamic compensation, and the assist-as-needed control action.

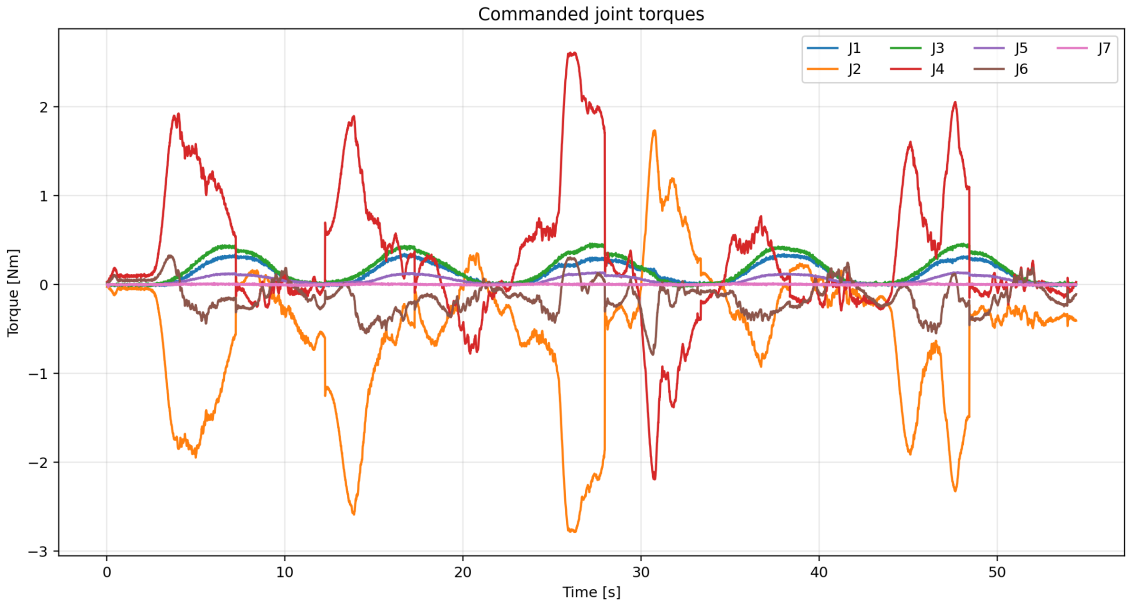

Figure 4.86: Commanded joint torques generated by the resistance-aware Cartesian impedance controller.

Figure 4.87 presents the estimated external joint torques and their Euclidean norm. Peaks in the external joint-torque signals are temporally consistent with the interaction-force events observed in the Cartesian force plots. These signals therefore provide a joint-space representation of the participant's physical interaction with the robot.

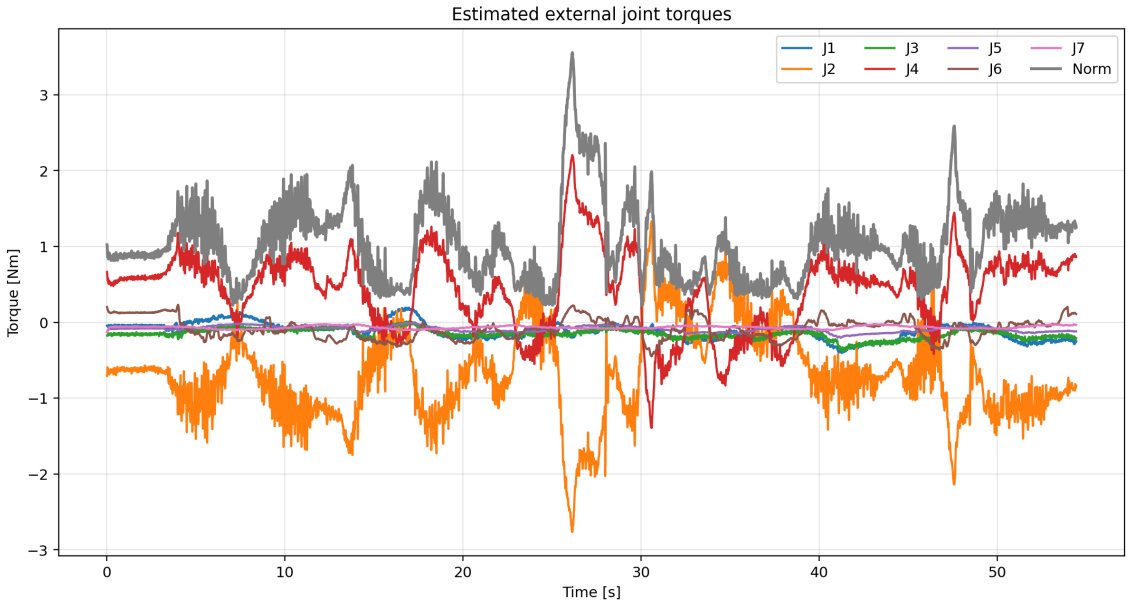

Figure 4.87: Estimated external torques of the seven FR3 joints and their Euclidean norm.

Figure 4.88 shows the progression of the exercise through the individual half-cycles. A total of ten half-cycles were completed, corresponding to five complete shoulder repetitions. Intervals with a reduced phase-progression slope correspond to periods during which the controller adaptively slowed the reference trajectory in response to detected resistance.

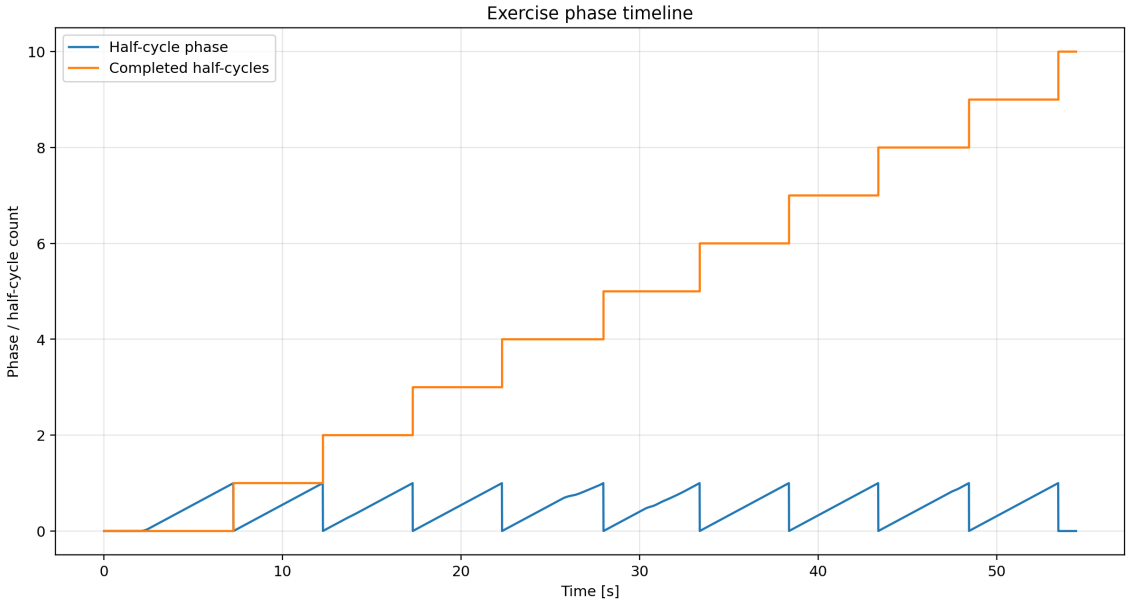

Figure 4.88: Half-cycle progression and number of completed half-cycles. Ten half-cycles correspond to five complete shoulder repetitions.

Finally, Figure 4.89 presents the normalized required assistance for each complete repetition. The third repetition required the greatest average robotic contribution, whereas

the fourth required the least. The non-monotonic variation suggests differences in participant resistance, active contribution, and movement execution between repetitions rather than a systematic improvement or deterioration over the five repetitions.

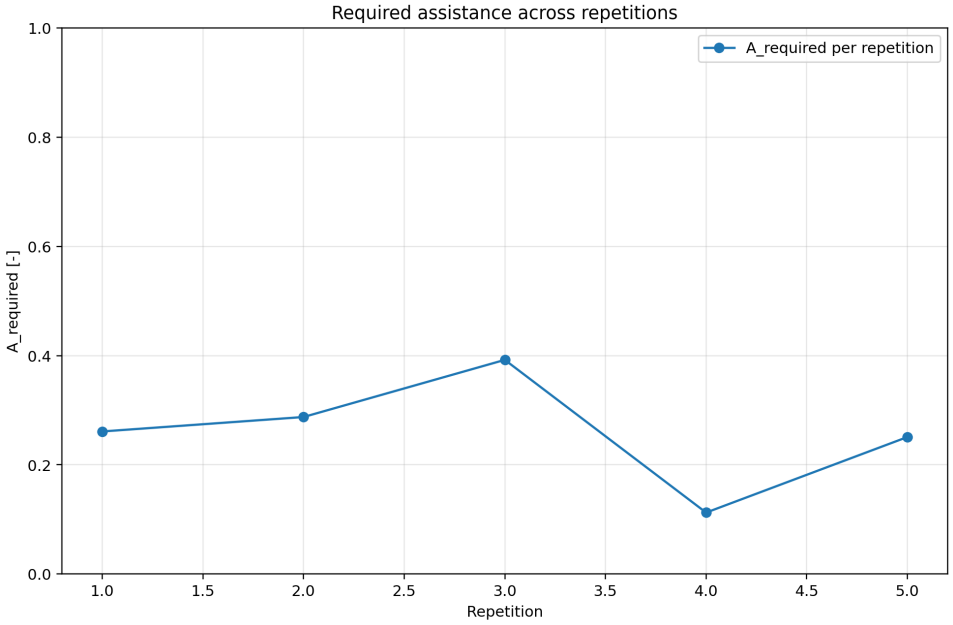

Figure 4.89: Normalized required assistance for each of the five complete shoulder repetitions.

Discussion and Conclusion

The results demonstrate that the FR3 completed five repetitions of a resistance-aware shoulder exercise while maintaining bounded and stable physical interaction. The arc-shaped Cartesian reference is more consistent with the kinematics of an approximately extended human arm than a purely vertical straight-line trajectory.

The tracking-error decomposition shows that the dominant deviations occurred along the tangent of the desired path. The perpendicular error remained lower, indicating that the virtual channel generally maintained the handle near the intended shoulder arc while allowing compliant lag along the movement direction. This behaviour is appropriate for assist-as-needed interaction, because a participant with reduced mobility may temporarily remain behind the reference trajectory without being forced away from the exercise path.

The resistance-detection results show that the controller identified periods of increased opposing force and reduced the trajectory progression speed. The assistive force was activated only when tangential lag was present and remained bounded by 4 N. The successful completion of all ten half-cycles shows that moderate resistance could be accommodated without premature termination of the exercise.

The measured, commanded, and externally estimated joint torques remained

bounded and did not exhibit sustained divergence. Their temporal variations were consistent with the Cartesian interaction-force events, supporting the internal consistency of the logged data.

The required-assistance metric varied substantially between repetitions and did not decrease monotonically. Therefore, the present experiment should not be interpreted as evidence of motor recovery. Instead, it demonstrates that the controller can detect changes in physical interaction and adapt its speed and assistance separately during each repetition. Evaluation of rehabilitation progress would require repeated sessions, multiple participants, consistent initial conditions, and statistical analysis of A_{required}, tracking error, interaction force, completion time, and movement smoothness.

Overall, the experiment validates the technical feasibility of using the FR3 for resistance-aware, assist-as-needed shoulder rehabilitation research. The system generated an anatomically motivated arc trajectory, maintained a virtual movement channel, reduced progression under resistance, provided bounded assistance, and completed the full protocol without unstable controller behaviour. These results support the use of the developed controller as a laboratory research platform, while any clinical application would still require formal medical supervision, ethical approval, risk assessment, and device-level safety validation.

4.1.8 Experiment 7: Resistance-Aware Assist-as-Needed Elbow Rehabilitation Using the Franka Research 3

Objective and Experimental Setup

The seventh experiment investigated a resistance-aware assist-as-needed rehabilitation exercise for elbow flexion and extension using the Franka Research 3 (FR3). The participant placed the elbow on a padded support on the table and maintained the elbow joint approximately fixed throughout the task. The participant held a cylindrical handle grasped by the Franka Hand, while the robot guided the hand along an arc representing the motion of the forearm around the elbow joint.

The end-effector started at an elevated position approximately 0.20 m above the table. During the first half-cycle, the handle moved downward to approximately 0.10 m, representing elbow extension. During the second half-cycle, it returned to the initial height, representing elbow flexion. This movement was repeated five times, corresponding to ten completed half-cycles.

The experiment was designed as an engineering evaluation of compliant physical human–robot interaction and not as a clinical rehabilitation trial. The robot was operated at low Cartesian velocity with bounded assistance, conservative interaction-force thresholds, and continuous supervision.

Elbow-Arc Trajectory

Let

$$\boldsymbol{p}_e \in \mathbb{R}^3$$

denote the estimated elbow-joint centre expressed in the robot base frame, and let

$$\boldsymbol{p}_0 \in \mathbb{R}^3$$

denote the initial handle position. The elbow-to-handle vector is

$$\boldsymbol{r}_0 = \boldsymbol{p}_0 - \boldsymbol{p}_e, \tag{4.51}$$

with corresponding radius

$$L = \|\boldsymbol{r}_0\|_2. \tag{4.52}$$

Assuming that the elbow remains approximately fixed, the desired handle position is generated by rotating $\boldsymbol{r}_0$ around the elbow flexion–extension axis $\hat{\boldsymbol{a}}$:

$$\boldsymbol{p}_d(t) = \boldsymbol{p}_e + \boldsymbol{R}(\hat{\boldsymbol{a}}, \phi(t))\, \boldsymbol{r}_0, \tag{4.53}$$

where $\boldsymbol{R}(\hat{\boldsymbol{a}}, \phi) \in SO(3)$ is the corresponding rotation matrix. Using Rodrigues' formula,

$$\boldsymbol{R}(\hat{\boldsymbol{a}}, \phi) = \boldsymbol{I}_3 + \sin\phi\, [\hat{\boldsymbol{a}}]_\times + (1 - \cos\phi)[\hat{\boldsymbol{a}}]_\times^2. \tag{4.54}$$

Each flexion or extension movement was treated as one half-cycle. A fifth-order minimum-jerk scaling function was used:

$$s(\sigma) = 10\sigma^3 - 15\sigma^4 + 6\sigma^5, \qquad 0 \le \sigma \le 1. \tag{4.55}$$

For half-cycle h, the elbow angle was defined as

$$\phi_h(\sigma) = \phi_{h,0} + (\phi_{h,f} - \phi_{h,0})\, s(\sigma). \tag{4.56}$$

For the implemented geometry, the first half-cycle produced an approximate Cartesian displacement

$$\Delta\boldsymbol{p} \approx \begin{bmatrix} 0.04 & 0 & -0.10 \end{bmatrix}^T \text{m}, \tag{4.57}$$

so that the handle moved downward by approximately 0.10 m, while also moving horizontally according to the natural elbow arc.

Tracking-Error Decomposition

The Cartesian position error was defined as

$$\boldsymbol{e}_p = \boldsymbol{p}_d - \boldsymbol{p}, \tag{4.58}$$

where $\boldsymbol{p}$ is the measured end-effector position. Its Euclidean norm was

$$e_{\text{norm}} = \|\boldsymbol{e}_p\|_2. \tag{4.59}$$

The unit tangent vector of the desired trajectory was calculated from the desired Cartesian velocity:

$$\hat{\boldsymbol{t}} = \frac{\dot{\boldsymbol{p}}_d}{\|\dot{\boldsymbol{p}}_d\|_2}. \tag{4.60}$$

The tangential tracking error was

$$e_{\parallel} = \hat{\boldsymbol{t}}^T \boldsymbol{e}_p, \tag{4.61}$$

whereas the perpendicular error vector was

$$\boldsymbol{e}_{\perp} = \left(\boldsymbol{I}_3 - \hat{\boldsymbol{t}}\hat{\boldsymbol{t}}^T\right) \boldsymbol{e}_p, \tag{4.62}$$

with magnitude

$$e_{\perp} = \|\boldsymbol{e}_{\perp}\|_2. \tag{4.63}$$

The tangential component quantifies lag along the desired elbow trajectory, whereas the perpendicular component measures deviation away from the virtual movement channel.

Resistance Detection and Adaptive Assistance

The interaction-force bias was estimated during an initial stationary calibration interval:

$$\boldsymbol{F}_{\text{bias}} = \frac{1}{N_b} \sum_{k=1}^{N_b} \boldsymbol{F}_{\text{ext}}[k]. \tag{4.64}$$

The bias-corrected human–robot interaction force was

$$\boldsymbol{F}_h = \boldsymbol{F}_{\text{ext}} - \boldsymbol{F}_{\text{bias}}. \tag{4.65}$$

The force opposing the desired motion was estimated by projecting the interaction force onto the path tangent:

$$F_{\text{opp}} = \max\left(0, -s_F \hat{\boldsymbol{t}}^T \boldsymbol{F}_h\right), \qquad s_F \in \{-1, +1\}. \tag{4.66}$$

A first-order low-pass filter reduced high-frequency measurement noise:

$$F_{\mathrm{opp},f}[k] = \beta F_{\mathrm{opp},f}[k-1] + (1-\beta)F_{\mathrm{opp}}[k]. \tag{4.67}$$

The trajectory speed scale was adapted according to the filtered opposing force:

$$\rho_{\mathrm{cmd}}(F) = \begin{cases} 1, & F \le F_{\mathrm{free}}, \\ \dfrac{F_{\mathrm{pause}} - F}{F_{\mathrm{pause}} - F_{\mathrm{free}}}, & F_{\mathrm{free}} < F < F_{\mathrm{pause}}, \\ 0, & F \ge F_{\mathrm{pause}}. \end{cases} \tag{4.68}$$

The half-cycle phase evolved according to

$$\dot{\sigma} = \frac{\rho(t)}{T_h}, \qquad 0 \le \rho(t) \le 1, \tag{4.69}$$

where T_h is the nominal half-cycle duration.

The controller used direction-dependent resistance thresholds and vertical support forces because elbow flexion and extension produce different interaction loads. In particular, the upward return motion required additional support to compensate partly for the weight of the participant's forearm and hand.

The bounded assist-as-needed force was calculated from the positive tangential lag:

$$F_{\mathrm{assist}} = \mathrm{sat}_{[0,F_{\mathrm{assist,max}}]}\left[k_a \max\left(0, e_{\parallel} - e_{\mathrm{db}}\right)\right], \tag{4.70}$$

and applied along the desired trajectory:

$$\boldsymbol{F}_{\mathrm{assist}} = F_{\mathrm{assist}}\hat{\boldsymbol{t}}. \tag{4.71}$$

The maximum assistive force was limited to

$$0 \le F_{\mathrm{assist}} \le 4\,\mathrm{N}. \tag{4.72}$$

Cartesian Impedance and Joint-Torque Command

The virtual-channel Cartesian force was

$$\boldsymbol{F}_{\mathrm{vc}} = k_{\parallel} e_{\parallel}\hat{\boldsymbol{t}} + k_{\perp}\boldsymbol{e}_{\perp}, \tag{4.73}$$

where $k_{\parallel}$ controls guidance along the path and $k_{\perp}$ maintains the handle near the desired elbow arc.

Including Cartesian damping, bounded assistance, and direction-dependent vertical

support, the commanded Cartesian force was

$$\boldsymbol{F}_{\text{cmd}} = \boldsymbol{F}_{\text{vc}} + \boldsymbol{D}_p\,(\dot{\boldsymbol{p}}_d - \dot{\boldsymbol{p}}) + \boldsymbol{F}_{\text{assist}} + F_z^{\text{support}}\hat{\boldsymbol{z}}. \tag{4.74}$$

The complete commanded wrench was

$$\boldsymbol{W}_{\text{cmd}} = \begin{bmatrix} \boldsymbol{F}_{\text{cmd}} \\ \boldsymbol{M}_{\text{cmd}} \end{bmatrix}, \tag{4.75}$$

and the desired joint torques were obtained through the Jacobian transpose:

$$\boldsymbol{\tau}_{\text{des}} = \boldsymbol{J}(\boldsymbol{q})^T \boldsymbol{W}_{\text{cmd}} + \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}), \tag{4.76}$$

where $\boldsymbol{J}(\boldsymbol{q}) \in \mathbb{R}^{6\times 7}$ is the FR3 Jacobian and $\boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ is the Coriolis compensation term.

Performance Metric

For repetition n, the average assistive force was calculated as

$$\overline{F}_{\text{assist}}^{(n)} = \frac{1}{T_n} \int_{t_n}^{t_n+T_n} F_{\text{assist}}(t)\,dt. \tag{4.77}$$

The normalized required-assistance metric was

$$A_{\text{required}}^{(n)} = \frac{\overline{F}_{\text{assist}}^{(n)}}{F_{\text{assist,max}}}, \qquad 0 \le A_{\text{required}}^{(n)} \le 1. \tag{4.78}$$

Experimental Results

Figure 4.90 compares the measured and desired Cartesian end-effector positions. The desired z-coordinate varies periodically between approximately 0.19 m and 0.09 m, corresponding to the intended vertical displacement of approximately 0.10 m. The simultaneous variation in the desired x-coordinate confirms that the end effector follows an elbow arc rather than a purely vertical path. The measured trajectory follows the periodic reference pattern but exhibits compliant tracking lag, particularly near the lower endpoints of the motion.

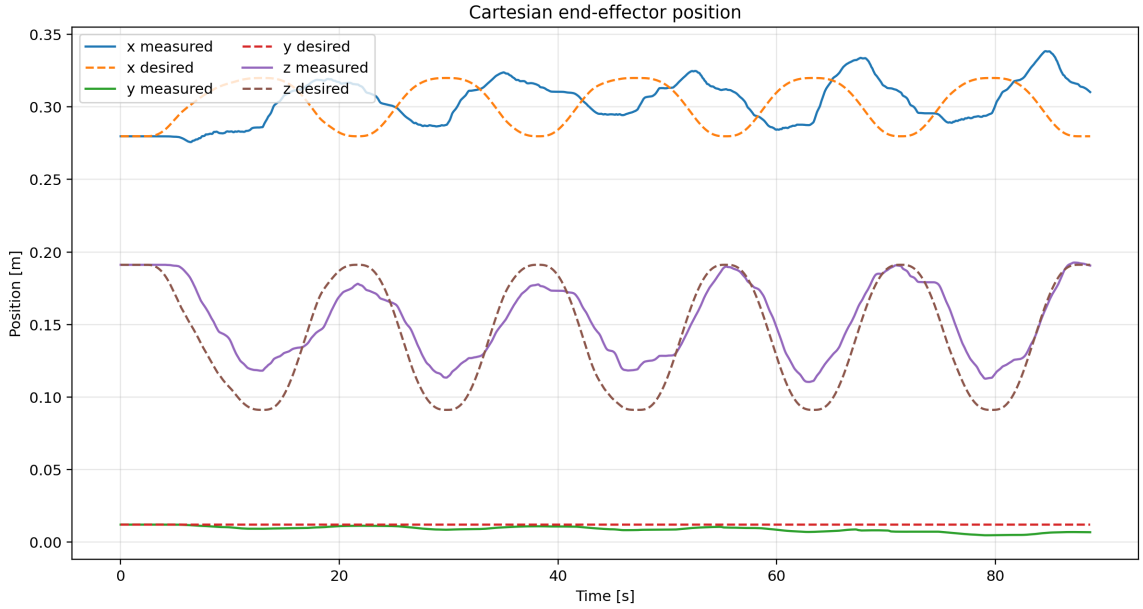

Figure 4.90: Measured and desired Cartesian end-effector positions during the elbow rehabilitation exercise. Five complete repetitions are visible. The desired x- and z-coordinates describe the elbow arc, whereas the desired y-coordinate remains approximately constant.

Figure 4.91 presents the Cartesian tracking-error components. The x- and z-errors show the largest variations because the exercise is performed primarily in the X-Z plane. The comparatively small y-axis error indicates that the movement remained close to the intended plane and exhibited limited out-of-plane displacement.

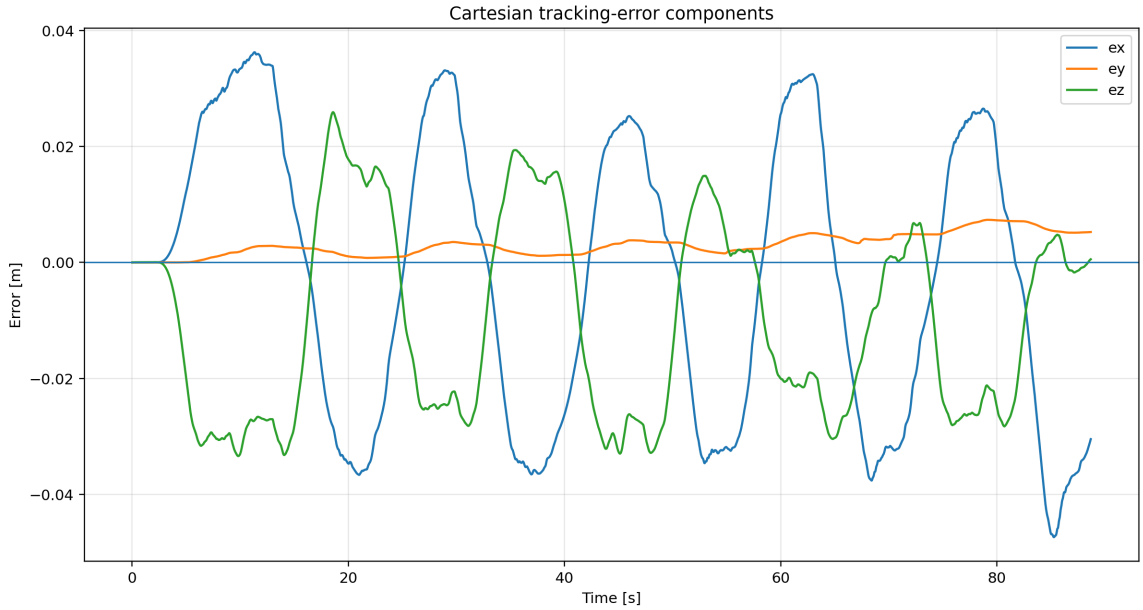

Figure 4.91: Cartesian tracking-error components e_x, e_y, and e_z. The comparatively small e_y component indicates that the movement remained largely within the intended X-Z plane.

Figure 4.92 presents the total, tangential, and perpendicular tracking errors. The total tracking error remains below approximately 0.05 m. Both tangential lag and perpendicular deviation contribute to the total error, indicating compliant human–robot interaction rather than rigid enforcement of the desired trajectory.

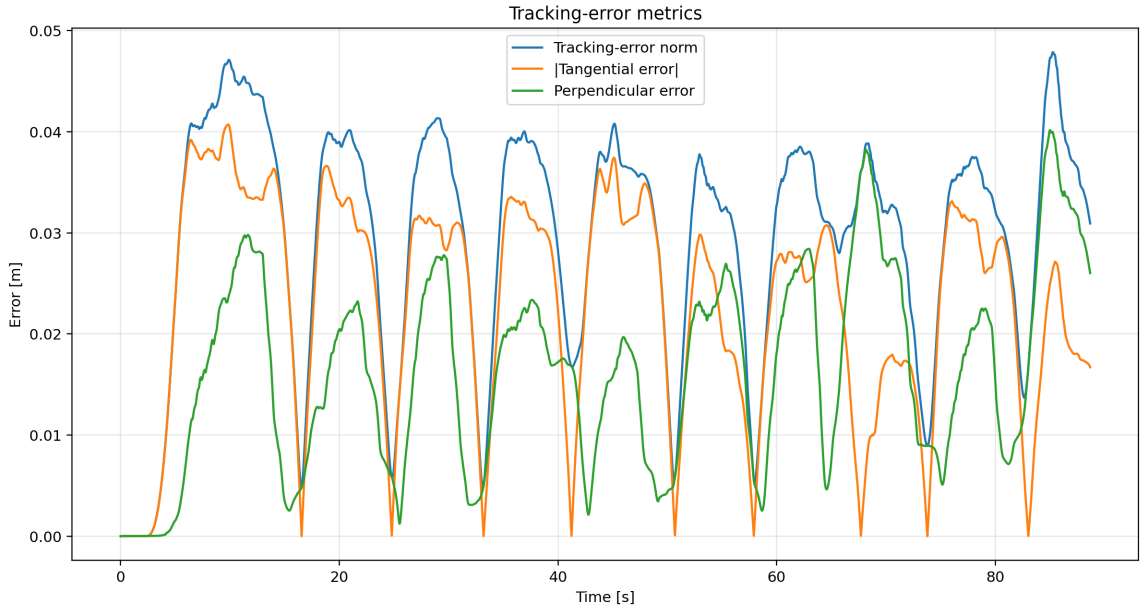

Figure 4.92: Tracking-error norm, absolute tangential error, and perpendicular error during the five elbow repetitions.

Figure 4.93 presents the bias-corrected Cartesian interaction-force components. The dominant variations occur in the x- and z-directions, which is consistent with the primary plane of the elbow movement. The smaller y-component represents interaction forces acting outside the intended movement plane.

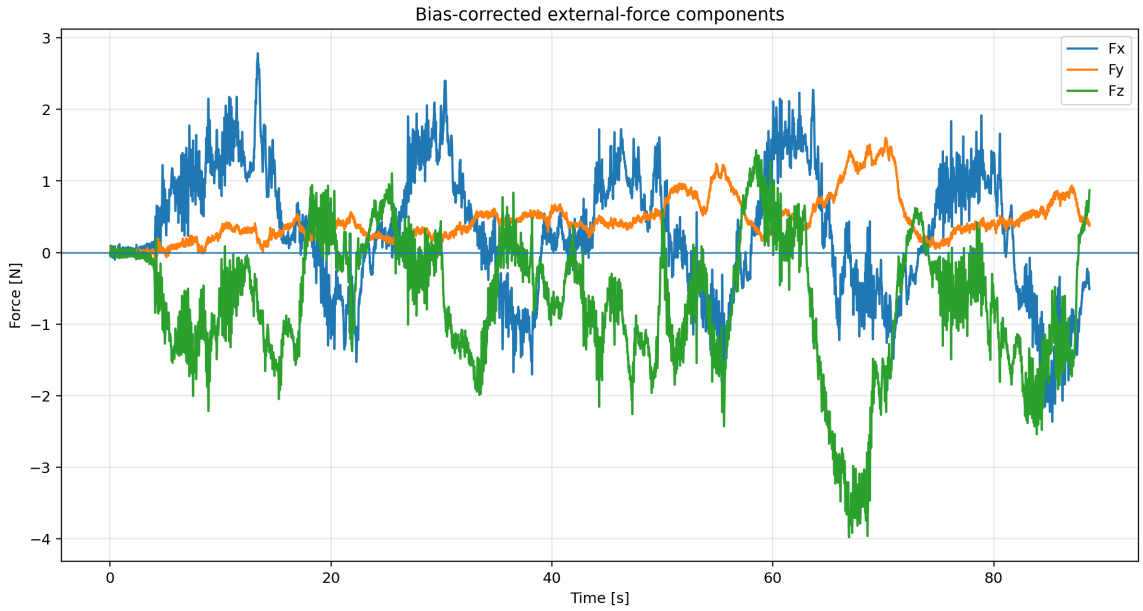

Figure 4.93: Bias-corrected Cartesian external-force components during physical human–robot interaction. The dominant force variations occur in the X-Z movement plane.

Figure 4.94 shows the absolute external-force norm and the force change relative to the initial calibration bias. The interaction forces remain bounded throughout the experiment. The largest change in force occurs during one of the later repetitions, indicating an interval of increased participant resistance or physical contribution.

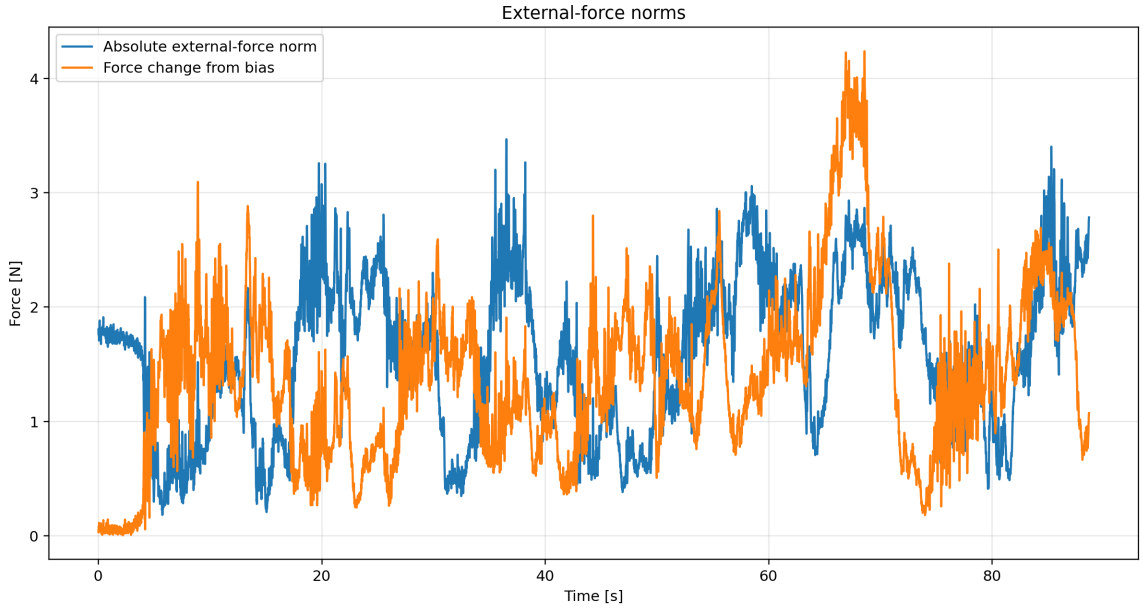

Figure 4.94: Absolute external-force norm and force variation relative to the initial calibration bias.

Figure 4.95 presents the raw and filtered opposing-force estimates together with the resistance thresholds displayed by the plotting script. The filtered signal follows the

general trend of the raw estimate while reducing high-frequency fluctuations. Intervals during which the filtered force exceeds the free-motion threshold indicate increased resistance and cause the controller to reduce the progression speed. The implemented controller additionally uses direction-dependent thresholds and an upward opposing-force offset for the flexion and extension phases.

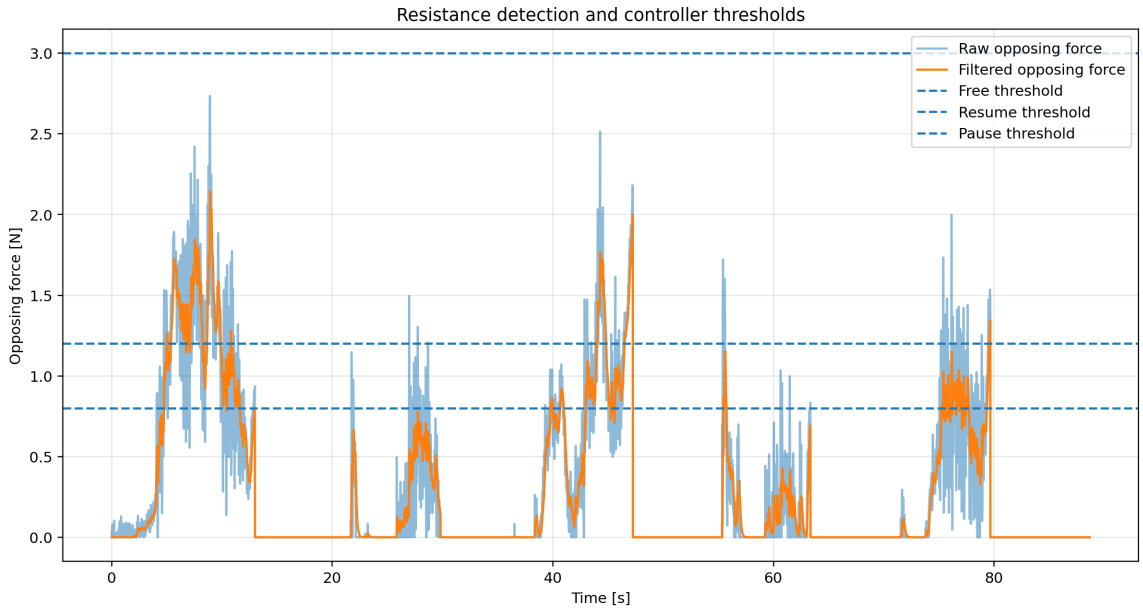

Figure 4.95: Raw and filtered opposing-force estimates together with the plotted free-motion, resume, and pause thresholds. The implemented controller uses direction-dependent thresholds for elbow extension and flexion.

Figure 4.96 presents the bounded assistive force and the running normalized required-assistance metric. Assistance is activated when positive tangential lag indicates that the participant is falling behind the desired trajectory. The assistive force remains bounded by the configured maximum value of 4 N. The normalized metric indicates the relative robotic contribution required as the exercise progresses.

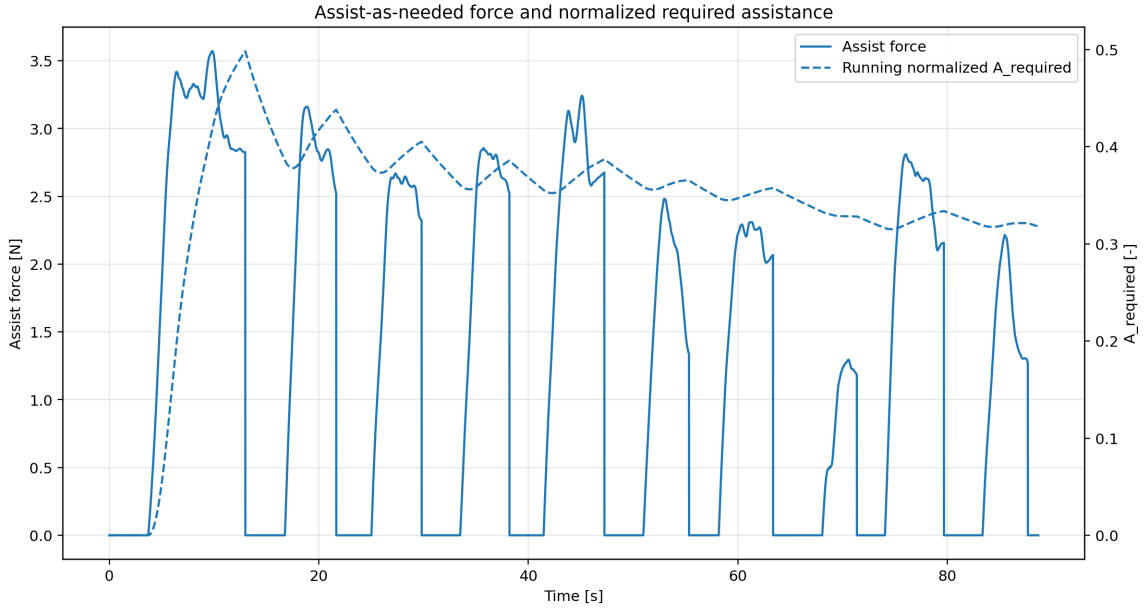

Figure 4.96: Assist-as-needed force and running normalized required-assistance metric. The assistive force is activated according to tangential tracking lag and remains bounded by 4 N.

Figure 4.97 demonstrates the adaptive trajectory- progression mechanism. Under low resistance, the speed scale remains close to one, allowing the reference trajectory to progress at approximately its nominal rate. During intervals of increased resistance, the speed scale decreases and the controller enters the `SLOW_ASSIST` mode. The controller operates mainly in the `MOVE` and `SLOW_ASSIST` modes before transitioning to `FINAL_HOLD` after completion of the exercise.

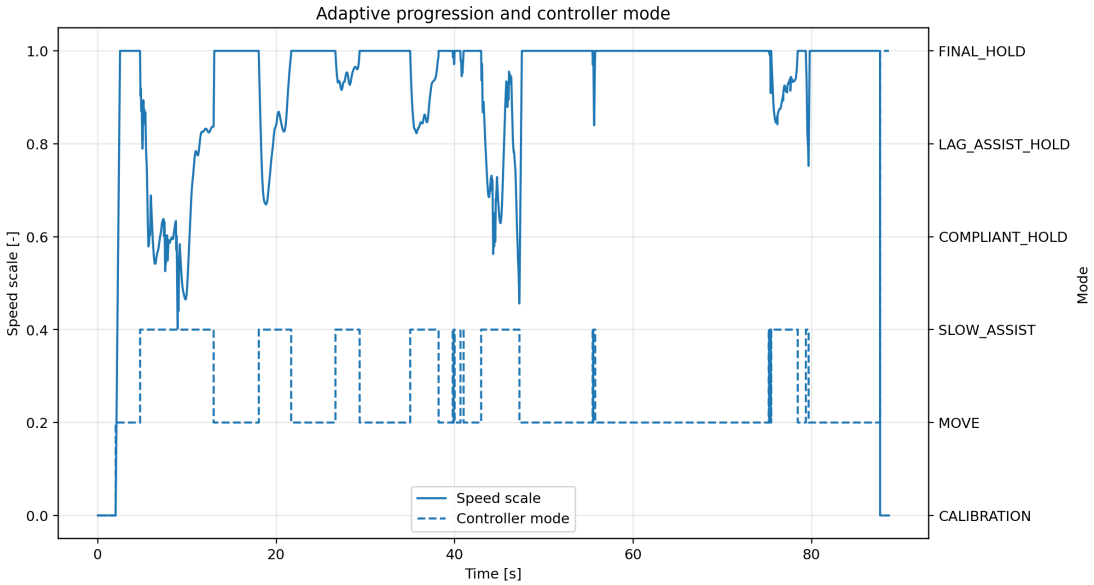

Figure 4.97: Adaptive trajectory speed scale and controller operating mode. Reduced speed-scale values indicate resistance-dependent slowing of the desired elbow trajectory.

Figure 4.98 compares the measured and desired end-effector speeds. The desired speed follows the periodic smooth trajectory profile and approaches zero near each flexion and extension reversal. The measured speed follows the same general periodic structure but exhibits additional variations caused by compliant human–robot interaction, adaptive slowing, and changes in participant contribution.

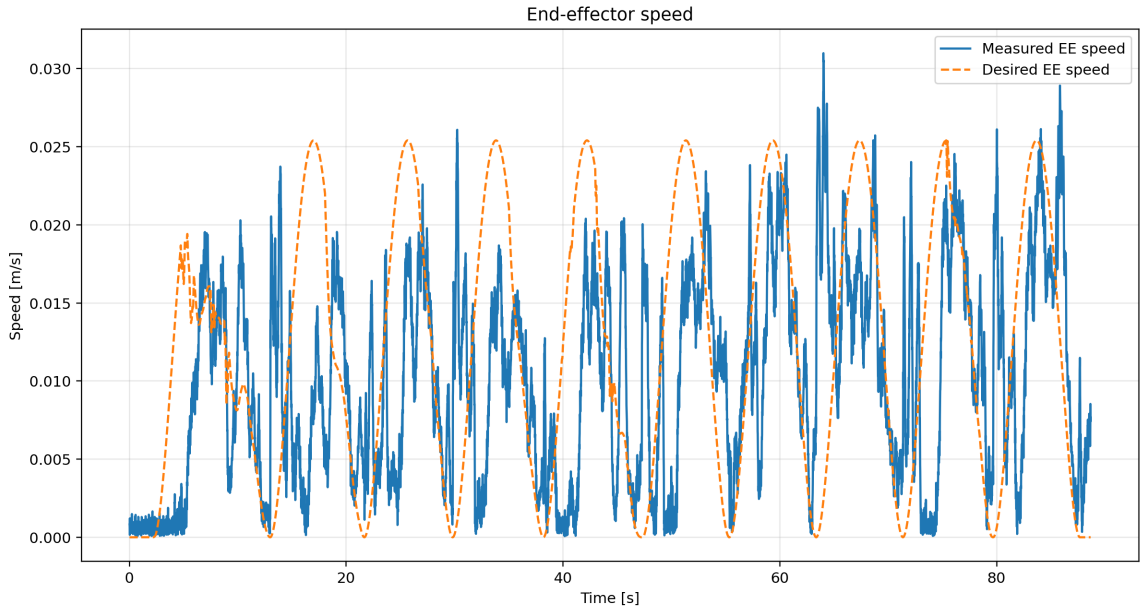

Figure 4.98: Measured and desired end-effector speeds during the elbow exercise. The desired speed approaches zero at the flexion and extension endpoints.

Figure 4.99 presents the desired and measured end-effector trajectories in the X-Z plane. The desired path represents the theoretical elbow arc around the estimated elbow centre. The measured trajectory follows the general arc geometry but exhibits repetition-dependent deviation and hysteresis. These differences may be attributed to controller compliance, uncertainty in the estimated elbow centre, small movements of the elbow on the supporting surface, and variation in the participant's active contribution.

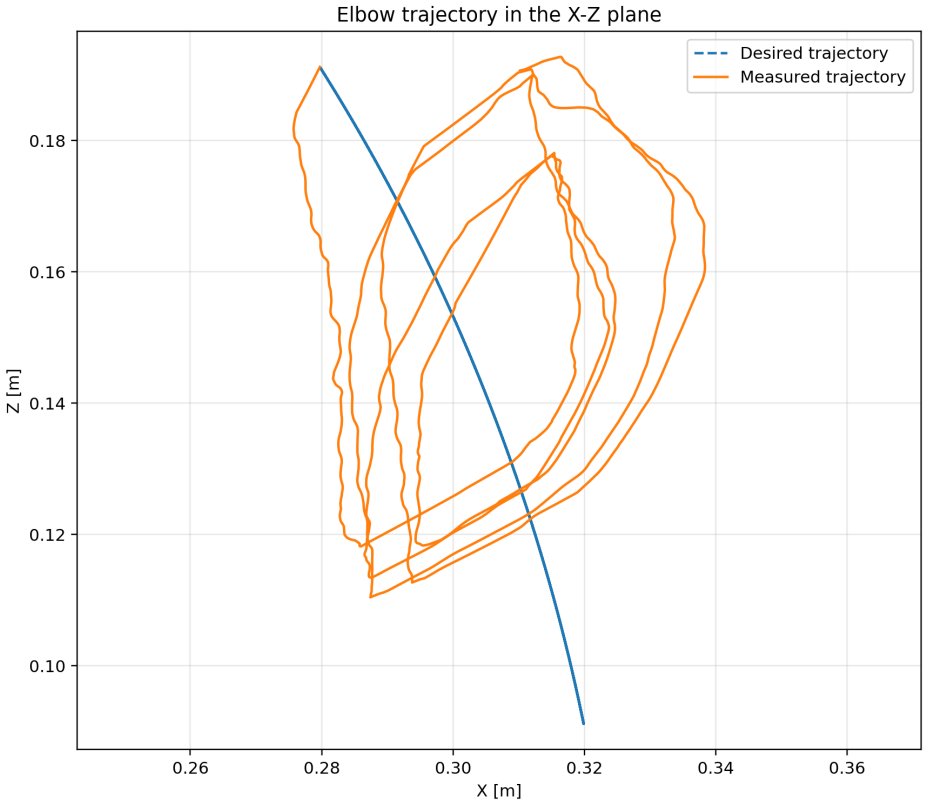

Figure 4.99: Desired and measured end-effector trajectories in the Cartesian X-Z plane. The measured trajectory follows the general elbow-motion region but exhibits compliant deviation and repetition-dependent hysteresis.

Figure 4.100 presents the measured torques of the seven FR3 joints. Their static offsets are associated primarily with the robot configuration and gravitational loading. The periodic time-varying components are related to the repeated end-effector motion and the physical interaction between the participant and the robot.

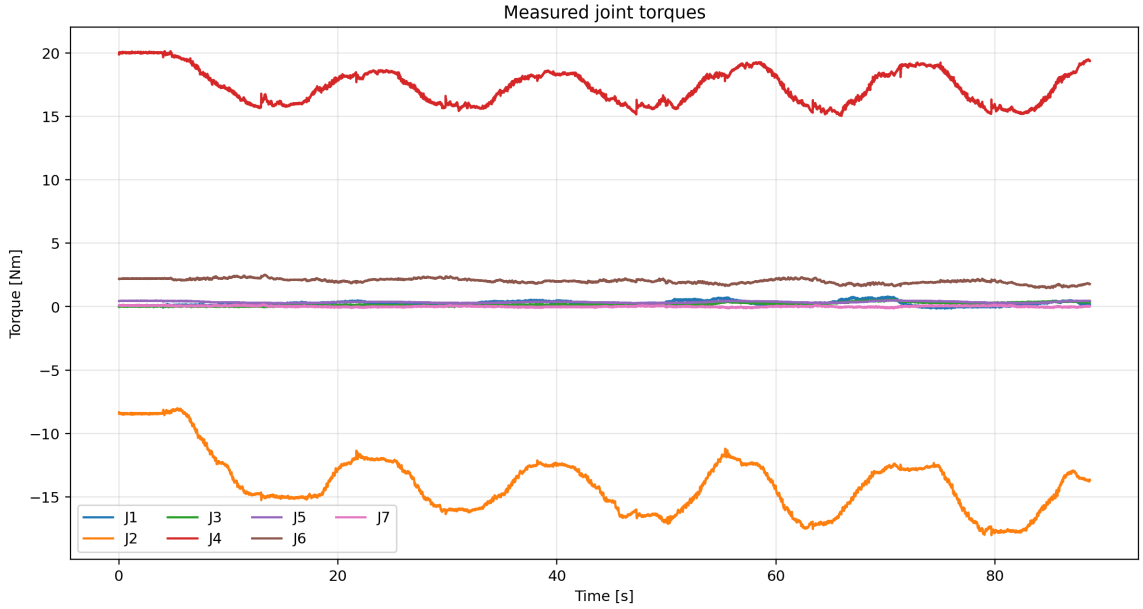

Figure 4.100: Measured torques of the seven FR3 joints during the five elbow repetitions.

Figure 4.101 presents the commanded joint torques generated by the Cartesian impedance, damping, support, and assist-as-needed control terms. The torque commands remain bounded throughout the exercise. Short-duration variations appear near movement reversals and during transitions between the direction-dependent controller states.

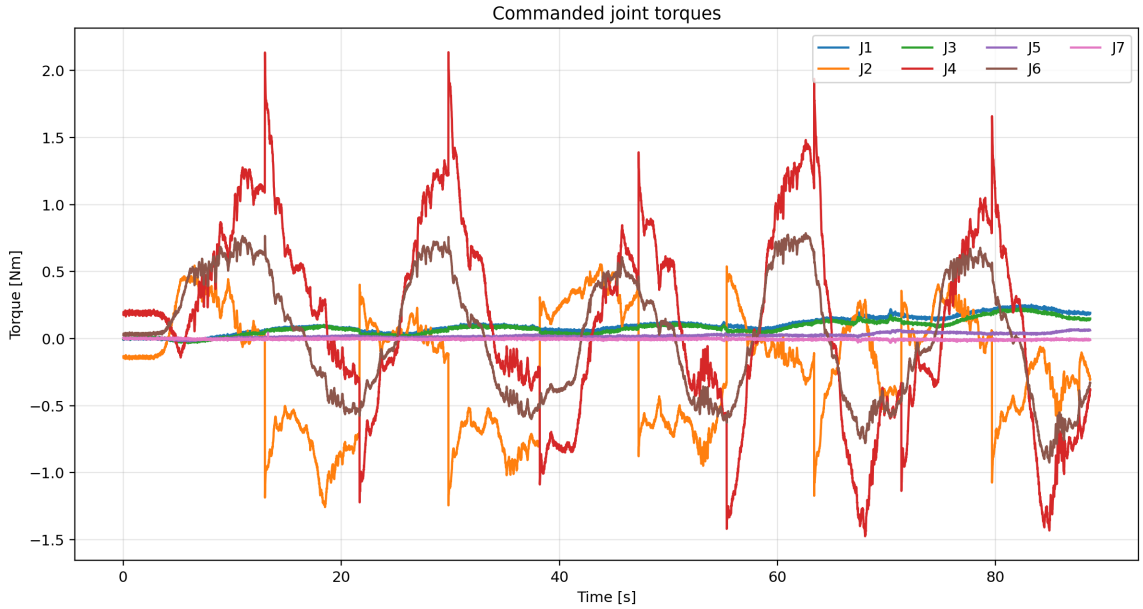

Figure 4.101: Commanded joint torques generated by the resistance-aware Cartesian impedance and assist-as-needed controller.

Figure 4.102 presents the estimated external joint torques and their Euclidean norm. Their temporal variations are consistent with the interaction events observed in the

Cartesian-force plots. These signals therefore provide a joint-space representation of the participant's physical interaction with the robot.

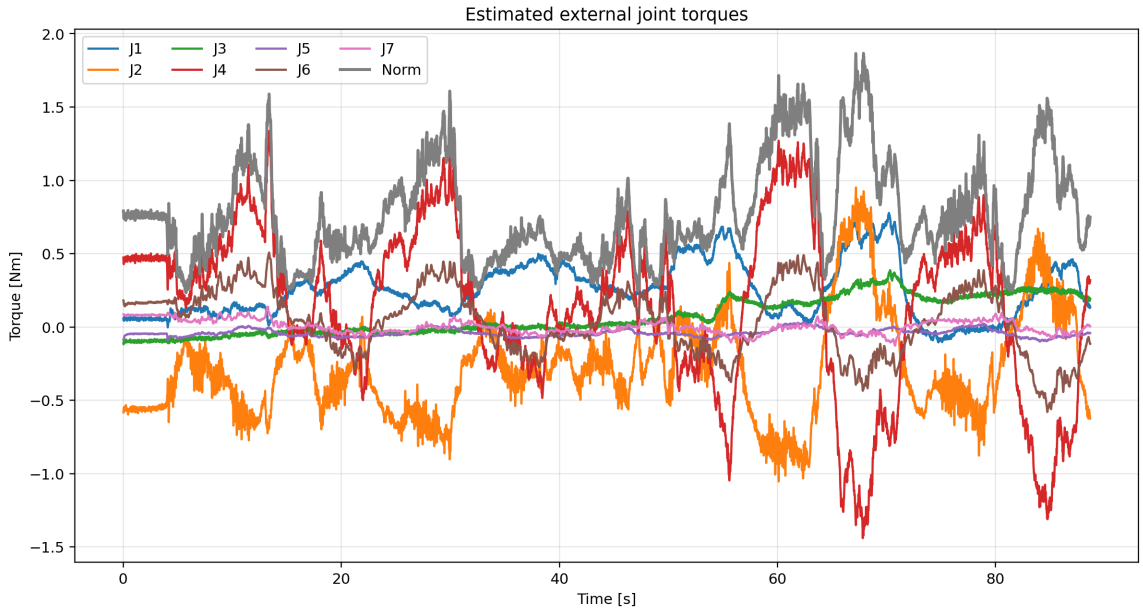

Figure 4.102: Estimated external joint torques and their Euclidean norm during the elbow rehabilitation exercise.

Figure 4.103 shows the progression of the exercise through its individual half-cycles. Ten half-cycles were completed, corresponding to five complete elbow flexion–extension repetitions. Variations in the phase-progression slope indicate intervals during which the reference trajectory was slowed in response to increased resistance.

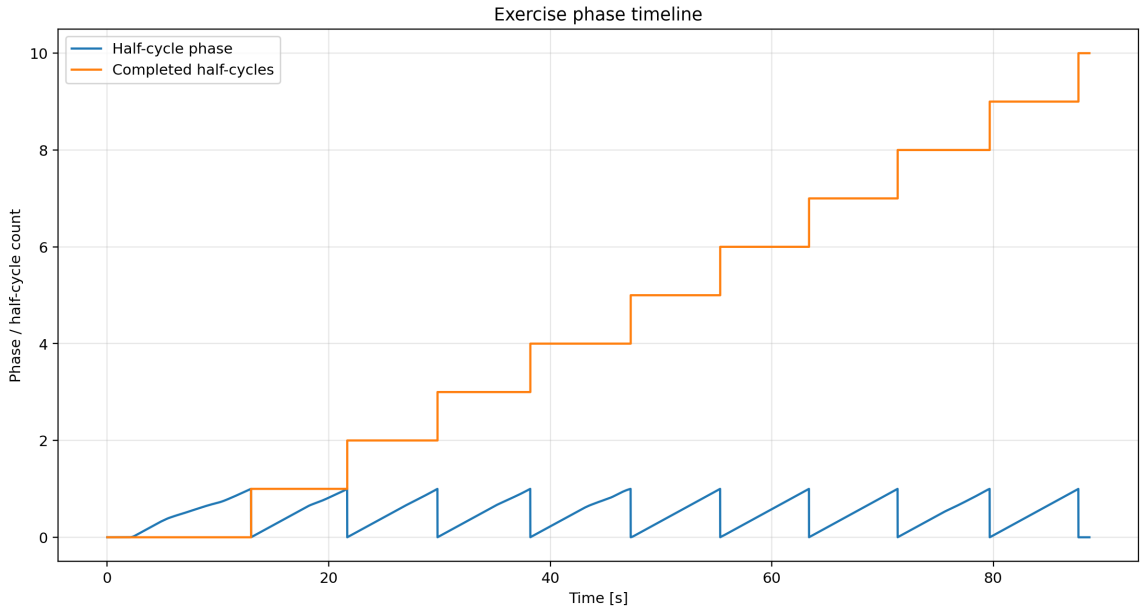

Figure 4.103: Exercise phase timeline and completed half-cycle counter. Ten half-cycles correspond to five complete elbow repetitions.

Finally, Figure 4.104 presents the normalized required assistance for each complete repetition. The approximate values are

$$A_{\text{required}} \approx [0.48,\ 0.32,\ 0.32,\ 0.20,\ 0.29].$$

The first repetition required the greatest average robotic contribution, whereas the fourth required the least. The assistance requirement generally decreased after the first repetition, although the variation was not strictly monotonic. This suggests differences in participant resistance and active contribution between repetitions rather than a uniform improvement throughout the exercise.

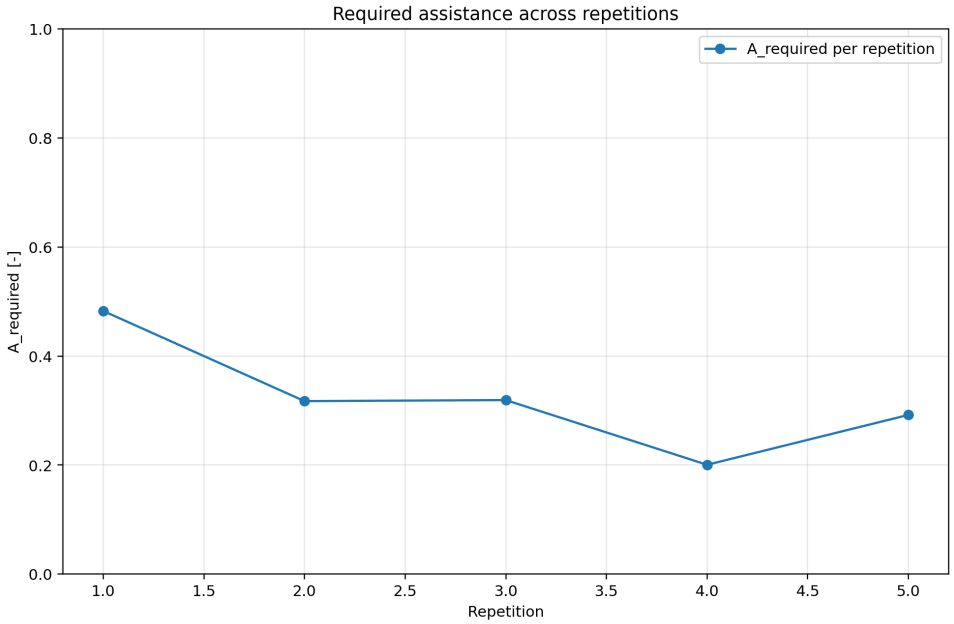

Figure 4.104: Normalized required assistance for each of the five complete elbow repetitions. The assistance requirement decreases after the first repetition but does not follow a strictly monotonic trend.

Discussion and Conclusion

The experiment demonstrated that the FR3 successfully completed five resistance-aware elbow flexion–extension repetitions while maintaining bounded interaction forces, low movement velocity, and adaptive assist-as-needed behaviour. The desired trajectory reproduced the geometric characteristics of an elbow arc, including simultaneous horizontal and vertical end-effector motion.

The tracking results show that the robot preserved the general periodic movement but did not rigidly enforce the reference trajectory. The maximum tracking error was approximately 0.05 m, and both tangential and perpendicular components contributed to this deviation. In particular, the measured X-Z paths exhibited hysteresis between flexion and extension. This behaviour is consistent with compliant human–robot

interaction, but it also indicates sensitivity to the estimated elbow centre, movement-axis alignment, elbow displacement on the table, and participant posture.

The force and torque results remained bounded and did not exhibit sustained oscillation or divergence. Periods of increased opposing force were accompanied by reductions in the trajectory speed scale and activation of assistive force. The assistance remained below the configured maximum of 4 N, demonstrating that the controller adapted its contribution without applying unbounded force.

The required-assistance metric was greatest during the first repetition and lower during the subsequent repetitions, although the variation was not monotonic. Therefore, the results should be interpreted as evidence that the controller adapted to repetition-dependent changes in participant contribution, not as evidence of clinical motor recovery.

Overall, the experiment validates the technical feasibility of using the FR3 for resistance-aware assist-as-needed elbow rehabilitation research. The system completed the full protocol, adjusted its progression according to interaction resistance, and provided bounded assistance while preserving compliant physical interaction. Further work should improve elbow-centre calibration, movement-axis estimation, direction-dependent force visualization, and participant support in order to reduce perpendicular trajectory error and repetition-to-repetition hysteresis.

Chapter 5

Conclusions and Future Work

5.1 Conclusions

This thesis investigated force-sensitive and interaction-aware control strategies for the Franka Research 3 (FR3), with emphasis on safe physical human–robot interaction, compliant manipulation, and rehabilitation-inspired assistance. The developed controllers were implemented directly through the Franka Control Interface using `libfranka`, allowing access to the robot state, joint-torque measurements, dynamic model, and real-time torque-control capabilities.

The experimental evaluation progressed from structured manipulation tasks to direct human interaction. The experiments included a free-space trajectory, a pick-and-place demonstration, human push and disturbance rejection, hand-guiding, hybrid position/force interaction with a compliant object, and resistance-aware assist-as-needed exercises for the shoulder and elbow. Together, these experiments demonstrated that the FR3 can support a wide range of force-aware behaviors using a common torque-control framework.

The main conclusions are summarized below.

- **The FR3 is a suitable platform for physical human–robot interaction.** The robot's seven degrees of freedom, joint-torque sensing, dynamic model, and direct torque-control interface provide the necessary capabilities for implementing compliant and force-aware controllers. The experimental results showed smooth motion, bounded interaction forces, and stable controller behavior across manipulation, contact, and human-interaction tasks.

- **Cartesian impedance control provides an effective balance between trajectory tracking and compliance.** The robot could be represented as a virtual mass–spring–damper system, allowing its response to external disturbances to be adjusted through stiffness and damping parameters. The comparison between soft and stiff configurations demonstrated the expected trade-off: the

soft controller allowed larger displacement under lower interaction forces, whereas the stiff controller reduced displacement but generated higher forces and external joint torques.

- **Admittance control enables intuitive hand-guiding.** In the hand-guiding experiment, measured external forces were converted into bounded Cartesian velocity and position commands. This allowed the operator to physically guide the robot while velocity, acceleration, workspace displacement, and interaction-force limits were maintained. The resulting behavior was compliant and suitable for lead-through programming and collaborative manipulation.

- **Hybrid position/force control supports compliant contact tasks.** During the soft-object experiment, the robot combined position regulation in the tangential motion directions with force-aware behavior in the contact direction. The FR3 detected contact, compressed the compliant object, and completed a horizontal dragging motion while maintaining bounded external force and joint-torque values. This demonstrated the usefulness of hybrid control for tasks that require simultaneous motion and contact regulation.

- **Internal joint-torque sensing is sufficient for estimating physical interaction.** Bias-compensated external-force and external-joint-torque estimates were successfully used for contact detection, disturbance identification, hand-guiding, and resistance estimation. This reduced the need for an additional wrist force/torque sensor and simplified the experimental setup. However, the quality of the estimates remained dependent on accurate bias compensation, robot-model accuracy, and the configuration of the manipulator.

- **Resistance-aware assist-as-needed control can adapt the robot contribution during rehabilitation-inspired exercises.** In the shoulder and elbow experiments, the controller monitored the interaction force opposing the desired motion, reduced trajectory progression when resistance increased, and provided bounded assistance when the participant lagged behind the reference trajectory. The assistance was therefore applied according to task performance rather than continuously at its maximum value.

Overall, the results confirm that the FR3 can execute stable and bounded force-aware behaviors ranging from object manipulation to direct physical collaboration and adaptive rehabilitation-inspired assistance. The thesis therefore establishes a complete experimental framework for studying Cartesian impedance, admittance, hybrid position/force, and assist-as-needed control on a real torque-controlled manipulator.

5.2 Challenges and Limitations

Despite the successful experimental implementation, several technical and methodological limitations were identified.

- **External-force estimation and bias sensitivity.** The external forces were estimated from the robot's internal joint-torque measurements and dynamic model. Residual bias, joint friction, payload uncertainty, cable effects, and small model inaccuracies can affect these estimates. Force-bias calibration was therefore required before each interaction experiment, and the estimates were filtered before being used by the controller.

- **Dependence on robot configuration.** The mapping between joint torques and Cartesian forces depends on the manipulator Jacobian. Near poorly conditioned configurations, small joint-torque errors can produce larger Cartesian-force estimation errors. Careful initial positioning was therefore necessary before each experiment.

- **Manual controller tuning.** Cartesian stiffness, damping, virtual mass, force thresholds, assistance gains, and safety limits were selected experimentally. Parameters that performed well in one configuration were not always directly transferable to another task, participant posture, or movement direction.

- **Trade-off between tracking accuracy and compliance.** Increasing Cartesian stiffness improves tracking and disturbance rejection but also increases the forces applied during physical interaction. Reducing stiffness improves comfort and compliance but allows larger trajectory deviations. This trade-off was clearly observed in both the human-push and rehabilitation-inspired experiments.

- **Uncertainty in the biomechanical joint centres.** The shoulder and elbow trajectories depended on estimates of the corresponding joint centres and movement axes. Small calibration errors produced differences between the desired geometric arc and the measured human–robot trajectory. In the elbow exercise, additional deviation could also result from small movements of the elbow on the table.

- **Direction-dependent behavior during elbow motion.** Elbow flexion and extension produced different loading conditions because the robot had to support part of the weight of the participant's hand and forearm during the upward motion. This required additional support and direction-aware controller behavior.

5.3 Future Work

The results of this thesis provide several opportunities for extending the proposed framework.

- **Direction-dependent assistance and gravity compensation.** The rehabilitation controllers could use separate force thresholds, assistance limits, and support terms for different movement directions. In particular, the elbow controller could estimate the gravitational contribution of the forearm and provide only the additional support required during flexion.
- **Variable impedance control.** The Cartesian stiffness and damping could be adjusted online according to interaction force, tracking error, movement phase, participant confidence, or estimated fatigue. Low stiffness could be used during free interaction, while higher stiffness could be activated temporarily when stronger trajectory guidance is required.
- **Improved resistance classification.** Future controllers could distinguish between voluntary resistance, passive limb weight, involuntary movement, fatigue, sudden disturbance, and mechanical obstruction. This could be achieved by combining force, velocity, trajectory error, and temporal features.
- **Vision-based human-state estimation.** Cameras or depth sensors could be used to estimate limb posture, joint centres, movement range, and compensatory body motion. Vision could also detect unsafe postures or predict movement intention before significant physical contact occurs.
- **Sensor fusion with an external force/torque sensor.** Although the internal torque sensors were sufficient for the experiments, a wrist force/torque sensor could be used to validate the estimated Cartesian interaction forces. Combining both measurements could improve force accuracy and reduce sensitivity to model uncertainty.
- **Multi-contact detection and localization.** The torque-residual framework could be extended to distinguish contacts occurring at different links of the robot. This would improve safety when the participant interacts with the robot body rather than only with the end-effector.
- **Learning-based parameter adaptation.** Reinforcement learning, Gaussian processes, or other data-driven methods could be used to identify suitable impedance and assistance parameters. Any learning-based method should operate within predefined safety constraints and should not directly replace deterministic hard-force, speed, and workspace limits.

Chapter 6

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